DEVELOPMENT IN PRONUNCIATION ACCURACY THROUGH VISUAL FEEDBACK AND DRILLS: EVIDENCE FROM STOP-RHOTIC CLUSTERS IN LEARNERS OF SPANISH AS L2

by

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To my God and to my beloved family

-Psalm 18-

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LIST OF ACRONYMS

ACCESS: Automatization in Communicative Contexts of Essential Speech Segments

AE: American English

ALM: Audio-Lingual Method

CALL: Computer-Assisted Language Learning

CG: Control Group

DG: Drills Group

DM: Direct Method

ESL: English as a Second Language

EVF: Electronic Visual Feedback

F1, F2, F3: Formant 1, Formant 2, Formant 3

iR: Intensity Ratio

L1: First language

L2: Second Language

NAE: North American English

NNS(s): Non-native Speaker(s)

NS(s): Native Speaker(s)

NSE: Native Speaker(s) of English

NSS: Native Speaker(s) of Spanish

OC: Operant Conditioning

OT: Optimality Theory

PLATO: Program Logic/Learning for Automated Teaching Operation

SD: Standard Deviation

SRC(s): Stop-Rhotic Cluster(s)

V(CV): Vowel (Consonant Vowel)

VFP: Visual Feedback Paradigm

VFG: Visual Feedback Group

VOT: Voice Onset Time

ABSTRACT

Author: Hernández- Morales, Ana, J. MA
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Title: Development in Pronunciation Accuracy through Visual Feedback and Drills: Evidence from Stop-Rhotic Clusters in Learners of Spanish as L2.
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Previous literature argues that native-like pronunciation is impossible in non-primary language acquisition related to the age of L2 onset (Scovel, 1988; Long, 1990). Studies provide evidence that different types of instruction have a positive effect on L2 learners' productions (i.e., Derwing, Munro & Wiebe, 1998; González-Bueno & Quintana-Lara, 2011) and L2 learners may achieve native-like mastery of the target language phonetic features (Bongaerts, 1995). Some studies provide evidence that drills are effective for producing more native-like speech (Paulston, 1970; Yoshida & Fukada, 2014). Other studies provide evidence that visualization of the speech contours has a significant effect on improving suprasegmentals (i.e., Chun, 1989; James, 1979) and segmentals (i.e., Motohashi-Saigo & Hardison, 2009). For stop rhotic clusters (SRC), Spanish and English differ in: short vs. long lag stop voicing; rhotic manner (tap vs. retroflex), and epenthesis (presence vs. absence). Studies suggest that L2 learners may achieve native-like mastery of target language phonetic properties of SRC (Colantoni & Steele, 2006) but no known studies have compared the use of a visual feedback paradigm (VFP), Drills, and traditional instruction in their production. Three groups of native English speaker learners of Spanish recorded a pre-test, 3 trainings and post-test throughout one semester. The visual feedback group and drills group performed a visual and auditory discrimination task respectively, comparing their productions with those of NSS. The control group performed cultural activities. Target tokens, containing SRCs, were embedded in novel phrases and carrier sentences. While both trainings were equally effective for the durational parameters of the plosive and the rhotic segment, the other two features (rhotic manner and epenthesis) showed more resistance to be mastered. Results show that the acquisition of SRC seems to follow a difficulty acquisition hierarchy, but there is a strong co-dependence between the rhotic manner and the presence of an epenthetic vowel.

CHAPTER 1. INTRODUCTION

1.1 Introduction & Goals of the Present Study

The holistic linguistic competence of the L2 learner has been the main point of research for many years in the field of second language acquisition. Different approaches towards acquiring a native-like pronunciation in second and foreign languages have been established and proposed. Even though the role of pronunciation has not been entirely excluded from research in the second language classroom, it has been left behind, "mistreated" (Dutra & Costa, 2012, p. 62) and marginalized (Derwing & Munro, 2005) due to teachers' lack of confidence to teach pronunciation as a result of lack of training as well as lack of an appropriate incorporation into the curriculum (Celce-Murcia et al., 2010). For instance, some of the greatest difficulties that L2 learners find in their learning process is being identified as non-native speakers (NNS) due to their foreign accent (Lord, 2005). If teachers would be able to spend the same amount of time and give the same importance to both segmental and suprasegmental features in the L2 classroom; acquiring and producing accurate, comprehensible, intelligible and fluent speech could be achieved despite existing age-related constraints (Long, 1990; Scovel, 1998). Even if L2 learners produce less fluent and heavy accented productions, it does not necessarily impede having a comprehensible and intelligible enough communication. However, a feature such as accentedness can cause a negative perception by the side of the native listener towards the non-native speakers (NNS; Derwing & Munro, 2009; Lippi-Green, 1997; Munro, 2003) and it can also be responsible for breakdowns in the communication (Agostinelli, 2012; Isaacs, 2009).

Pronunciation teaching has been approached by different methods throughout the years. For instance, the Audio-lingual Method (ALM) gave priority to memorization, drills, repetitions and learning language patterns aiming at having effective and more accurate L2 pronunciation. Because these techniques seemed to be effective in helping students to acquire 'correct' L2 pronunciation habits or behaviors, this method used the majority of the instruction in the Audio-lingual classroom, taking the attention away from other skills such as reading and writing (Saville-Troike, 1973). This method was based on behavioral conditioning in which a learner might acquire a correct behavior after repeated 'trials' and

even after drilling has ceased, that learner should be able to keep those 'good behaviors' (VanPatten & Williams, 2012). "Continuous repetition, therefore, is an important factor in developing new behaviors (VanPatten & Williams, 2012, p. 18). Due to the difficulty experienced by L2 learners in acquiring real communicative skills and more fluent speech patterns by means of this kind of explicit learning methods (Oxford, Lavine & Crookall, 1989), researchers in second language acquisition and pronunciation turned their focus to more implicit methods (Isaacs, 2009). With the Communicative Approach (CA), meaning, context and authentic language were the main focus (Oxford, Lavine, & Crookball, 1989) and the importance of meaning overshadowed the importance of form. This implied that producing intelligible and comprehensible enough speech (rather than accurate and fluent) became more important than having accurate and fluent speech.

Teaching second language pronunciation has increasingly been recognized as one of the most important components of the L2 classroom (Silveira, 2002) as it encompasses not only linguistic but also discourse, strategic and sociolinguistic competences (Morley, 1994; Silveira, 2002). With the upcoming of new technologies, methods based on providing L2 learners not only an auditory but a visual representation of their speech have gained a lot of territory in the pronunciation teaching. The addition of a visual component into pronunciation instruction enable L2 learners to identify the differences between their two language sound systems and produce more native-like speech while comparing them with native speakers' (NSs) productions. For instance, Visual Feedback Paradigms (VFP), speech recognition programs or a combination of both have been used in the second language classroom in the last forty years to teach both suprasegmental (Chun, 1989, 1998; De Bot & Mailfert, 1982; De Bot, 1983; Hardison, 2004; Weltens & De Bot, 1984) and segmental (Motohashi-Saigo & Hardison, 2009; Carey, 2004; Olson, 2014) features. Some studies focusing in suprasegmental features have provided evidence that the perception of intonation patterns using visual and audio-visual feedback is beneficial for L2 learners (de Bot, 1982, 1983). Anderson-Hsieh (1992) claimed that because the learners have access to real-time representations of both, native targets and their own productions, "they can more easily replicate" (p.61) the pattern they are looking at. Furthermore, studies in segmental features have tested voiceless plosives while implementing a VFP (Olson, 2013, 2014) with the target features in isolation (Saito, 2007) or embedded in novel phrases (Olson &

Offerman, 2013). However, no studies have compared the effectiveness of using a VFP and drills in the production of complex segments such as SRC. The production of voiceless plosives (/p/ /t/ /k/) in stressed syllables (Lord, 2005), the realization of the rhotic in specific contexts (Hualde, 2005) and the acquisition of an epenthetic segment, present an obstacle for English native speakers (ENS) learners of Spanish. As such, realization of the voiceless plosives in English implies having longer VOT values due to the aspiration of these segments. In contrast, Spanish produces voiceless stops with a short-lag due to lack of aspiration (Lisker & Abramson, 1964). The transfer from English phonetic properties into Spanish productions are noticeable when ENS learners of Spanish produce speech with longer VOT values (English-like). Additionally, English native speakers do not have the difference between the flap and trill 'r'. In fact, they produce the voiced apico-alveolar rounded retroflex non-lateral liquid /ɹ/ (Hammond, 2001) while Spanish native-speakers produce these same complex segments with the Spanish tap /ɾ/. The third phonetic feature that Spanish learners must acquire to produce more native-like speech regarding SRC is the production of the epenthetic vowel between the stop and the rhotic. This entirely involuntary process happens in the Spanish native speakers' speech and seems to be the most difficult phonetic feature to acquire among the three that are being observed (Colantoni & Steele, 2006).

For the present study, participants were trained using a VFP or drills in order to compare the effectiveness of visual input versus auditory input. The Visual Feedback group performed a self-assessment task consisting in comparing their own spectrograms and sound waves with those of native speakers of Spanish (NSS); participants who received these Visual Feedback trainings were expected to raise their awareness regarding the differences between their productions and those of NSS at the end of each session and in the overall training. By the other side, the Drills group completed a self-assessment task to help them realize differences between the auditory perception of their own speech and that of a NSS. As a result, they would be able to mimic the production of the segments to be more Spanish-like and acquire the following three phonetic features for SRC in English and Spanish: long-lag vs. short-lag, retroflex /r/ vs. flap /ɾ/ and presence or absence of an epenthetic vowel. The study had four aims: first and second, to find out if the proposed trainings (VFP and drills) are effective to improve NNS's accuracy in the production of

SRC phonetic features; second, to see if one of the trainings outperform the other; third, to understand if the phonetic features behave similarly after instruction and if they seem to follow the acquisition difficulty hierarchy proposed by Colantoni and Steele (2006).

1.2 Outline of the Study

In chapter 2, a detailed review of the previous literature regarding the importance of teaching pronunciation in the L2 classroom is presented. Three main sections include: a description of some issues that L2 learners face when producing L2 speech and their difficulties attaining a native-like attainment pronunciation and how these problems lead to and highlight the importance of teaching pronunciation in the L2 classroom.; second, two practices and methods that are used throughout L2 pronunciation instruction are described: from the first methods used in L2 teaching (Drills and Repetitions) to the state-of-the-art methodologies that use technology as part of the instruction (use of Visual Feedback Paradigms); third, this chapter provides a definition and discussion of the existing differences in both English and Spanish regarding the three phonetic features aimed to attain through the trainings: long-lag vs. short-lag values, articulation of the flap /r/ and presence or absence of an epenthetic vowel. In order to conclude Chapter 2, the research questions and hypotheses of the present project are posed, summarizing how the two trainings used in this project are expected to positively influence the attainment of the three phonetic features in participants' L2 speech, making their pronunciation closer to that of a NSS. Chapter 3 presents a detailed description of the participants for each experimental group (Visual Feedback Group and Drills Group) and the Control Group followed by the discussion of the methodology and procedures carried out before, during and after each one of the trainings. Chapter 4 contains the results obtained after eliciting the data for each experimental group and the control group. Each subsection on this chapter addresses results per type of phrase: Carrier Sentences (subsection 4.1) and Novel Phrases (subsection 4.2). Each one of the subsections subdivides itself to present the results for each one of the phonetic features targeted (VOT length, rhotic manner and epenthesis). Chapter 5 summarizes the significant findings of the study and provides the main conclusions. Finally, Chapter 6 includes the pedagogical implications, limitations of the study, future directions and a general conclusion.

CHAPTER 2. LITERATURE REVIEW

This chapter addresses the use of Visual Feedback Paradigms (VFP) and Drills (and repetitions) as two principal approaches for teaching pronunciation to L2 learners. These approaches have emerged from research and pedagogical practice over the last 70 years and have been widely used to provide pronunciation instruction in the L2 classroom and contrasts the advantages and disadvantages of each technique compared with no-instruction methods.

Section 2.1 describes the importance of producing intelligible, comprehensible and less accented L2 speech, and exhibits the potential benefits for learners if these issues are addressed in the L2 classroom. Also, it provides an overview on the issue of incorporating effective pronunciation teaching techniques into the L2 classroom curriculum and the creation of adequate materials that meet the upcoming challenges of the L2 classroom as well as learners' needs. Section 2.2 highlights the need for L2 pronunciation instruction research due to the fact that pronunciation has been under-studied, left behind and marginalized (Derwing & Munro, 2005). Section 2.3 discusses two approaches (Drills and Visual Feedback Paradigm) that have been used in the L2 classroom and presents the most important concepts of each. Section 2.4 presents and describes three phonetic features that are difficult to achieve for native English speaking learners of Spanish when producing segments composed by a voiceless stop followed by a rhotic segment. These three features that differ in English and Spanish are: long-lag vs. short-lag, retroflex vs. flapping of the rhotic segment and absence vs. presence of an epenthetic vowel respectively. Section 2.5 closes the chapter with a summary of the approaches presented, details gaps present in previous research, and poses questions and hypotheses for the present research.

2.1 Pronunciation: Intelligibility, Comprehensibility & Accentedness

Communication relies on a speaker's ability to produce and perceive speech, and as such pronunciation is a crucial component of communication. With respect to L2 learners, pronunciation can be described in terms of three crucial concepts: intelligibility, comprehensibility, and accentedness. These three intertwined concepts relate to the degree

with which a message is successfully interpreted, the difficulty with which that message is decoded, and how the production of that message relates to the local variety (Derwing & Munro, 1995a; Derwing & Munro, 2005; Derwing & Munro, 2009). In L2 learning, both the production and perception of speech play an important role for native and non-native speakers (NNSs) because learning a language involves the interaction of not only phonetic but also semantic, syntactic, pragmatic and discourse features (Cenoz & Lecumberri, 1999).

According to Derwing and Munro (2012), intelligibility, comprehensibility and accentedness are constructs necessary to understand that L2 pronunciation is learnable (Derwing & Munro, 2009). In the first place, intelligibility has been defined as "the apprehension of the message in the sense intended by the speaker" (Nelson, 1982, p. 63). Munro and Derwing (1995a) have provided a similar definition. They define intelligibility as the extent to which it is understood by a NS. Miller (2012) notes that "the speaker needs to physically produce the sounds of the target language with enough accuracy to be understood" (p. 69). Some authors have proposed that language learners need just a "comfortably intelligible pronunciation" in order to communicate (Abercrombie, 1949, p. 120) (see also Gilbert, 1980; Pennington & Richards, 1986; Crawford, 1987; Morley, 1991). However, other authors have argued that an improved intelligibility should be the main objective of all pronunciation teaching (Derwing & Munro, 1995).

Researchers have tried to assess intelligibility in many different ways, although none of the methods used seem to provide a fully adequate measurement of how much someone actually perceives and understands in a communicative act (Derwing & Munro, 2009). Throughout the years, native listeners have been asked to transcribe NNSs productions (usually recordings) in order to count how many words they can transcribe correctly (Lane, 1963) or recognize key words to calculate the percentage of success during a recognition task (Barefoot, Bochner, Johnson & von Eigen, 1993). Other attempts to measure intelligibility rely on seeing how accurate a NS can paraphrase a message produced by a NNS (Brodkey, 1972), looking at the effectiveness of speech synthesizers (Greene, Logan & Pisoni, 1986; Logan, Greene & Pisoni, 1989) or using a whole set of activities that include dictations, indication of true/false sentences, comprehension questions and a summary of the content of the message that has been uttered (Derwing &

Munro, 2009, p. 479). Munro and Derwing (2011) have also discussed other linguistic and paralinguistic factors such as rate of speech, speech clarity, voice quality, word choice, ambient noise which may affect the assessment of intelligibility in a non-native production.

In some cases, along with errors in other linguistic elements, an exchange of sounds in the target language (found within minimal pairs for example) can lead to misunderstandings and confusion in a conversation between a NS and a NNS, stopping the NNS from conveying the message intended or making it less understandable for the NS. For example, a NS of Spanish would pronounce the word 'todo' *all* ([to.ðo]) with the fricative /ð/; however, when a NS of English pronounces the same word, s/he would be heard as pronouncing 'toro' *bull* ([to.ɾo]). This effect is due to the fact that in English the alveolar tap [ɾ] can be produced as an allophone of North American English (NAE) phonemes /t/ and /d/ as in '*butter'* or *'ladder'* (Face, 2006). Similarly, the contrast between the two Spanish rhotics (trill and flap <r>) in intervocalic position may present difficulty for NS of English as the trill /r/ is non-existent in English. Therefore, the Spanish learners must readjust their parameter where the English allophone [ɾ] has its own phoneme /ɾ/ in the Spanish inventory. As a result, a NS of Spanish could hear 'pero' *but* ([pe.ɾo]) instead of 'perro' *dog* ([pe.ro]) or 'caro' *expensive* ([ca.ɾo]) instead of 'carro' *car, cart* ([ca.ro]) (Face, 2006). Simões (1996) provides another example of interchanging sounds that can happen within vowels in which a NNS were to produce the word 'mesa' *table* ([me.sa]) but instead produces 'misa' ([mi.sa]) *Catholic mass*; this, because in some contexts the English sound for 'e' is the Spanish sound /i/.

One of the conclusions of a study conducted by Flege (1992) about intelligibility of English vowels produced by Dutch speakers was that one would not expect a perfect intelligibility for NNSs due to the fact that not even native speakers of English (NSE) could identify with a 100% of accuracy other native speakers' productions. However, it was found that if the acoustic properties of the Dutch vowels were closer to the acoustic properties of the English ones, L2 speakers were more efficient producing the targeted vowels. Flege and Munro (1994) focused their study in the word '*taco*' produced by Spanish and English monolinguals and Spanish-English bilinguals. In the perceptual task, bilinguals were able to recognize the native language of a speaker based on listening how they produced the word '*taco*'. In the production task, the most important finding was that

the VOT for /t/ (as well as for the other three segments) was more Spanish-like for bilinguals because they acknowledged that this word was a Spanish word borrowed in English contexts. Later on, Flege (1995) (see also Flege & Munro, 1994) argued that it seems unlikely that a native speaker of one language can simply substitute one segment to the closest one in the L2 because there is still a motor inability to produce sounds that are not found in the L1 and that even the closest approximation to that segment will still carry L1's phonetic characteristics. Even when the context in which an utterance is produced might help the NS to understand the message a L2 learner is trying to convey, and minimize the influence of the L1, a constant mix of sounds can cause not only more confusion but it can arise negative attitudes towards the learner or even discrimination (Lippi-Green, 1997; Munro, 2003; Derwing & Munro, 2009).

Second language learners are not only challenged by having to produce intelligible but also comprehensible speech in order to communicate their ideas in an understandable way. In this particular case, comprehensible speech refers to how understandable is the production of one NNS when producing speech in the L2 and the degree of difficulty the NS has to understand what is being said. To this effect, comprehensibility is the degree of difficulty with which the native speaker (NS) can understand the utterances of a NNS (Derwing, Munro & Wiebe, 1998) (See also Munro & Derwing, 1995a, Derwing & Munro, 1995b, 1997, 2005; Derwing, Thomson, Foote & Munro, 2012). In other words, comprehensibility refers to how easy or difficult it is for a NS to understand a NNS production. According to Derwing and Munro (1997), comprehensibility has been more difficult to measure because its definition has been used interchangeably with that of intelligibility (Gass & Varonis, 1984) or it has been referred as "being at a higher level of understanding than intelligibility" (Derwing & Munro, 1997, p. 2).

The measurements are usually judgment ratings from both native and non-native listeners (Munro & Derwing, 1995, 1999; Derwing & Munro, 1997, 2005, 2009; Derwing, Munro & Wiebe, 1998) to assess how easy or difficult is to recognize or understand native speakers. There are several studies that look how learners of English as L2 and different L1 language backgrounds understand NS's speech (Derwing, Munro, & Wiebe, 1998; Varonis & Gass, 1982). It has been claimed that there are a wide variety of factors that can affect judgments of comprehensibility (Derwing & Munro, 1997) such as the familiarity

with the topic of the conversation, particular talkers or foreign accents (Varonis & Gass, 1984) or even a 'poor' grammar proficiency. These findings are subject to some ongoing debate, as some have found that grammatical errors did not interfere with comprehension as much as non-native-like pronunciation (Gynan,1985), while others have shown that grammar was actually more important (Ensz,1982). Furthermore, Politzer (1978) found that neither grammar nor pronunciation affected listening comprehension in English-accented German but vocabulary errors did. However, Fayer and Krasinski (1987) found that non-native patterns in pronunciation as well as hesitation contributed to disturbance in listeners' attention and listeners' displeasure. Following this, Agostinelli (2012) argued that errors made in pronunciation tend to create more communicative breakdowns than morphosyntactic errors. It is worth mentioning that as comprehensibility is a feature assessed merely by the subjectivity of the listeners, biases and great variation could be found in the ratings (Derwing, Munro & Wiebe, 1998) Moreover, results can be influenced by the differences in methodologies used to collect the data (Schairer, 1992) or could be even conditioned by the languages under study (Munro & Derwing, 1995). In spite of the ongoing debate regarding the nature of comprehensibility, it has been attested that teaching pronunciation has a direct and positive effect on both intelligibility and comprehensibility (Derwing, Munro & Wiebe, 1997, 1998; Derwing & Munro, 2009). This idea is supported by many researchers (e.g., Derwing & Munro, 2005; Field, 2005; Levis, 2005) and English as a Second Language (ESL) instructors (e.g., Foote et al., 2012) who claim that intelligibility and comprehensibility should be the goal of language classrooms, rather than the attainment of native-like speech, which, for most adult learners, is an unrealistic goal (Flege, Munro & MacKay, 1995).

As shown in the sections above, comprehensibility and intelligibility are essential factors for communicating in a second language. A message must be not only understandable but also take little effort to be understood. However, there is still a third factor that plays an important role in transmitting a message even more efficiently; that is accentedness. Accentedness is defined as "how different a pattern of speech sounds to a local variety" (Derwing & Munro, 2009, p. 478), in other words, the phonetic difference between a L2 leaner (or any NNS) in regards to a specific group of speakers. Non-native

speakers may face positive and negative consequences for having a foreign accent (Flege, 1988; Derwing & Munro, 2005).

On the positive side, having a foreign accent signals the NS interlocutor that the NNS may require modified and simplified input in order to comprehend the message that s/he is being communicated (Gass & Varonis, 1984; Derwing & Munro, 2005). It has been claimed that even though having an accent can cause some problems in the communication, it does not necessarily impede having an intelligible and comprehensible production (Derwing & Munro, 2009). On the negative side, accented productions may lead to the establishment of negative stereotypes, discrimination (Munro & Derwing, 1995a; Munro & Derwing, 1995b; Derwing & Munro, 1997, 2005, 2009), judgments or intolerance in some circles (specially employers) (Sato, 1991) because heavy accented speakers are usually associated to low levels of proficiency in the L2 or ignorance of the language (See also Gluszek & Dovidio, 2010; Purkiss, Perrewé, Gillespie, Mayes & Ferris, 2006; Kim, Wang, Deng, Alvarez, & Li, 2011). Several other studies have actually shown that NSs tend to downgrade NNSs because their accented productions (Anisfeld, Bogo & Lambert, 1962; Brennan & Brennan, 1981; Kalin & Rayko, 1978; Lambert, Hodgson, Gardner & Fillenbaum, 1960; Ryan & Carranza, 1975; Munro & Derwing, 1995a; Munro & Derwing, 1995b; Derwing & Munro, 2009). Gluszek and Dovidio (2010) expanded this argument explaining that NSs will determine if a NNS can be considered non-fluent in the target language by means of his/her accent, even if the NNS uses correct syntactic and grammatical rules. For these reasons, accentedness has been viewed as a "bad thing" (Munro & Derwing, 1995a, p. 74), "a kind of disorder" (Derwing & Munro, 2009, p. 477), "a problem" (Munro & Derwing, 1995a, p. 74), and even "as a language pathology" (Munro & Derwing, 1995a, p. 74) that needs to be *treated, intervened* or *eradicated.* Not only that, but according to Derwing and Munro (2009), native and nonnative listeners are very sensitive in recognizing speech that is different from their own variety and usually provides NSs with unwanted information about the L2 learner's geographical and socio-cultural background (Cenoz & Lecumberri, 1999). For instance, NSs could identify non-native productions by listening to short speech samples such as words, phrases, individual segments or speech samples of just 30-millisecond (ms) (Flege & Hillenbrand, 1984). Munro, Derwing and Burgess (2003) found that NSs could detect an accented production

even if the sample provided was a single word backwards. Previous research has proposed that NSs rely on cues to detect accented speech; segmental variations, prosodic factors, and even voice quality help listeners to do so (Esling & Wong, 1983; Flege & Hillenbrand, 1984; Derwing & Munro, 2005).

To contrast some of these negative effects, many L2 learners feel that receiving pronunciation instruction is crucial to improve their speaking skills (Levis & Grant, 2003; Saito, 2001, 2012, 2013) and are willing to receive instruction (See Section 2.2.2) in order to have more native-like productions (Bongaerts et al., 2000; Birdsong, 2003; Colantoni & Steele, 2006). Even when communication does not merely depend on the accent of the L2 learner, a heavy accent may interfere in the production of an intelligible and comprehensible message with a native interlocutor (Munro & Derwing, 1995a, 1995b; Derwing & Munro, 1997, 2009; Derwing, Munro & Wiebe, 1998). This means that even though highly intelligible and comprehensible productions are not specifically rated as less accented, highly accented productions are indeed perceived as less intelligible and less comprehensible (Munro & Derwing, 1995a; Munro & Derwing, 1995b). In sum, while it is necessary to address pronunciation instruction to help L2 learners achieve not only less accented productions but also more comprehensible and intelligible speech, the literature has shown that addressing these factors could improve (if not totally eliminate) the negative effects of having a non-target pronunciation by means of miscommunications, misunderstandings, negative attitudes, or even discrimination towards the L2 learners. Recognizing the importance of having intelligible, comprehensible and less accented speech could help instructors and learners to better understand why pronunciation instruction is necessary in the L2 classroom and what other external factors affect the acquisition of ‘good’ pronunciation.

2.2 Pronunciation instruction: The Need, The Desire, and The Lacks

2.2.1 The Need

Providing L2 learners a communicative competence by means of pronunciation (Morley, 1996) is essential for developing an efficient system of communication that allow speakers to have intelligible and comprehensible productions, especially if one of them is

a NNS (Gonzalez-Bueno, 2001, p. 135). Three positions could be delineated to highlight the importance of having pronunciation instruction within the L2 classroom: first, as pronunciation instruction has been under-studied (Derwing & Munro, 2009), and more research in this field would approach this "problem" despite the subjectivity of these type of studies. Second, even when "common views on pronunciation teaching have been pessimistic" (Derwing & Munro, 2009, p. 481), it has been shown that pronunciation is learnable (Derwing & Munro, 2009). Finally, studies have shown that there are several benefits for L2 learners in receiving pronunciation instruction (Gilbert, 1984; Brown, 1991; Celce-Murcia et al.,1996; Morley, 1996).

In the first place, pronunciation instruction has been neglected for years, marginalized (Derwing & Munro, 2005), and overshadowed by the study of other aspects of language such as grammar and vocabulary. An increasing body of studies have recognized L2 pronunciation instruction as one of the most important components in the L2 classroom teaching (Silveira, 2002), despite claims such as the 'difficulty to teach this skill' (Morley, 1994). Elliot (1995, 1997) also highlights that pronunciation and pronunciation teaching is an important factor for L2 speech. Although it has been left behind as a mere linguistic competence instead of being considered as a "communicative competence" (Morley, 1994, p. 90), L2 pronunciation teaching encompasses not only these 'mere linguistic skills' but also discourse, sociolinguistic and strategic competences (Morley, 1994; Silveira, 2002). In other words, receiving pronunciation instruction involves the use of not only many new constructs (from different linguistic fields such as phonetics or syntax) but it also comprises a whole set of extra linguistic skills to effectively convey an intended message.

Second, pronunciation is learnable (Derwing & Munro, 2009), despite being identified as one of linguistic domains most vulnerable to critical period effects (Birdsong, 1999, p. 17). This reinforces the idea that acquiring native-like pronunciation is possible despite L1, and age-related constraints (Scovel, 1988; Long, 1990). Flege (1992) proposes that having a non-native accent is not a result of lacking the ability to pronounce segments in the L2 but it is an indirect consequence of having a developed L1 phonological system at the time the L2 learning process began (Birdsong, 1999). Therefore, non-native speech is mainly due to an increasing difficulty in establishing new L2 phonetic categories rather

than a deficiency in the L2 learners' abilities. As stated by Silveira (2002), the teaching of pronunciation is vital in order to provide the L2 learners with perceptual and productive skills that help them reconfigure some of the parameters of their "interlanguage phonology" (p. 105). Thus, the need to incorporate pronunciation instruction into the curriculum and the increasingly number of accent reduction programs have allowed L2 learners to "learn to speak the language as naturally as possible, free of any indication that the speaker is not a clinically normal native" (Griffen, 1980, 1991, p. 182). Although there is no indication of studies linking pronunciation instruction and the elicitation of accents (Derwing & Munro, 2005); it has been shown that reducing the accentedness improves the perception of intelligibility and comprehensibility by NSs (Munro & Derwing, 1995a, 1995b, Derwing & Munro, 1997, 2005, 2009) and achieving native-like speech patterns is possible (Bongaerts et al., 2000; Birdsong, 2003; Colantoni & Steele, 2006).

In addition, pronunciation instruction can help to reduce negative perceptions and attitudes towards the NNSs (Munro & Derwing, 1995a, 1995b, Derwing & Munro, 1997, 2005, 2009) as a consequence of having a foreign accent and causing difficulty while communicating. Derwing (2003) presented some of the comments obtained in this study which clearly display negative attitudes towards L2 learners. For example, "people get angry if they don't understand me", "sometimes people chose not to understand", or "the police didn't pay attention to what I said" (p. 558). Then, pronunciation instruction is a valuable tool for L2 learners to minimize these negative attitudes (Munro & Derwing, 1995a, 1995b; Derwing & Munro, 1997, 2003, 2005, 2009) and be included in a conversation without any problems or restrains. Lastly, there are a number of clear benefits associated with pronunciation instruction and decreased accentedness. As discussed above, a less accented production does improve listeners' perceptions regarding L2 speech, rating L2 speakers as more intelligible and comprehensible (Munro & Derwing, 1995a, 1995b; Derwing & Munro, 1997, 2009). Due to the quasi-categorical dependent relationship among intelligibility, comprehensibility and accentedness (Derwing & Munro, 1997, p. 1; Munro & Derwing, 1995, Derwing, Munro & Wiebe, 1998), more fluent and more native-like speech would mean that NSs would comprehend and understand non-native speech giving learners confidence to speak and not feel frustrated (Yates & Zielinski, 2012; Alghazo, 2015). As Schmidt (1990) claimed that "learning without some form of attention

(or noticing or detection) is not possible" (p. 7), explicit pronunciation instruction is necessary and required to reduce accentedness and consequently, comprehensibility and intelligibility of L2 speech.

2.2.2 The Desire

There is an increasing desire among L2 learners to receive instruction to improve their pronunciation (Derwing & Munro, 2005, 2009; Makarova & Ryan, 2000; Peacock, 1999; Yates, 2010) and sounding less accented, more fluent, more intelligible and therefore, more comprehensible. L2 learners do not only want to produce syntactically accurate and 'correct' utterances that show an enriched vocabulary but they also want 'more native-like, intelligible, and comprehensible productions', but in order to achieve that, L2 learners would need a 'high level of fluency' (Simões, 1996). From the researcher perspective, although some have argued against expectations that learners will sound like native speakers (Celce-Murcia et al., 2010), most argue that "learners should be encouraged to reach their full potential" (Derwing & Munro, 2005, p. 384), and that it would be "churlish to deny them [L2 learners] such an objective" (p. 184).

In a study performed by Derwing (2003), the results showed just a 3% of the ESL adult learners considered pronunciation important as part of the classroom instruction. However, the remaining 97% expressed their desire to achieve better speaking skills which would be reflected in having a perfectly native-like pronunciation (Derwing & Munro, 2003, 2005, Scales et al., 2006). For instance, many L2 learners (for EFL learners see Makarova & Ryan, 2000; Peacock, 1999) think that learning pronunciation techniques or strategies are vital to improve their speaking skills and such learners are eager to receive pronunciation instruction (Levis & Grant, 2003). As such, it is clear that pronunciation instruction is important for language learners because they want to improve their speaking skills in order to be perceived as more fluent, less accented and sound more native-like. Even if the learners are aware that they might not achieve a complete native-like pronunciation, receiving instruction in this matter seems to have more benefits than setbacks. First, it may help L2 learners to find a connection between what they learn in the classroom with real life situations (Celce et al., 2010); second, it would contribute to them in being easily understood by NSs who at the same time will avoid simplifying the language

and finally, NSs will perceive their interlocutors as speakers with an increasing oral proficiency in the L2 which would positively impact the perception the NS has towards the NNS.

2.2.3 The Lacks

Some of the problems of teaching pronunciation within the L2 classroom stem from two issues described below: first, lack of teacher training and second, lack of materials to instruct pronunciation.

2.2.3.1 The Lack of Teacher Training to Instruct Pronunciation

Even when many teachers are constantly looking for ways to implement pronunciation instruction within the L2 classroom (Levis, 2007), there are still two unresolved issues related to the lack of teacher training to instruct pronunciation: first, the lack of formal training teachers receive on how to teach and incorporate pronunciation into the classroom agenda and second, teachers' lack of confidence to teach pronunciation.

First, Levis (2005) highlighted that many teachers rely on their own intuitions when required of them to explain pronunciation techniques. In other words, when learners need a thorough explanation of how sounds are produced or how adjacent segments affect the place or manner of articulation of each other (especially in their L2), teachers usually say what their intuition and their own experience tell them, not what research has found. Even if second language instructors have the pedagogic competence in the L2, it does not necessarily imply that the instructor has the linguistic competence to explain other aspects of the language since teaching pronunciation "involves the ability to identify specific aspects of language" (Dalton & Seidlhofer, 1994, p. 67). This supports Derwing and Munro's (2005) claim that many L2 teachers do not possess the skills, or the training to explain formal aspects of the language related to pronunciation (Breitkreutz, Derwing, & Rossiter 2002; Burgess & Spencer, 2000; Derwing, 2010; MacDonald, 2002; Olson, 2014). Isaacs (2009) also claims that teachers lack the skills to incorporate pronunciation techniques in the classroom agenda and usually fail to provide a consistent pronunciation instruction and accurate guidance for the L2 learners. Olson (2014) for example, shows that language instructors spend about eight minutes teaching pronunciation per week

(*M*= 8.1; *SD*= 4.6) which means that they spend an average of 2.6 minutes per class teaching pronunciation to their L2 learners. In the same study, Olson (2014) provides evidence that at least a 77% of the instructors feel that they were spending "less than the just right amount" (p. 51) of time on pronunciation.

This leads to the second issue related to teachers' lack of confidence to teach pronunciation. Derwing (2010) states that "L2 teachers are often worried that they aren't well prepared to teach pronunciation or even to incorporate some pronunciation activities into their regular language classrooms. They feel as though they don't know where to start" (p. 27) so they do not incorporate pronunciation instruction at all or they leave it aside to teach it when there is 'extra time'. In addition, while a growing body of literature has begun to address L2 speech production and perception (i.e., *Journal of the Acoustical Society of America, Journal of Phonetics, Language and Speech)*, it is "rarely cited or interpreted in teacher-oriented publications" (p. 382). If L2 teachers are not aware of this literature or have difficulty understanding it (because it is focused to a specific and specialized public) (Derwing & Munro, 2005), it is not surprising that it is not incorporated in the classroom.

2.2.3.2 The Lack of Materials to Instruct Pronunciation

Despite the "popularity of accent reduction programs" (Munro & Derwing, 1995a, p. 74), the increasing desire of L2 learners to receive pronunciation instruction (Derwing, 2003; Derwing & Rossiter, 2002; Harlow & Muyskens, 1994; Foote, Holtby, & Derwing, 2011; Timmis, 2002), and teachers' will to provide this type of training (Derwing & Munro, 2005, 2009); not having a systematic way of deciding what, when and how to teach pronunciation (Derwing & Foote, 2011) seems to present the biggest challenges for language teachers and materials designers in three ways: nature of speech, integration into the curriculum and the content of these resources.

Also, even if a wider variety of contexts that expose learners to more real contents and experiences with the L2, successful learning depends on meeting the learning objectives which at the same time relies on how the curriculum is designed and learners' existing abilities and needs (Alghazo, 2015; Nation & Macalister, 2009; Richards, 2001). Therefore, not incorporating pronunciation instruction into the curriculum of the L2 classroom (Alghazo, 2015) would more likely result in creating materials (textbooks,

course books, CDs, etc.…) that are based on materials developers' intuition (Derwing & Munro, 2015) and do not provide appropriate, effective, assertive and engaging activities focused on improving learners' aptitudes. The materials that are usually included in the curricula consist of textbooks, software, computers, projects, visual aids, and homework sheets among others (Alghazo, 2015, p. 318). However, nowadays textbooks still represent the main source of reference in the language instruction and set the framework for what learners and teachers should follow (Alghazo, 2015; Jones, 1997; Maley, 2016; Ur, 1999). According to Alghazo (2015), even if those textbooks address phonetics and phonology, they lack the pedagogical part (Alghazo, 2015, p. 319) except for a few that have incorporated this component (e.g. Celce-Murcia et al., 2010). It has been suggested that resource materials should be used instead of course materials (Maley, 1995, 2016; Prabhu, 1989), the use of teacher-made or student-made materials (Maley, 2016). Others have suggested to teach segments with higher functional load (Catford, 1987; Munro & Derwing, 2006) or that are considered more important for successful communication as suprasegmentals (Derwing, 2008; Field, 2005; Hahn, 2004), primary sentence stress (Hahn, 2004), word stress (Field, 2005; Zielinski, 2008) and speaking rate (Munro & Derwing, 2001) which are known to affect intelligibility and comprehensibility (Derwing & Munro, 2005; Munro & Derwing, 2001; Derwing, 2008).

Finally, and despite the existing confusion on what, how and when to teach pronunciation (Derwing & Foote, 2011); a wide variety of materials are available for language teachers (Celce-Murcia et al., 2010; Gilbert, 2004; Grant, 2010; Hewings, 2004; Yates & Zielinski, 2009). Textbooks, course books, CDs have been designed by material developers usually focusing in one or two aspects of L2 pronunciation such as rhythm or intonation (for intonation see Levis, 1999) because they are relevant to conveying meaning in communication (Morley, 1987; Gilbert, 1993). However, these resources do not usually provide 'enough' content (in terms of practice) to teach pronunciation (Arteaga, 2000; Derwing, Diepenbroek & Foote, 2012; Ellis, 1997; Thomson & Derwing, 2004) or "authentic, context-rich activities that provide focused practice for the specific area of pronunciation to be targeted" (Isaacs, 2009, p. 4). According to a survey conducted by Foote et al. (2011), a total of 18.3% of instructors reported not using the pronunciation activities proposed in their textbooks in contrast to a 60.2% that indicated the use of these

activities. However, a 21.5% of the instructors indicated that their textbooks did not provide any source of pronunciation instruction which may indicated that pronunciation instruction may represent the greatest challenge for material developers. Textbooks for instance, are highly valuable in the L2 classrooms because for many language learners (if not the majority) this is the only context where they can obtain explicit pronunciation instruction and feedback (for ESL learners see Derwing, Diepenbroek & Foote, 2012) from 'someone' who has been trained in this matter.

In order to include pronunciation, material developers have to seek which aspects of speech should be included, what kind of material would fit *most* of the learners, what are the segments that should be targeted and how all these could be included into a very strict curriculum constrained by time and plenty of content to be taught. To sum up, the correlation between having more native-like speech, and mastering L2 features seems to rely not only in the instructors' or learners' skills but also in the importance of having adequate materials that allow language instructors to meet their students' needs in the best possible way, sometimes limited by the fact that many of the materials are decontextualized, focused on one specific skill; a lot more are insufficient or for the fact that there is no agreement on what, how and when to teach pronunciation, many of the materials or research done is "almost completely divorced from modern language teaching" (for intonation see Levis, 1999, p. 37) or is inaccessible to professionals "without specialized knowledge in phonetics" (Derwing & Munro, 2005, p. 382) which causes instructors' lack of motivation to teach pronunciation or by the contrary, if pronunciation is taught, the activities are inefficient or inappropriate for the learners causing confusion or even disengaging learners from this learning process.

2.3 Approaches in Pronunciation Instruction in the L2 Classroom

Given the importance of pronunciation instruction for intelligibility, comprehensibility and accentedness; pronunciation has been addressed in the L2 classroom through different approaches, more specifically for the present paper: drills and repetitions and Visual Feedback Paradigms.

2.3.1 Drills and Repetitions

In modern language teaching, drills and repetitions are most commonly linked to behaviorist theories, especially operant conditioning. Operant Conditioning (OC) is based on B.F. Skinner and Pavlov's presumptions that language is a behavior that can be acquired, modified and conditioned by means of reinforcement of the appropriate behaviors and discouragement of the bad ones. In the following section the background, claims and contributions that still being taken into account in the L2 classrooms of the Audiolingual Method (ALM) will be described.

The first approach that focused on drills and repetitions was the Direct Method (DM). This approach was developed at the end of the 19th century and beginning of the 20th and consisted of acquiring a foreign language using the learners' L1 natural learning process. Learners were not required or allowed to produce any speech at the beginning of the instruction so they could internalize the target language sound system (Celce-Murcia et al., 1996) but later on, learners were expected to produce speech accurately because they had supposedly used their intuition to compartmentalize the new sounds and imitate the model provided. This method seemed to improve learners' perception of the new sound system and to be more self-conscious about their learning process. However, this method failed to include many aspects of the language that are considered crucial to acquire when learning another language. For example, deficiencies in providing meaningful vocabulary that fit in the little class time or real situations, not teaching grammar systematically, and consequently ignoring reading and writing skills caused that both language teachers and researchers looked for new methods to improve not only L2 learners' pronunciation but also their overall proficiency and have a balance between form and function.

Taking into account the learners' awareness to recognize and imitate L2 sounds, a new teaching method was proposed in the middle of the 20th century, the ALM. This method was based on behaviorist theories (as the DM) which proposed that language could be acquired, modified and conditioned as any other conduct by means of reinforcing that behavior. Thus, drills and repetitions play an important role in the ALM because is through these exercises that the learners could develop a native-like, less-accented, and more accurate pronunciation; furthermore, favoring deviant L1 tendencies (Celce- Murcia et al., 1996; Isaacs, 2009; Lightbown & Spada, 2006). By imitating good models in the target

language, the learner would receive positive feedback if the production was accurate and 'correct' or negative feedback if the pronunciation was 'incorrect'. Unlike the DM which provides a variety of settings to the same element to favor real associations, the ALM was more focused on producing new vocabulary within the same grammatical structure (or setting), so that way the accuracy was 'guaranteed' for that specific structure and segment. During the awakening of the ALM, drills and repetitions were defined as the action of "getting learners to repeat words, expressions or sentences to help them remember and develop more accurate pronunciation"[1]. These practices were considered a key piece in teaching pronunciation because they helped learners to create 'good' pronunciation habits through producing speech multiple times until the targeted form has been accurately acquired. Similarly, Del Castillo (1990) explains that the main focus of the ALM "is on traditional notions of pronunciation, minimal pairs, drills and short conversations" (p. 3). The ALM is focused on improving L2 learners' listening and speaking skills (Saville-Troike, 1973) rather than in any other ability aimed in second language classrooms (reading or writing) because the results were merely focused in acquiring a native-like pronunciation by means of having a 'good pronunciation' (if there is such a thing in languages) through habit creation. There is a growing body in research describing the benefits of drills and repetitions in L2 pronunciation instruction. Paulston (1970) for example, distinguishes between the initial teaching phase (where memorizing through drills and repetitions exercises are required) and the testing-teaching phase where she differentiates among mechanical, meaningful and communicative drills. Abbs, Gupta and Khetarpal (2008) suggested an articulatory-phonological planning of the word form and execution of that plan as two functional components of performance of overt spoken repetition suggesting that the articulatory-phonological planning is an internal repetition while the execution of that planning is more explicit.

A more recent interpretation of the ALM has been called ACCESS (Automatization in Communicative Contexts of Essential Speech Segments). It focuses on form-focused instruction, repetition, and processing, as well as in target expressions that may be reused in future communicative situations (Gatbonton & Segaloqitz, 2005, 2006). Some other authors such Oxford, Lavine, and Crookball, (1989) or Saville-Troike (1973) have stated

[1] Teaching+Tips_Pron_01_Drilling.pdf

that as the ALM and behaviorism theory 'resulted in little success in providing L2 learners with actual language knowledge' and that language aspects such as pronunciation should subsequently be disregarded as an important construct in L2 teaching/learning and language proficiency. The implementation of technologies consisting of drill-and-practice materials marked the beginning of a new phase in computer-assisted and technology-based pronunciation teaching.

According to Derwing and Munro (2008), the decline of the ALM led to a marginalization of pronunciation teaching and research due to the belief that pronunciation instruction was not effective or it had a negligible effect on the L2 learners' pronunciation. Additionally, drills and repetitions have also been tagged as a non-communicative type of instruction, as "they are not necessary or beneficial for foreign language acquisition or the development of fluency and should be discarded from instructional practice" (Wong & Van Patten, 2003, p. 403). Despite those misconceptions, drills and repetitions are still a very useful approach in teaching pronunciation nowadays. Research have provided evidence that using drills and repetitions have several benefits for the L2 learners. For instance, some studies in perception found that listening repeatedly to L2 sounds significantly enhances not only the phonetic perception of the L2 but it has a direct impact in the production of segments (Bradlow, Akahane-Yamada, Pisoni & Tohkura, 1997; Rochet, 1995; Wang, Jongman & Sereno, 2003). In this sense, pronouncing and being exposed to the targeted segments more frequently, and for longer periods of time (within and outside the classroom) is beneficial for the L2 learners even if they do not produce accent-free (more native-like) speech. Other studies have had similar findings in this regard: that drills and repetitions not only help to create correct and more accurate pronunciation but also, they improve learners' perception of the L2 (Senel, 2006; Paulston, 1970; 1971; 1976) plus they have measurable benefits for processing L2 (Gu & Johnson, 1996; Trofimovich & Gatbonton, 2006) even without receiving explicit instruction (Yoshida & Fukada, 2014).

2.3.2 Visual Feedback Paradigm

While some computer-assisted systems such as CALL have been used as part of pronunciation instruction few decades ago, recent developments have provided tools such as Praat (Boersma & Weenink, 2001) that are more accessible, versatile, easy to work with

and functional for both researchers and L2 learners. There is a growing body of research regarding how this type of technology can be incorporated in the L2 classroom and how can language teachers take full advantage of these tools.

For instance, the first landmark in the development of CALL refers to the PLATO (Program Logic/Learning for Automated Teaching Operation) project "for language teaching and learning at the computer" (Azarnoosh, et al., 2016, p. 135; Beatty, 2010; Marty, 1981) proposed as early as 1950s and 1960s. In the 1970s and 1980s many more programs highlighted the importance of receiving pronunciation instruction giving the learner more autonomy in the learning process and encouraging critical thinking to identify the new information that was now being provided (Azarnoosh, et al., 2016). In the late 70s, with the advent of new technologies, many researchers tried to produce a number of "machines" (Weltens & de Bot, 1983, p. 79) that could allow them to visualize human speech as accurate as possible and that could provide feedback that could facilitate the learning process (Weltens & de Bot, 1983). While most of these 'machines' were "hardly applicable or even testable" (Weltens & de Bot, 1983, p.80) (see also León & Martín, 1970); the *Visi-Pitch* software (Kay Elemetrics, 1986) provided researchers with enough tools to investigate the extent to which visual analysis of self-produced speech versus native speaker productions had in the L2 learner's perception of their own speech (Abberton & Fourcin, 1975; Schmidt, 1998; Molholt, 1988,1990; Olson, 2014). The use of VFPs began over four decades ago (Abberton & Fourcin, 1975; James, 1976, 1979; Vardanian, 1964). The main difference with the development of computerized software lies largely in the awareness of the learners towards their own oral productions in order to improve their pronunciation skills through a more autonomous understanding of their mistakes/errors in the production of suprasegmental features such as pronunciation or intonation through graphic (visual) representations of their speech. Other experiments conducted by James (1979) provided evidence that visualization of the speech contours can have a significant effect on improving the intonation of second language learners and is more effective than drillings and in a greater extent than no-pronunciation instruction. The use of a VFP within the L2 language classroom has been proved to work efficiently towards the improvement and acquisition of certain phonetic features in the L2 (Olson, 2014) and also in the retaining of those features (Olson & Offerman, 2013).

For instance, suprasegmentals have been among the most studied features in using VFP and have also been a cornerstone for its development because "suprasegmentals provide the framework for utterances, they direct the listener's attention to important information in the discourse" (Anderson-Hsieh et al.,1992, p. 51). There are several studies that refer to the use of VFPs to teach intonation contours and discourse intonation (Anderson-Hsieh et al., 1992, 1994; Chun, 1989, 1998, 2002, Molholt, 1988, 1990) and that provide evidence of the potential benefits of this type of instruction (de Bot, 1983; Weltens & de Bot, 1984). For example, De Bot (1983) showed at least three stages in the use of this type of pronunciation instruction: (1) a visual representation and its auditory cue are presented, (2) the non-native attempts to produce the utterance presented following the cues, (3) both visual representations of the intonation contour are presented (De Bot, 1983). The results obtained by Weltens and De Bot (1984) showed that even when feedback delay was not effective; the visual information received was a critical factor when using the pitch visualizer for intonation teaching (p. 80) because learners could identify differences in intonation patterns, especially by voice and recording quality. Similarly, Anderson-Hsieh et al. (1992) carried out an experiment with Chinese and Korean speakers learning English and used an electronic visual feedback to teach stress, rhythm, linking, and intonation. Results showed that the English learners could more easily replicate the targeted suprasegmental features and performed relatively better with time because "electronic visual feedback is an excellent tool for raising the students' awareness" (p. 61) and obtain more-native-like targets.

Some authors have mentioned that the benefits of using a VFP to teach prosodic features such as suprasegmentals (Chun, 1989; 1998; de Bot & Mailfert, 1982; de Bot, 1983; Hardison, 2004; Weltens & de Bot, 1984) can also be extended to teach segmental features (i.e., Lambacher, 1999; Olson, 2013; 2014) as the 'visualization of the intonation pattern allows for the "establishment of an automatic judgment system based on theories of pattern recognition" (León & Martín, 1972, p. 141). As such, the establishment of this judgment system would not only help the learners to recognize patterns in suprasegmental as intonation and stress but it could also help them to focus their attention in segmental features such as segment length, intensity and loudness (to mention some), features which have been understudied (Lambacher, 1999). Using an Electronic Visual Feedback (EVF)

paradigm, Lambacher (1999) helped L1 Japanese learners of English to improve both perception and production of English consonants that are difficult for Japanese speakers to acquire (oral and nasal voicing, liquid and fricative contrasts). The results obtained revealed that using an EVF paradigm has in fact a positive impact on the learners' perception and production of English consonants; the use of the EVF paradigm helped learners to recognize the targeted pattern and to objectively assess their own productions throughout the instruction by comparing the given and their own pattern.

Similarly, Hardison (2004) employed an auditory-only (A-only) and an audio-visual experiment in order to provide evidence that 'real-time' visualizations are useful to improve L2 pronunciation. The results obtained in his study indicated that the learners not only improved in their production of segmental but also in the accuracy of suprasegmental features. Interestingly, the learners even became more aware of other aspects of their pronunciation and even assigned attentional hierarchical levels (Hardison, 2004) for the features seen, claiming that it is difficult to process all the elements at the same time. More recently, positive results have been obtained in pronunciation accuracy when implementing VFPs within the classroom. The use of waveforms and spectrograms seems to provide L2 learners with the reality of what they do and help them to be aware to what is expected for them to do. Carey (2004) studied the benefits of using a VFP on using the speech software *Kay Sona-Match* in the acquisition of English vowels by L1 Korean speakers. Results showed that L1 Korean learners of English are eager to receive pronunciation training but moreover, learners expressed that they used their own strategies to modify the articulation of their words to be more English-like based on the visual representations they were looking at. Saito (2007) obtained positive results related to accuracy when L1 Japanese learners of English used the speech software Praat to acquire the vowel /æ/ by looking at the differences between F1 but particularly, F2. Similarly, Motohashi-Saigo and Hardison (2009) and Okuno (2013) worked with L2 Japanese leaners speakers of English in gaining accuracy in vowel length and geminates. Their results support the benefits of using speech software and conclude that the use of speech visual cues are more effective than providing A-only input. Moreover, the speakers were able to better perceive and identify tokens with difference in vowel length, geminates, etc... Okuno (2013) also concludes that the effects of a perceptual training may be transferred to production; this, due to the results he obtained

where the participants did not receive explicit instruction on how to pronounce the targeted words but they did improve in the production task. Lord (2010) studied the effects of providing explicit pronunciation instruction to L2 learners of Spanish as well the benefits of being exposed to an immersion program on the overall production of voiced occlusive and fricative allophones in Spanish. Results obtained in this study showed that learners benefited from pronunciation instruction and the immersion program, especially in the production of the fricative segments; however, she highlights that the gains may be even higher if learners would have pronunciation instruction and an immersion program experience. Although no known studies have compared the effects of drills and repetitions versus the effects of using a VFP in the acquisition of SRC, studies (i.e., Chun, 1989; Olson, 2014) have provided evidence that using a VFP has positive in the acquisition of individual segments such as voiceless and voiced stops at the word and sentence level. The results are compelling for various reasons: first, the positive effects the VFP for producing intervocalic stops; second, it confirms other claims about the recognizing L2 patterns by intuition and awareness (i.e., León & Martín, 1972) which may be overlooked by other training methods such as listening and repeating (Miller, 2012). More important, the benefits of the training indicated that the gains are transferable across segments.

2.4 Targeted Phonetic Features of the Study

In the following sections, the three phonetic features that form a stop- rhotic cluster will be described and then contrasted between English and Spanish segments. Colantoni and Steele (2006) propose that voice onset time (VOT), rhotic realization and the epenthetic vowel, especially in a cluster, present a relative difficulty of acquisition for English speakers, learners of Spanish as L2. These phonetic features are presented below.

2.4.1 Voice Onset Time (VOT) in NSs of Spanish and English and Learners of Spanish as L2

One of the phonetic features that the current study focuses on is VOT of voiceless stop consonants. Ladefoged (1975) defines VOT as “the release of the stop closure to the to the start of the oscillating line indicating the vibrations of the vocal folds in the vowel” (p. 126). In other words, VOT is the release of the closure produced by a stop consonant

and the onset of voicing represented by a vowel, to be considered as “positive” VOT (Whalen, Levitt & Goldstein, 2007).

For the interest of this study, VOT values in Spanish language are considered to have a short lag-prevoiced contrast (Lisker & Abramson, 1964, Borzone de Manrique & Gurlekian, 1980; Quilis, 1993; Díaz-Campos, 2003; Hualde, 2011) compared with the considerably much longer VOT in English for the same phonemes. The retiming of articulatory gestures (Hualde, 2005, p. 113) is represented in the contrast between voiced and voiceless stops Spanish and English According to Vicente (1986) and Lisker & Abramson (1964) the average length in milliseconds (ms) of the phonemes /p/ /t/ /k/ in initial position are 4, 9 and 29ms for Spanish and 58, 70 and 80ms for English. The English realization of these long-lag stops are represented as [p^h], [t^h], [k^h] in the International Phonetic Alphabet (Hualde, 2011) due to a much longer VOT by means of the aspiration of the consonants in the onset position (Lisker & Abramson, 1964; Flege, 1991; Gonzalez-Bueno, 1997). The difference in the VOT can be seen in Table 1 (Hualde, 2005, p. 150) for the segments /ba/ and /pa/. In numeral (1), the voice lead refers to the vibration of the vocal folds before and after the release of occlusion due to voicing (/b/). Numeral (2), shows the short voice lag characteristic of Spanish[2] when the vocal folds vibrate soon after the occlusive and finally, numeral 3, refers to the long voice lag particular of English productions. This long lag is the result of a great aspiration and the vocal folds’ vibration for a longer period of time after the release of the occlusive segment.

[2] Note that it also refers to English /b d g/ as containing as short voice lag; this is context specific. According to Hualde, 2005, “after the stressed vowel these consonants are realized as unaspirated” (p. 150)

Table 1. Spanish and English VOT Differences for /pa/ (Hualde, 2005, p. 150)

(1) Voice lead
^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ‖ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ b a Spanish/ English /b d g/
(2) Short voice lag
- - - - - - - - - - - - - - - - ‖ - - ^ ^ ^ ^ ^ ^ p a Spanish /p t k/; English /b d g/
(3) Long voice lag
- - - - - - - - - - - - - - - - ‖ - - - - - - - - - - ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ p^h a English /p t k/

The same contrast between a native speaker of Spanish and native speaker of American English VOT length can be seen in the syllable ‘pa’ in the work ‘paja’ ‘*hay’* in Figure 1.

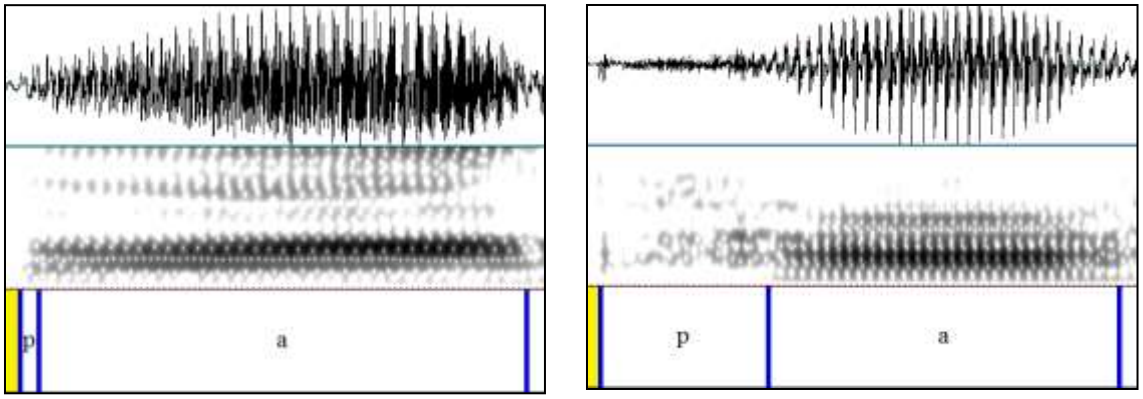


Figure 1. Spectrogram of the syllable ‘*pa*’ produced by a native speaker of Spanish (left) and by a native speaker of English (right).

Although remarkable differences between Spanish and English VOTs may not affect comprehensibility, it makes a foreign accent more noticeable (Lord, 2005, p. 559). As consequence, NNS may be negatively perceived by the NS and be attributed negative attitudes because of this (Munro, Derwing & Saito, 2006; Derwing & Munro, 2009; Gluszek & Dovidio, 2010; Kim, Wang, Deng, Alvarez & Li, 2011).

2.4.2 Rhoticity in NSs of Spanish and English and Learners of Spanish as L2

The second phonetic feature targeted in this study is the realization of < r >; more specifically a rhotic segment after the voiceless stop consonants /p/, /t/, /k/ which have a short prevoiced lag in Spanish (as mentioned before). The transfer of the English approximant /ɹ / to Spanish is a marker of a non-native like production (Hualde, 2005, p. 181). In this section, Spanish rhotic segments will be described phonetically and articulatory and then, will be compared with the English non-lateral liquid /ɹ /.

Rhotic segments are produced in the alveolar ridge in English and Spanish, they differ in the manner of articulation (approximant, flap or trill). According to Hammond (2001, p. 266), “the vast majority of traditional analyses of Spanish phonology assume *a priori* that there are two rhotic phonemes in Spanish, a flap /ɾ/ and a trill /r/”. Thus, they are contrastive (Hualde, 2005, p. 181), even though the flap /ɾ/ is restricted for word-initial position. Blecua (2001), Hammond (2001), Hualde (2005) and Docherty & Foulkens (2002) among many others have pointed out the similarity among the Spanish flap /ɾ/ and the English apico-alveolar allophone /ɾ/ for the alveolar stops /d/ and /t/ found in post-tonic intervocalic positions as in ‘*ladder*’ [læɾɹ] or ‘*butter*’ [bʌɾɹ]. American English has only one rhotic phoneme in its phonemic inventory, /ɹ/, which has been defined as a voiced apico-alveolar rounded retroflex non-lateral liquid (Hammond, 2001). The measurement of the /ɹ/ is usually marked by the reduction in F3 (Stevens, 1998, p. 535) as seen in Figure 2 for the word ‘*read’* (Docherty & Foulkness, 2002). Stevens (1998) noted that the tongue configuration for the approximant has an effect in the production on the rhotic, reducing the level of energy in the spectrum above the F2 and F3, prominence seen in the box in Figure 2 (ranges cover between 100 to 200 ms for length and 1 to 3 kHz for frequency).

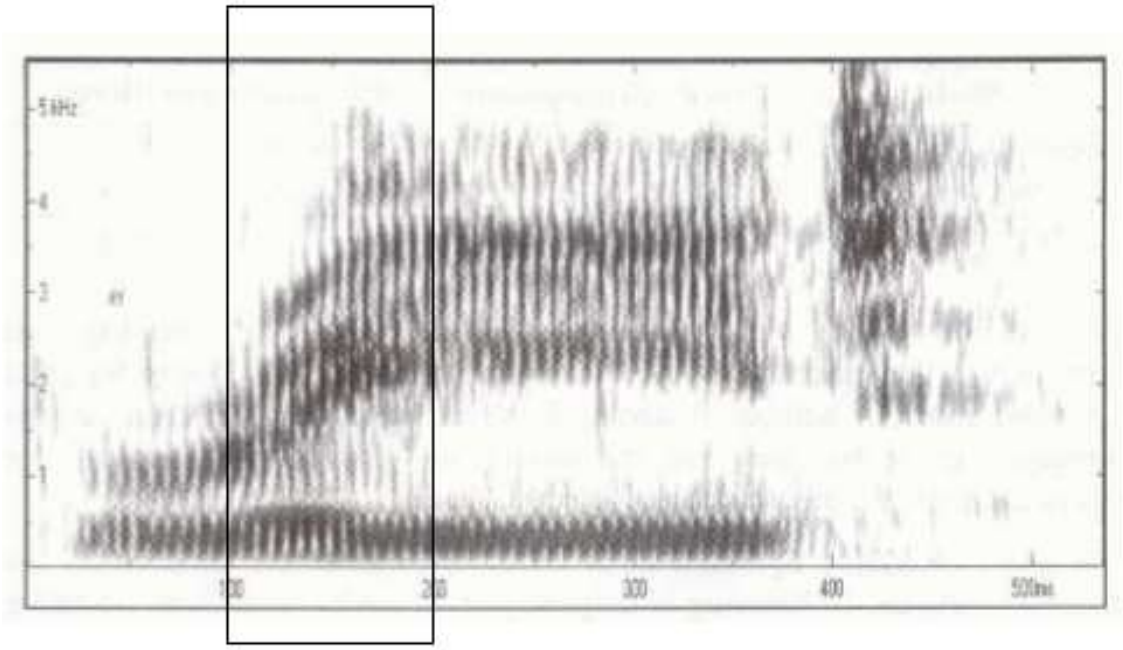


Figure 2. Spectrogram of the word '*read*' (Docherty & Foulkness,2002)

In Spanish, the trill /r/ is produced as a multiple vibrant, causing the tip of the tongue to move several times in the alveolar ridge. According to Blecua (2001), the trills in Spanish present *"vocalic elements"* (p. 29) that mirror the following vocalic segment. These vocalic segments are not the effect of orthography or a deliberate, conscious act of the speaker to produce them. Martínez-Celdrán (1997) claimed that "the production of the multiple vibrant /r/ takes advantage of the Bernoulli effect" (p. 94) as the closures produced by the tip of the tongue reaches the alveolar ridge and compresses the air rapidly until the pressure of the exhaled air is greater that the pressure on the tip of the tongue which consequently moves away from the place of greater pressure. Once the tip of the tongue moves away, the pressure in the alveolar ridge decreases leaving space for the air to flow but that same space is filled again when the tip of the tongue moves back to the alveolar ridge at least two or three more times (Hammond, 2001; Hualde, 2005; Ladefoged & Maddieson, 1996; Martínez-Celdrán, 1997). As the trill /r/ is restricted to complex onset segments, SRC require the articulation of the tap /ɾ/. Likewise, the flap /ɾ/ produced in the SRC is produced by a single contact of the tip of the tongue with the alveolar ridge, the airstream is temporarily blocked and the pressure moves the tip of the tongue away from the place of more pressure. In other words, the occlusion caused by the flap is seen in a

spectrogram as silence interval containing low frequency and periodic components (Almeida & Dorta, 1993; Blecua, 1996, 1997, 2001; Delattre, 1965; Martínez-Celdrán, 1984; Ladefoged, 1975, 2006; Borzone, 1980; Quilis, 1981, 1993; Lindau, 1985; Massone, 1988; Willis, 1999). Saito and Lyster (2012) following Flege et al. (1995) also took into account the decrease in the F3 at the beginning of the adjacent vowel (as seen in Figure 2) in order to measure and analyze the acoustic properties of /ɹ/. In Spanish, the /r/ and /ɾ/ also show low acoustic energy at the closure, and a reduction of energy in their F3 due to multiples closures or a single closure respectively as shown in Figure 3 (seen as blank spaces) (see also Demestre, Llisterri, Riera & Soler, 2006). Allophones of the phoneme /r/ show a different behavior or manner depending on the context (in this case in intervocalic position); trill /r/ show at least three clear closure and two vocalic segments in the word 'corro' *I run* while there is one closure in the word 'coro' *chorus* which contains a flap /ɾ/.

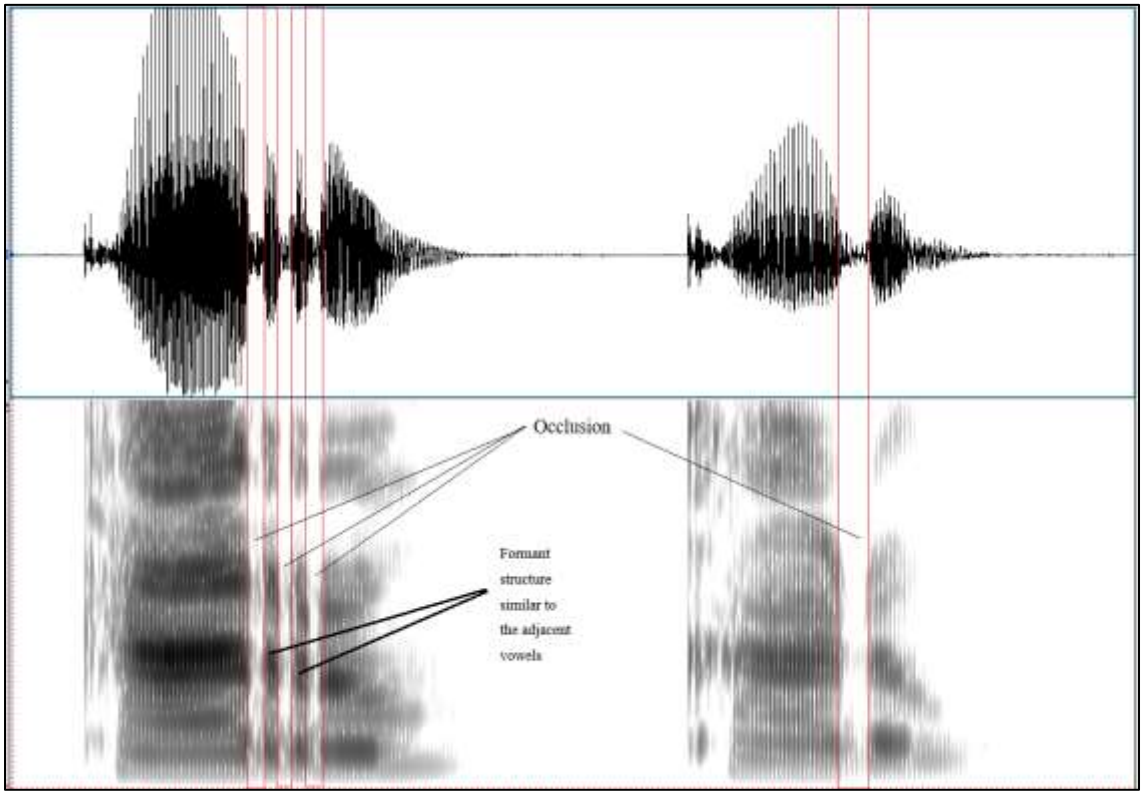


Figure 3. Spectrogram and sound waves of trill (left side) and flap (right side) rhotics in Spanish productions 'corro' and 'coro' (Demestre, Llisterri, Riera, & Soler, 2006)

2.4.3 Epenthesis in NSs of Spanish and English Learners of Spanish as L2 in SRC

Epenthesis is a phonological process of adding or inserting a segment (vowel or consonant) within the word as part of a retiming articulatory gesture (Hualde, 2005, p. 113) and are realized as "uninterrupted sequences" (Colantoni & Steele, 2006, p. 60). Malmberg (1965) claimed that in difficult clusters such as <r> + consonant or consonant+ <r>, many languages (especially romance with Latin influence) use a 'vocalic parasite element' to break the complex element. Important to note, is the presence of epenthetic segments before a flap is common cross-linguistically (Appendix A). For example, Olson (2003, 2006) explored the insertion of an epenthetic vowels in Mono, a Niger-Congo language. In his study, epenthetic vowels were inserted at the beginning of a monomoraic words in order to repair the marked structure (McCarthy & Prince, 1993). More pertinent and like Spanish, the vowel inserted "copies the tone and quality of the following vowel" (Hall, 2006, p.392) not only after a fricative but also liquids (lateral and rhotics) as seen in (1).

(1) Words in Mono that show insertion of a vowel in a CV context (Hall, 2006, p.392)

/gàfrū/ →	gàfūrū ~ gàfrū	'mortar'
/plézū/ →	pélézū ~ plézū	'bat'
/jãbrù/ →	jãbùrù ~ jãbrù	'goat'

Dutch also has an intrusive schwa between liquids and non- coronal consonants as in the word [hɛrəfst] 'autumn' (Booij, 1995; Hall, 2006). For example, in the case of words like 'horn' and 'to hear', the presence of the epenthetic schwa avoids the deletion of the <n> in coda position for 'horn' [horən] as /ə/ is not be considered an underlying or lexical schwa while it is in the word [horə]. The former word allows the deletion of the final <n>, helping to distinguish 'horn' from 'to hear' as seen in (2).

(2) Words in Dutch that show insertion of a vowel in a CV context (Hall, 2006, p.397)

hoorn	/horn/ → horn ~ horən, *horə	'horn'
horen	/horən/ → horən ~ horə	'to hear'

Other languages like Winnebago present also epenthetic vowels in Cr clusters in onset position and where <r> is a sonorant as in [hiperes] 'know' and the mirroring vowel

reduplicates throughout the entire sequence. In ‘pras’ ‘flat’, a vowel is inserted to break the Cr cluster and be pronounced as paras. The effect of the epenthesis can be seen at the word-level, as the CRV sequence gets reduplicated to become a CVRV sequence (Hall, 2006, p. 402; Miner, 1992, p. 29; Susman, 1943, p.33) as in the word parapparas ‘wide’. In Spanish, the production of the flap /ɾ/ in SRCs make an involuntary and unconscious articulation of an epenthetic segment which usually mirrors the vowel of located within the same syllable (Gaya, 1921; Lenz, 1892, 1893; Malmberg, 1965; Navarro, 1918b). For instance, the duration of the epenthetic (or svarabhakti) segment could either similar to the lexical vowel if the segment is positioned in an unstressed context or it could “give rise to a lexicalized copy vowel whose quality matches that of the underlying nucleus tautosyllabic with the complex onset” (Bradley & Schmeisser, 2003, p. 3) as in the Spanish words atrás> ataras ‘*back*’ or cobré> cobere ‘*I cashed ___’* (see the boxes). See also the words sacra> sacara ‘*sacred*’ or podré> podere ‘*I will be able*’ (see in Colantoni & Steele, 2006, p. 61)

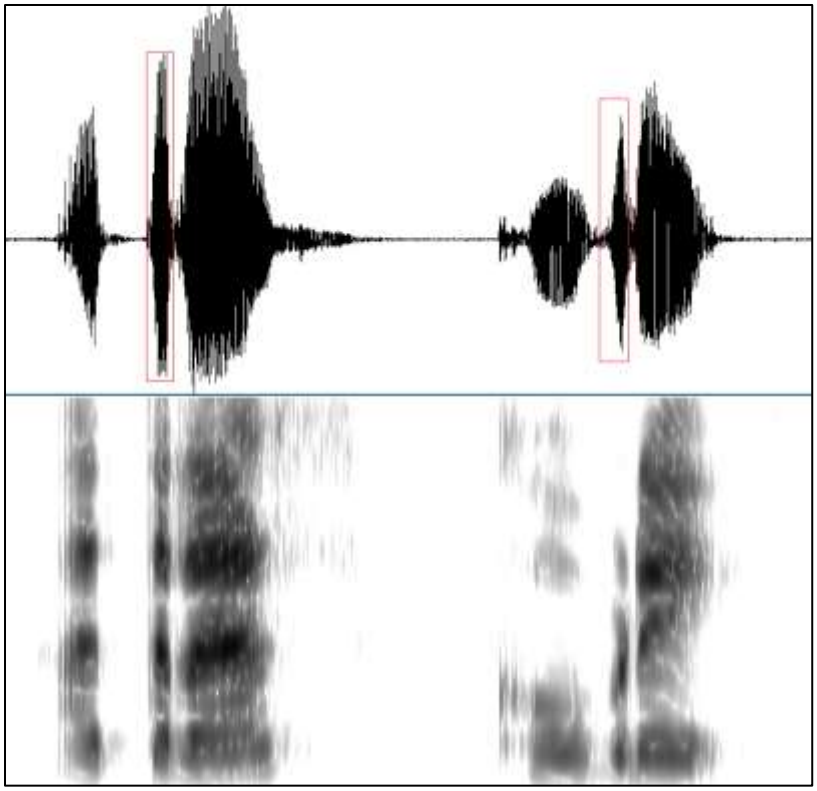


Figure 4. Spanish words '*atrás*' back and '*cobré*' I cashed (a check) (Bradley & Schmeisser, 2003)

In English, the SRCs are pronounced with the approximant [ɹ] which favors affrication, allowing a constant flow of the airstream making epenthesis unnecessary (Colantoni & Steele, 2006; Olsen; 2012). Olsen (2012) explained that because taps and trills are represented in the same way in English (<r>), L2 learners usually activate the approximant /ɹ/ instead of the tap already existent in their phonology. Ladefoged (2006) expressed that in English "alveolar stops become voiced taps when they occur between two vowels the second of which is unstressed" (p. 74) which would agree with the Spanish production of the tap [ɾ] and the presence of an epenthetic vowel if the English allophone for [t] and [d] would be produced in Spanish SRC. While Spanish speakers produce the tap [ɾ] by raising the apex of the tongue towards the alveolar ridge making a brief closure that promotes the articulation of the epenthetic vowel, apical taps (as allophones of [t] and [d]) and trills are what English native speakers are taught in their language classrooms (Face, 2006). With this respect, there are some considerations to make regarding the epenthetic segment in Spanish SRC: first, according to Hualde (2005, p.113), the origin of the

epenthetic plosive can be explained as a retiming of articulatory gestures due to the change of manner of articulation of the fricative segment. Recent studies such as Colantoni and Steele (2006) have researched the insertion of an epenthetic vowel in onset clusters including the structure *Cr* and *Cl (C* represents a consonant and the adjacent segment is either the rhotic '*r*' or the lateral '*l*'). They have argued that this epenthetic process is dissimilatory in order "to increase the articulatory and perceptual difference between the stop and rhotic". They realized that there is a quasi-categorical epenthesis in rhotic clusters compared to lateral clusters. Also, Ramírez (2006) studied this phenomenon in clusters formed by laterals and concluded that although the epenthetic vowel is not produced voluntarily, it is "part of the articulatory plan of the speaker" (p. 6). Second, as the trill /r/ is restricted to appear in complex segments, the production of the flap /ɾ/ is necessary (not to say mandatory) when producing Spanish SRC. At the same time, the articulation of the flap /ɾ/ is restricted to word initial position and is usually found in word-final or intervocalic position. Therefore, the articulation of the flap would produce the epenthetic segment which would account for the phonological constraint of the flap to be in a VɾV context. Third, while in second language learning, the presence of an epenthetic segment may be an undesirable repair strategy (Ramírez, 2006) or an effect of L1 influence as in the production of *sC* complex segments by NSS learners of English (as in 'school' [ɛsku:l) vs. [sku:l]); in the present study, the epenthetic vowel is a byproduct of the articulation of the flap /ɾ/ within the SRC. As such, the production of it implies the 'correct' articulation of the rhotic segment.

2.5 Research Questions and Hypotheses

Considering that there are no studies that compare visual feedback and auditory training (drills and repetitions) in complex segments composed by voiceless stop-rhotic clusters, testing the effectiveness of one training over the other or both trainings at the same rate, some research questions have aroused for the present study:

RQ1: Does visual feedback improve NNS accuracy on complex segments (e.g. Stop-Rhotics) more than "traditional" instruction? Specifically, does it improve for VOT, flapping and/or epenthesis?

RQ2: Do drills improve NNS accuracy on complex segments (e.g. Stop-Rhotics) more than “traditional” instruction? Specifically, does it improve for VOT, flapping and/or epenthesis?

RQ3: Does visual feedback or repetition lead to greater improvements for NNS accuracy on complex segments?

RQ4: Does accuracy change apply to each of the segments of the complex cluster similarly (e.g. VOT, Rhoticity, Epenthesis)? Do these changes follow the difficulty acquisition hierarchy?

For each research question, one hypothesis has been proposed:

- Both interventions, visual feedback and repetition groups will improve learners’ production of the targeted segments (e.g. Olson, 2014; Castillo, 1990).
- Learners will produce more accurate target segments (compared with the repetitions group) when a VFP is implemented, based on the multimodal cues and increased noticing (Schmidt, 1990, 2001).
- Most learners will produce closer VOT values compared to the Spanish stops in both groups. A small percentage of participants will produce the flap ‘r’.
- It is hypothesized that participants will follow the acquisition hierarchy in Spanish SL clusters: VOT> Rhoticity > Epenthesis (Colantoni & Steele, 2006)

CHAPTER 3. METHODOLOGY

To answer the above research questions, a classroom-based empirical study was conducted comparing the effectiveness of visual feedback, drills and repetitions, and traditional pronunciation instruction on the production of stop-rhotic clusters by English native speaker learners of Spanish. This chapter details the participants' profiles, stimuli, procedures, treatments and measurements used in order to collect and analyze the data.

3.1 Participants

For the current project, a total of 68 participants were initially recruited from three different Spanish classes taught at a Midwestern university in the United States. There were two experimental groups and one control group. The experimental groups were distributed as Visual Feedback Group (n = 24), and Drills Group (n = 20), and one Control Group (n = 24). Proficiency level tests were administered by the university prior to the start of the academic semester in order to place students in the appropriate level according to their knowledge of Spanish as L2. All participants were in their third semester of Spanish. The Visual Feedback Group (VFG), Drills Group (DG), and Control Group (CG) were between the ages of 18-36, mean age, 19.5 years old (SD= 2.43 years) and were enrolled in a 16-week Spanish course. The courses were taught by two native speakers of Spanish (one native speaker for the Visual Feedback Group and Drills Group, one for the Control Group). Both instructors were from the same region in Colombia, in order to reduce the possibility of any interference due to geographical variances in the input provided to the participants.

To know the amount of exposure to Spanish, language background and age of first contact or acquisition of Spanish as their L2 or L3, participants were given a Language Background Questionnaire to fill out at the end of the academic semester. This questionnaire was based on topics included in the Bilingual Language Profile (Birdsong, Gertken & Amengual, 2012) and was provided to the participants at the end of the academic semester. From a total of 68 participants, 9 participants reported significant exposure to a language other than English or Spanish, and were subsequently eliminated from analysis

(3 for Visual Feedback Group, 1 for Drills Group and 5 for the Control Group). In addition, 15 participants were eliminated for failing to complete all the activities (Visual Feedback Grou*p*= 5, Drills Grou*p*= 6, Control Grou*p*= 4) or poor recording quality (Visual Feedback Grou*p*= 0, Drills Grou*p*= 0, Control Grou*p*= 2)

At the end, a total of 39 participants (13 for each group) were taken into account in the present study. According to the Bilingual Language Profile, the mean age of acquisition of Spanish as L2 was 12.8 years old (*SD*=2.99). All of the participants had studied at least one year of Spanish at the college level and the average time the participants had been learning Spanish in academic settings was 6.8 years (*SD*= 4.8). As such, participants were English native speaker learners of Spanish in a university setting, with limited to no exposure to Spanish outside the classroom or to any other language. The Visual Feedback Group, Drills Group and Control Group differed in the type of training (or intervention)[3] that was used within the classroom in order to obtain the speech samples. The pronunciation activities were included as part of the course syllabus and there was no extra credit for completing the tasks. The participants were required to complete the pronunciation activities (tasks) as part of their course work and each task was graded upon completion and not by performance of the individual. Participation in the experiment (i.e., providing data for the current analysis) was voluntary. However, at the end of the training all learners received $5 as a compensation for their participation in the study independent of the completion or performance.

3.2 Stimuli

The stimuli used for this study consisted in 36 tokens with voiceless stop-rhotic clusters (henceforth SRC) (/pr/, /tr/ and /kr/) embedded in Carrier Sentences and Novel Utterances. Six of the tokens were embedded in the carrier sentence “Digo _____ de nuevo” (see Section 3.2.1) similar to the carrier phrase used in Fox, Flege, and Munro (1995) and were randomized in each training. Thirty tokens were embedded in novel phrases, which provided a less-controlled environment for the targeted tokens (see Section 3.2.2). All SRCs were in onset position and were in non-stressed syllables. Tokens were followed by the Spanish vowels /e/, /i/, /o/ and /u/ only. The vowel /a/ was not included as

[3] The terms ‘training’ and ‘intervention’ will be used interchangeably for the purposes of this study.

a significant number of the resulting lexical items (i.e., SRC + /a/) are cognates in English. In order to control for non-cognate status, the tokens were crossed referenced in English dictionaries (Merriam and Cambridge) to eliminate the possibility that they were cognates or orthographically close to the targeted words in Spanish. The researcher reduced as much as possible the use of nasal consonants after the vowel within the SRC to avoid that the total intensity of the oral vowel was reduced for the influence of an adjacent nasal consonant as described in Delattre (1954, p. 108; see also Delattre, 1955). There were not any fillers or distractors included as the feedback/auditory activities increased the chances that the participants found out the purpose/ targeted tokens in each activity (see Section 6.2 for a discussion of the limitations). Finally, to avoid any interference due to position of the cluster within the token, stress and following phonetic environments, were controlled as much as possible.

3.2.1 Carrier Sentences

For this task, six tokens (out of 36) were composed by targeted Spanish stop- rhotic clusters (/pr/, /tr/, /kr/) followed by the vowels /e/ and /i/. The vowels /o/ and /u/ were not used in this list to have a balanced number of tokens per list (6 tokens in the carrier sentence and 6 in the novel phrases). The tokens were randomized each time to avoid familiarization with stimulus order. The tokens were embedded in the carrier sentence '*Digo* _____ *de nuevo*', similar to the elicitation task presented in Fox, Flege, and Munro (1995) (*'Digo ahora* _____ ')[4]. Each token was read out loud before recording and after each intervention/ training, participants were asked to repeat and go over the list three different times. Table 2 provides a sample of the Pre-Test list and Training 1 list (*note that the tokens were randomized*) to control the degree of familiarization of the tokens in a specific position within the list.

[4] Placing the target token in sentence medial position allowed for the avoidance of any potential end of utterance effects and provided a better parallel with the novel phrases.

Table 2. Sample of tokens within Carrier Sentences for Pre-Test and Training 1.

Segment	Pre-Test	Segment	Training1
/pr/ + e	Digo *presupuesto* de nuevo	/kr/ + i	Digo *crisol* de nuevo
/kr/ + i	Digo *crisol* de Nuevo	/pr/ + e	Digo *presupuesto* de nuevo
/tr/ + e	Digo *tremola* de Nuevo	/kr/ + e	Digo *creciente* de nuevo
/pr/ + i	Digo *primera* de nuevo	/tr/ + i	Digo *triturar* de nuevo
/kr/ + e	Digo *creciente* de nuevo	/tr/ + e	Digo *tremola* de nuevo
/tr/ + i	Digo *triturar* de Nuevo	/pr/ + i	Digo *primera* de nuevo

3.2.2 Novel Phrases

As mentioned above, voiceless stop-rhotic clusters were embedded in Novel Phrases (30 in total). This type of task was used in order to provide short speech samples containing the targeted segments (Elliot, 1997) in a more natural, and less controlled context. Each of the novel phrase consisted of a variable context and contained one target lexical item with a SRC in utterance medial position (except for the word ‘trillizos’ *triplets* which was in word-final position). During each session (see Section 3.3) participants recorded 6 novel phrases: two containing each of the SRCs (/pr/, /tr/, /kr/). Tokens embedded in novel phrases changed with every new set that was provided to the participants after each training or the equivalent in time to the training (for Control Group). Expanding the data set, each novel phrase was repeated three times during each session. Briefly, the novel phrases consisted of target tokens embedded in a low-controlled environment that had not been subject to training in order to evaluate if participants were able to improve the oral production of the targeted segments when the context around them changed.

Table 3. Sample of tokens within Novel Phrases for Pre-Test and after Training 1.

Segment	Phrases	Segment	Phrases
/pr/ + i	La *primavera* es mi estación favorita	/pr/ + u	La sal de *prusiato* está compuesta de elementos químicos
/kr/ + o	El cuervo *crocita* en el atardecer	/tr/ + e	Si no *trepáis* el árbol, no alcanzaréis al gato
/tr/ + o	Hubo un grande *tropel* allá	/pr/ + o	La bebé *probeta* nació hoy
/kr/ + u	Ese tipo de *crucigramas* me gustan	/kr/ + u	La casa *crujió* en la tormenta
/pr/ + u	Un virus bastante *pruriginoso* causa una afección cutánea	/tr/ + o	Me *tronché* el dedo
/tr/ + e	No *trepemos* esa puerta	/kr/ + i	Uso la red para *cribar* el trigo

3.3 Procedures

Broadly, for each of the experimental groups (VFG, DG and CG), this study consisted of a Pre-Test, three interventions (or trainings), and a Post-Test. With the exception of the type of intervention (Visual Feedback, Repetitions or Cultural activities), the remainder of the course work was the same for all the participants in this study. The procedures were conducted over 12 weeks during Fall semester 2015. The average time between each training was 2.6 weeks. Throughout the academic semester, the participants were required to record their voices in 5 different opportunities corresponding to the Pre-Test, three training sessions and the Post-Test. In each recording, participants had to read out loud a list of carrier phrases and another list of novel sentences; each one of them containing 6 utterances which contained targeted tokens with SRC. Each list had to be recorded three times during its respective recording session. Only the first and last phase of the experiment were the same for the three groups that participated in the study. Participants were aware that this was a pronunciation activity and that it was aimed to help them improve their overall pronunciation; however, they were not explicitly told about the

specific segments (stop-rhotic clusters) that were object of the study or about the aimed phonetic features (VOT, rhoticity and epenthesis). Additionally, all groups spent the same amount of time on the take-home portion of the activities (*M* = 150.66 minutes, *SD* =57.3), as demonstrated by participant responses on a self-reported 'Time Control Form' (Appendix B). Students were trained in the stimuli they recorded previously to the training session to not make any kind of intervention over the new set of novel phrases provided to the participants at the end of each training.

3.3.1 Visual Feedback Group

3.3.1.1 Pre-Test

In the first stage of the experiment, second week of the Fall semester, the Visual Feedback Group (VFG) received an in-class demonstration, oral and written instructions on how to download the free speech software Praat (Boersma & Weenik, 2001). After receiving these guidelines, the instructor gave the participants an overview of the pronunciation activities and indicated that the grade received was not based on performance but rather in completion of the activity. The instructor explained to the participants the procedures to record, save, and submit electronically the sound files that belonged not only to the Pre-Test but also those of the upcoming trainings because the steps were the same for each recording and after- training session.

After instructing students on how to send the recordings via email, participants were given the Pre-Test lists which contained a six-item list with six *Carrier Sentences* and another six-item list consisting of six *Novel Phrases.* Participants recorded in two different sound files the tokens corresponding to 'Sentences' or 'Phrases'. Participants were also asked to go over each list and read them out loud three times. The tokens in the *Novel Phrases* list changed across the trainings while the *Carrier Sentences* were randomized each time.

Following the Pre-Test and before Training 1, the VFG was requested to bring to the first pronunciation class (as scheduled on the syllabus) the printed screenshots of the six words (tokens) that started with 'pr', 'tr' or 'cr' from the 'Novel Phrases' list given in the Pre-Test. Having the screenshots printed, each participant had to set the boundaries for

each one of the components of the word (letter by letter) by listening several times their recording so they could have a more accurate division of the word. They were told they had to do this as well as possible so they could compare their spectrograms with those of NSSs that were going to be provided in the pronunciation training session. They were instructed to follow these same procedures throughout the semester and for each training except the Post-Test.

3.3.1.2 Treatments

After the Pre-Test, three different interventions were made using a Visual Feedback Paradigm. Each training (or intervention) included three parts: Analysis of the learners' own spectrogram, a comparison with the native- produced spectrogram, and a visual discrimination task. These three parts of the treatments will be described below and one (Activity 1) out of the five activities performed during the academic semester will be found in Appendix C.

First, the analysis of the learner-produced spectrogram consisted of an in-class handout. Taking into account that none of the participants had an academic background in linguistics or any kind of phonetic training; the questions used in the handout avoided technical jargon. In order to participate in the treatments, and prior to each intervention, participants were asked to listen, select and print out the first six tokens from the list "Phrases" (discarding the second and third repetition for each token) which contained the stop-rhotic clusters /pr/, /tr/ and /kr/. The handout comprised a series of questions (See Table 4) leading the participants to identify the criteria used in setting the boundaries for each element (letter) in the target token; stablishing differences on how their /p/, /t/, and /k/ and /r/ differed from those of other native English speakers. The questions addressed one stop plus the rhotic at the time for one of the tokens in the list (one question for 1 token for /pr/, 1 for /tr/ and 1 for /kr/) so participants could make the analysis for each particular cluster. The participants were not explicitly told of the possibility of an epenthetic vowel between the stops (/p/, /t/ and /k/) and the rhotic segment (/r/).

Table 4. Sample Questions from In-class Handout for self-analysis.

Sample	English Translation
¿Cómo decidiste marcar los límites de cada sonido?	How did you decide to set the boundaries?
¿Qué características visuales tienen tu ‘t’ y tu ‘r’?	What are the visual characteristics of your ‘t’ and your ‘r’?
¿Es tu ‘t’ más larga o más corta que la de tus compañeros?	Is your ‘t’ longer or shorter than your classmates’?
¿Es tu ‘r’ más larga o corta que la de ‘r’ de tus compañeros? ¿Es tu ‘r’ más clara (lighter) o más oscura (darker) que la de ellos?	Is your ‘r’ longer or shorter than your classmates’? Is your ‘r’ lighter or darker?

Second, the comparison with the native- production consisted of presenting the participants segmented spectrograms produced by a native speaker of Spanish (NSS) of three targeted tokens from the list of “Phrases” (See Sample of one token in Figure 5). Participants were instructed to answer a series of questions (See Table 5) contrasting the differences between their own pictures and those that were being shown in the handout.

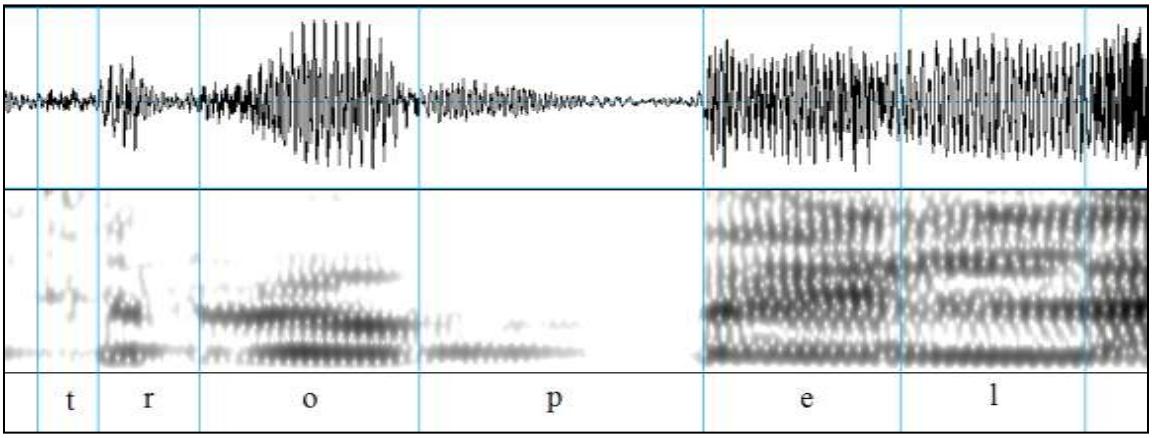


Figure 5. Word ‘*tropel*’ pronounced by a NSS.

Table 5. Sample Questions from In-class Handout for analysis of the NSS production.

Sample	English Translation
¿Qué características visuales tienen la ‘t’ y la ‘r’ del hablante nativo?	What are the visual characteristics of the ‘t’ and ‘r’ produced by a native speaker of Spanish?
¿Es tu ‘t’ más larga o más corta que la del hablante nativo?	Is your ‘t’ longer or shorter than the one produced by the native speaker of Spanish?
¿Es tu ‘r’ más larga o más corta que la del hablante nativo? ¿Es más clara o más oscura que la del hablante nativo?	Is your ‘r’ longer or shorter than the one produced by the native speaker of Spanish? Is it darker or lighter than the native’s?

The participants could compare their spectrograms and the soundwaves using expressions such ‘darker or lighter, longer or shorter, bigger or smaller’. Questions such as “¿Es tu ‘p’ más larga o más corta que la del hablante nativo? ¿Es más clara o más oscura que la del hablante nativo?” (*Is your ‘p’ longer or shorter than the one produced by the NS? Is it darker or lighter?)* participants had responses like “Mi ‘p’ es más corta y más clara” (“*My ‘p’ is shorter and lighter”,* VF08*).* To the question “¿Es tu ‘r’ más larga o más corta que la del hablante nativo? ¿Es más clara o más oscura que la del hablante nativo?” (*Is your ‘r’ longer or shorter than the one produced by the NS? Is it darker or lighter?)* there was more variation in the responses; for example, VF08 answered “Mi ‘r’ es más larga y oscura” (“*My ‘r’ is longer and darker”)* whereas participant VF21 responded “Mi ‘r’ es más corta y más oscura que la del hablante nativo” (“*My ‘r’ is shorter and darker than the one from the NS”).* Once the participants completed the handout, the researcher had the participants to ask questions for clarification if confusion arouse. The researcher guided participants’ attention towards certain features that were different or similar in their spectrograms with respect to those of the NSs in terms of how ‘dark’ or ‘light’, how long or short a feature was. Participants were asked if they could guess or find out why this could be happening and be different (or similar) in each one of the productions.

Finally, the visual discrimination task consisted in a quiz type activity performed at the end of the training session (See Appendix D). Participants observed two spectrograms of the same word, one produced by a native speaker of Spanish (NSS) and

the other one by a native speaker of English (NSE), and they had to decide which one had been produced by a NSE. There were a total of six spectrograms for this visual discrimination task distributed by word-pairs. This step was to provide feedback to the participants regarding their perception of the differences between speakers and segments and corroborate if they were able to identify visual differences between the oral productions of a native speaker of Spanish and another native speaker of English. In effect, participants were provided a feedback shortly after they took the mini-test and were asked to take into account what they saw and learned from the pronunciation activity when recording the new list of 'Phrases' and 'Sentences' provided to the participants at the end of the treatment.

3.3.1.3 Post-Test

The Post-Test consisted on recording a final list of 'Carrier Sentences' and 'Novel Phrases' once Training 3 was completed. The same recording procedures were followed to record these tokens and no further in-class analysis was required.

3.3.2 Drills Group

3.3.2.1 Pre-Test

Paralleling the procedure for the Visual Feedback group (VFG); the Repetitions group (DG) performed a Pre-Test, three interventions and a Post-Test. The procedures for the Pre-Test were identical to those used for the VFG. Participants were asked to record the same list of 'Carrier Sentences' and 'Novel Phrases' that was given to the VFG in the Pre-Test and were asked to send their recordings via email as well. However, they were asked to practice and repeat each one of the lists for at least 25 minutes prior to make the recordings. The purpose of doing this was to ensure equal amounts of time were spent on the activities by the VFG and DG.

3.3.2.2 Treatments

As with the VFG, the treatments in the DG consisted on three parts. Namely, participants had a listening-production task, an assessment- comparative analysis of the learner- production with a native speaker's, and an auditory discrimination task. Each one

of these tasks were in a handout (Appendix E) divided by cluster type (one section for /pr/, one for /tr/ and one for /kr/). The procedures will be described below.

First, the participants were asked to carefully listen a recording produced by a native speaker of Spanish; this recording contained just one of the targeted clusters they had recorded for the Pre-Test; Then, they were asked to repeat out loud what they were listening and for as many times as they listened to them (10 times). Participants did this same procedure three different times during the class time assigned for the completion of the pronunciation activity; each time the targeted cluster was different. The assessment-comparative analysis of the learner-production with a native speaker's contained assessment and comparison sections. In the assessment part, participants were given a handout with a series of questions to evaluate their self-awareness and self-assessment regarding their overall pronunciation and more segment-focused productions in Spanish (See Table 6). Participants were asked to describe the segments (or sounds) that were more difficult to pronounce and their auditory characteristics using words than did not present any higher level of difficulty such as "más larga que" (longer than), "más fuerte que" (stronger than), "se repite más que" (it repeats more than), etc…

Table 6. Sample Questions for Self-assessment activity

Sample	English Translation
¿Cuál es el sonido/ segmento que más se dificulta pronunciar? ¿Por qué?	What is the sound/ segment that is more difficult to pronounce? Why?
¿Qué características auditorias tiene tu 'p'? ¿Es larga o corta?	What are the auditory characteristics of your 'p'? Is it long or short?
¿Qué características auditorias tiene tu 'r'? ¿Cómo la describes?	What are the auditory characteristics of your 'r'? Is it long or short?

After the self-assessment section, the comparative analysis focused on participants having to answer similar questions to those answered for the Visual Feedback group (see Table 7). In the comparison exercise, participants compared their productions with those of a native speaker of Spanish. Participants were required to rely in their auditory skills and their short time memory because the stimuli was presented a limited amount of time

(10 times per cluster type). Students had to describe their productions and those of the native speaker in terms of loudness, length, presence or absence or vibration in the ‘r’ and any other features they considered important for producing a ‘correct word’.

Table 7. Sample Questions for Comparison activity

Sample	English Translation
¿Qué características auditorias tiene la ‘p’ del hablante nativo?	What are the auditory characteristics of the native speaker ‘p’?
¿Qué características auditorias tiene la ‘r’ del hablante nativo?	What are the auditory characteristics of the native speaker ‘r’?
¿Cuál es la diferencia entre tu ‘pr’ respecto a la del hablante nativo? ¿Cómo reconoces esta diferencia?	What is the difference between your ‘pr’ and the one of the native speaker? How can you recognize this difference?
¿Crees que tu ‘p’ es igual o diferente que la del hablante nativo? ¿Cómo?	Do you think that your ‘p’ is the same as the one of the native speaker? How do you know?
¿Crees que tu ‘r’ es igual o diferente que la del hablante nativo? ¿Cómo?	Do you think that your ‘r’ is the same as the one of the native speaker? How do you know?

At the end of this exercise, participants were asked to take into account their answers and suggest ways for improving their pronunciation (see Table 7, upper question). Then, they were told to listen once again a recording from a native speaker of Spanish and to repeat the tokens and as many times as they listened to them, taking into account the differences and similitudes they found in their analysis. Some of the answers obtained for these questions included “Puedo usar más aire y puedo decir ‘r’ más suave” (*I can use more air and say the ‘r’ softer”,* participant D07) and “Puedo poner mi lengua arriba y por la ‘t’. puedo pronunciar más corta” (*I can place my tongue up and for the ‘t’, I can shorten it”*, participant D17*)*. This shows that the participants were trying to identify how they could articulate the segments more accurately. After this, participants had to respond a self-assessment question (See Table 8, bottom question) in order for the researcher to see if the

participants thought the treatment could have been helpful to improve their pronunciation. The stimuli were taken from the list of 'Novel Phrases' that they had already recorded so they were familiarized with the tokens.

Table 8. Sample Questions after Self-assessment and Comparison activities.

	English Translation
¿Qué crees que podrías hacer para mejorar la pronunciación de tu 'tr' ¿Cómo?	Do you think that you could improve the pronunciation of your 'tr'? How do you know?
¿Crees que pronunciaste la 't' y la 'r' más similar a las del nativo esta vez? ¿Por qué?	Do you think that this time you pronounced the 't' and the 'r' more similar to the one of the native speaker? Why do you think you did?

Similar to the Visual Feedback Group, the final activity consisted of a mini- test (See Appendix E) where participants listened some novel phrases produced by either a native speaker of Spanish or a native speaker of English and they had to recognize if the utterance they listened to was from a native or a non-native speaker of Spanish. To have a record of these answers, the participants were asked to check the corresponding box in their handout. Recordings were randomly selected in each exercise/test so participants had a variety of combinations.

3.3.2.3 Post-Test

Close to the end of the Fall semester participants were required to record a final list of tokens listed as 'Phrases' and 'Sentences' as in the previous sessions. No further interventions or treatments were performed after the last in-class pronunciation activity.

3.3.3 Control Group

3.3.3.1 Pre-Test

Paralleling the procedure for the Visual Feedback Group (VFG) and the Drills Group (DG), the Control Group (CG) performed a Pre-Test, three interventions and a Post-

Test. The procedures for the Pre-Test were identical to those used for the VFG and for the DG. Participants were asked to record the same list of ‘Sentences’ and ‘Phrases’ provided to the VFG and to the DG in the Pre-Test and were asked to send their recordings via email as well.

3.3.3.2 Treatments

This group had traditional instruction which means that participants did not have any kind of treatment focused on the targeted stop-rhotic clusters. For some researcher, traditional instruction refers to the implementation of common, recurring and seemingly ‘without innovation’ methods to teach pronunciation within the L2 classroom (i.e., Thomson & Derwing, 2014). For the purposes of the present study, this concept will be defined as the fact of not receiving specific training to enhance the improvement of pronunciation skills except through the exposure participants had with the L2 by means of their instructor and curricula activities. This type of instruction is useful because while many researchers have found that receiving pronunciation instruction is effective and leads to significant improvement in the L2 learners’ pronunciation skills (e.g., Chun 1998; De Bot 1980; De Bot & Mailfert 1982; Derwing, Munro & Wiebe, 1998; Elliott, 1997; González-Bueno, 1997 Gu & Johnson, 1996; Lord, 2005; Molholt 1988; Olson, 2014; Thomson & Derwing, 2014; Trofimovich & Gatbonton, 2006; Yoshida & Fukada; 2014), it could give a glance on how likely or unlikely is for L2 learners to acquire a native-like pronunciation without specific pronunciation training but being exposed to both kinds of input used in the experimental groups (auditory and visual) even when the focus is different.

Even though there was no training focused in pronunciation, there were three in-class activities equivalent in time to the in-class intervention of the two experimental groups. These interventions were focused on cultural presentations that students prepared for class as addressed in the syllabus of the course. In order to balance the time that participants from VFG and the DG groups spent out of the classroom (identifying the tokens and segmenting them in the spectrograms and performing repetitions); this group was given an extra activity (Appendix F) that they had to complete after watching a video of ‘Don Quijote’ which would provide a visual and auditory input to the participants but

as mentioned above, there was no intentional, direct or specific mention of the targeted stop-rhotic clusters. Participants were asked comprehension questions in Spanish about the topic of the video which would usually aim to help participants use the vocabulary learned in class (i.e., physical characteristics, personality, types of constructions…)

3.3.3.3 Post-Test

As in the VFG and the DG, the Post-Test consisted in producing and recording a final list of tokens from a list of 'Phrases' and 'Sentences'. No further action was required from the participants after the researcher received the final recording set.

3.4 Measurements

For all tokens, the measurements taken correspond to the VOT of the plosive consonant, the rhoticity (length and intensity ratio (iR)) and presence or absence of an epenthetic vowel. Measurements were made using Praat Version 6.0.21 (Boersma & Weenink, 2001).

3.4.1 Voice Onset Time (VOT)

The VOT for each one of the voiceless stops contained in the targeted SRC was analyzed from the beginning of the aperiodic wave at zero crossing with the horizontal axis and just before onset of the vocalic segment of the adjacent syllable (Díaz-Campos & Lazar, 2003). Native speakers (NSs) of American English (AE) tend to aspirate their voiceless plosives [p^h] [t^h] [k^h] resulting in long-lag productions contrasted by the short-lag of Spanish voiceless stops [p] [t] [k] (Lisker & Abramson, 1964; Hammond, 2001; Hualde, 2011). Figure 6 and 7 show the difference with respect to VOT for the word 'perro' *dog,* produced by a NS of AE and by a NSS. Note that the yellow and red box highlight the voiceless stop /p/ in each spectrogram. The spectrogram shows that the production of the NS of AE is longer than the one from the NSS (50ms vs. 13ms).

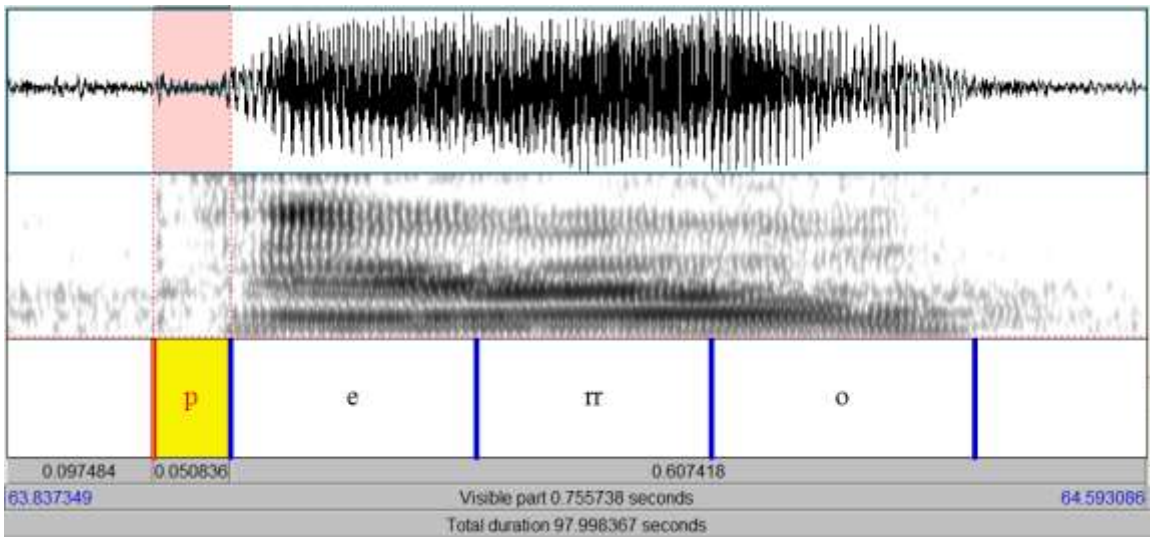


Figure 6. Word '*perro*' dog produced by a NSE.

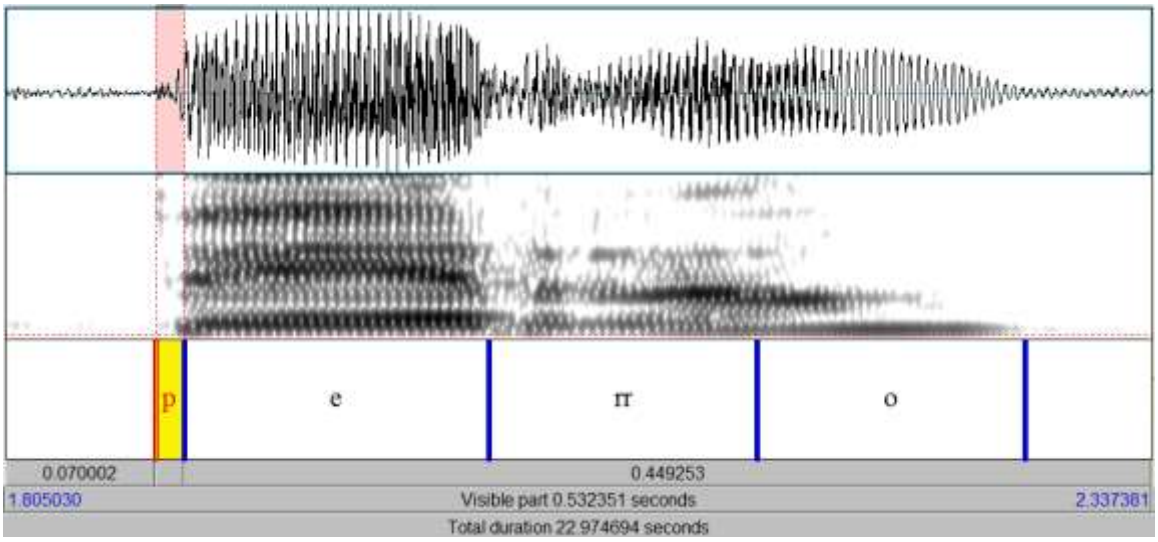


Figure 7. Word *'perro'* dog produced by a NSS.

3.4.2 Rhoticity

For the rhotic segment, there were two types of measurements: first, the intensity Ratio (iR) and second, the length of the rhotic segment.

First, the iR was calculated as the difference between the minimum intensity of the rhotic segment relative to the peak intensity of the following vowel. The realization of the retroflex /ɹ/ produced by native speakers of English (NSE) is voiced throughout and does not contain a full occlusion, therefore such productions generally have a higher intensity.

As the measurement of the /ɹ/ is usually marked by the reduction of energy in the F2 and F3 (Stevens, 1998, p. 535), the boundaries for the rhotic segment were determined by the beginning of the rise of F2 and F3 until the stabilization of the formants of following vowel. In contrast, the flap /ɾ/ produced by native Spanish speakers (NSS) is realized with a full occlusion, resulting in a lower intensity during the production of the target phoneme (Blecua, 2001; Bradley, 1999, 2001, 2002; Colantoni & Steele, 2005, 2006, 2007; Gili-Gaya, 1921). The intensity measurement was then compared to the following vowel in order to account for individual differences in volume and recording quality. In short, an iR closer to 1.00 is more representative of an American English /R/ because the presence of energy is higher while iR closer to 0 is more representative of a native Spanish speaker's /R/ because the occlusion prevents the production of energy. Thus, the louder the rhotic segment, the less Spanish-like it is. Similar analysis has been applied to measure Spanish consonant clusters containing /ɾ/ (Blecua, 2001; Bradley, 2005; Colantoni & Steele, 2005; 2006; 2007; Parker, 2012; Weissglass, 2014). Other studies have also focused in measuring percentage of voicing of this segment (Colantoni & Steele, 2005, 2006, 2007), accuracy in the production (Major, 2012, 2001; Reeder, 1998, Tarone, 1979; 1983; Webb & Bradley, 2009) and other studies have also focused in intervocalic stops in English and Spanish (e.g., Olson, 2014).

In Figure 8, the word '(mar)garita' *daisy* is presented to show how does the flap /ɾ/ occurs in a normal VCV sequence which would be expected to happen if the L2 learners produce the rhotic segment closer to Spanish due to the also expected epenthetic vowel. In other words, if the SRC produced by the NSE is more Spanish-like, an occlusion similar to this one (white stripe between vowels 'a' and 'i') should be noted in the learners' productions.

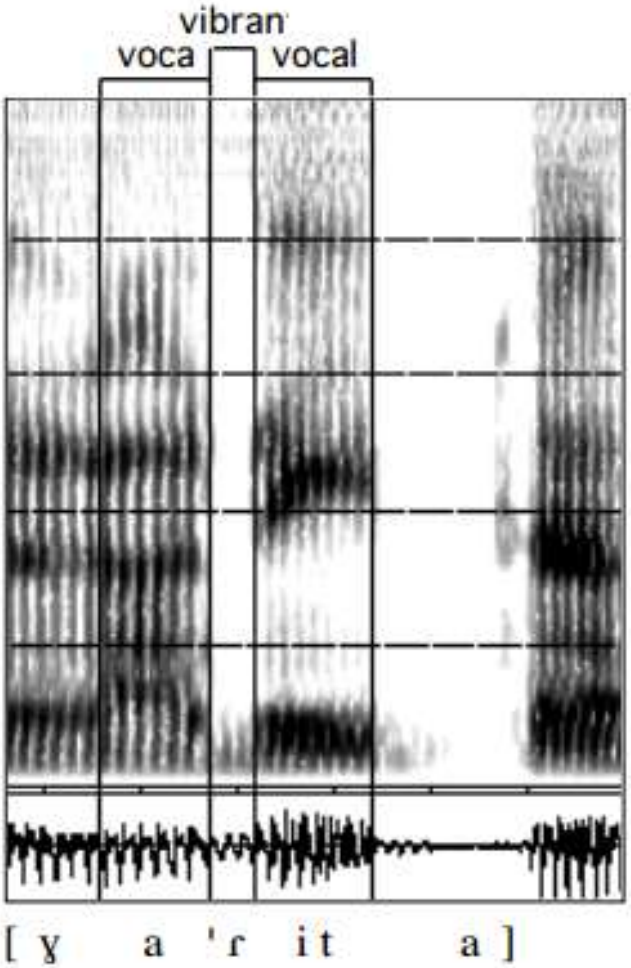


Figure 8. Spectrogram of the sequence (mar)garita *daisy* showing the intervocalic /ɾ/. (Blecua, 2001)

Taking into account this observation, Figure 9 and 10 show the word ‘crucigrama’ *crossword* produced by a NSS and compares it to the same word produced by a ENS. Note how the intensity (yellow line) tends to decline in the production of the NSS and how the segment increases intensity in the production of the NSE. Also note, that the duration of the rhotic segment shows the long-lag that identifies English stops. In the Spanish production, it can be seen that the rhotic segment is visibly shorter than the English rhotic as well. This constituted the second type of measurement although it was not as remarkable (or better to say, reliable) as the iR measurement to show the effect of the trainings over this segment.

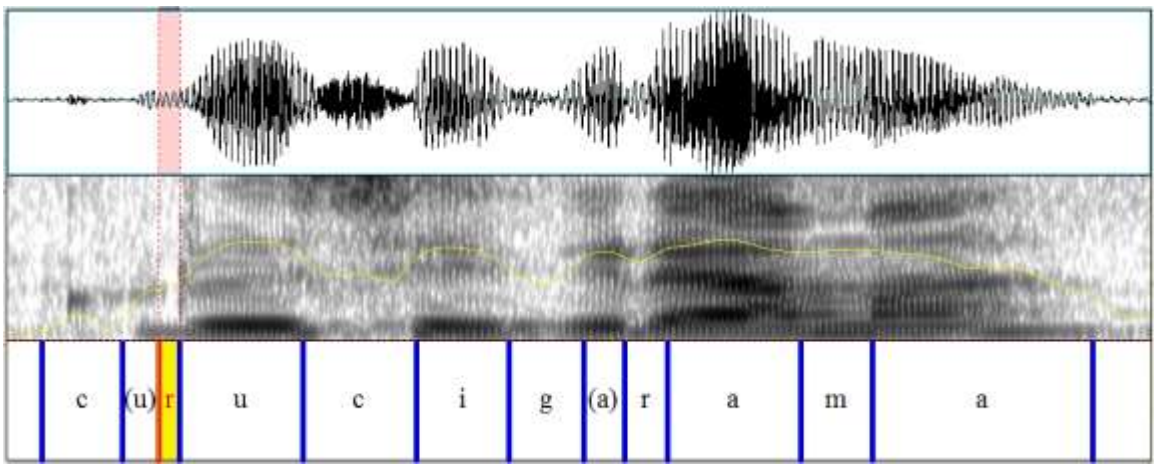


Figure 9. Spectrogram showing the intensity in a VCV sequence for a NSS production.

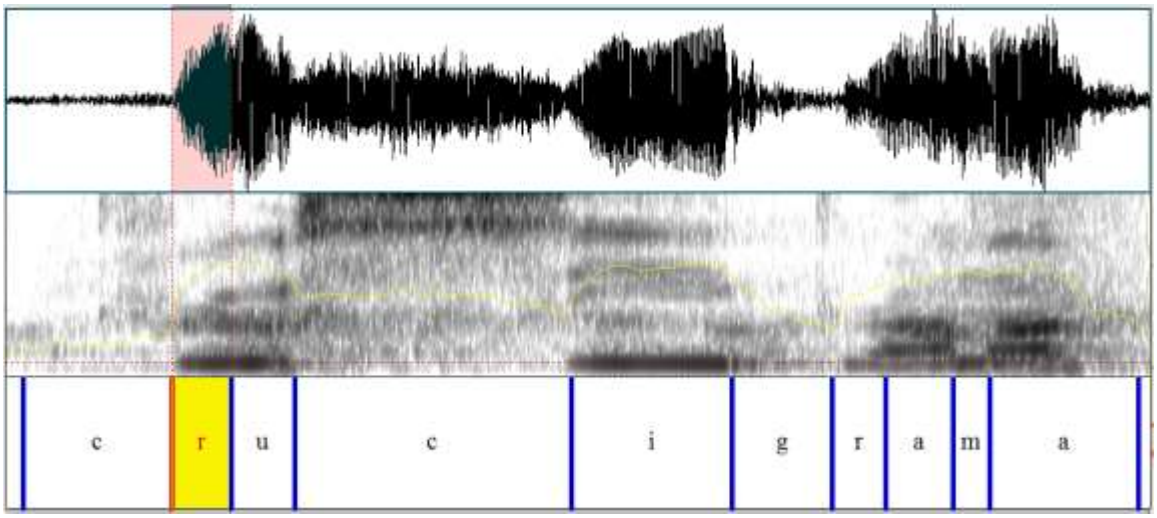


Figure 10. Spectrogram showing the intensity in a CV sequence for an NSE production.

3.4.3 Epenthesis

In the present study, the epenthetic vowels were counted. In order to do so, the boundaries of the epenthetic vowel were determined via vocalic complex patterns in the waveform and if present, it would be the vocalic segment between the stop and the rhotic segment. In other words, the presence of vowel-like segments with the same characteristics of the vowel produced after the rhotic segment (in terms of length and intensity) as well as

the patterns of the formants of the vowel segment the epenthetic vowel mirrors[5]. The methodology to measure the insertion of a vowel-like segment (or epenthesis) was similar to the one used by Ali, Lahrouchi, and Ingleby (2008). "The acoustic segments were identified from both waveform and spectrograms (wideband) by positioning the cursor at time points in the waveform as well as the segment onset and offset on spectrograms" (p. 1179). They investigated the insertion of epenthetic vowel within clusters and geminates in Moroccan Arabic (which has been influenced by French and other Berber dialects) in contrast with Standard Moroccan Arabic. In the first place, they stablished if there was vowel material in the onset consonant cluster (p. 1179) and second, they measured the duration of each segment in milliseconds (ms), the epenthetic and lexical vowel as well as the vowels before and after consonants in the clusters as in the geminated segments.

Figures 11 and 12 show one spectrogram of the word 'tropel' *fight* produced by a NSS and the latter one by a NSE. Note that in the first spectrogram there is an additional box between the consonant 't' and the 'r' showing the presence of an epenthetic vowel whereas in Figure 12, there is no intermediate segment.

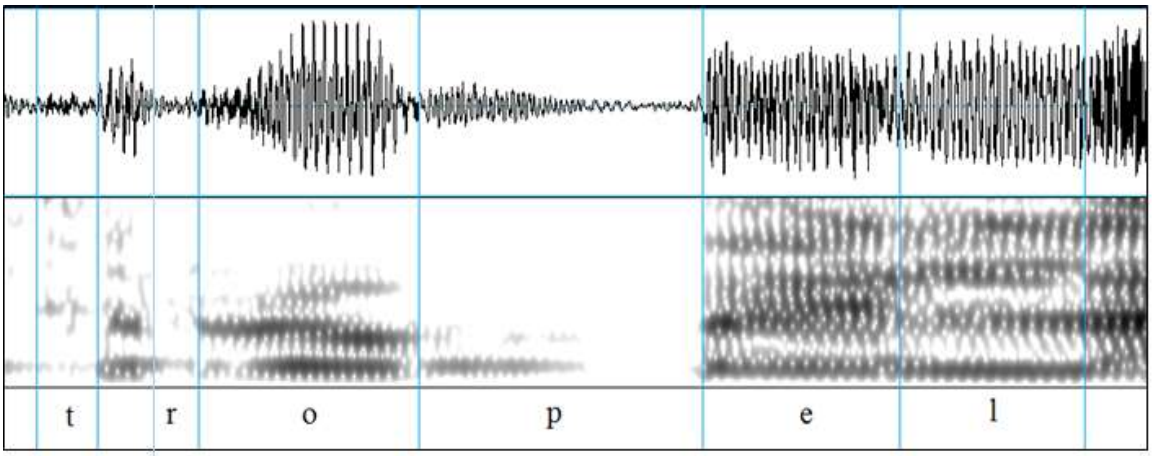


Figure 11. Spectrogram and waveforms of the word '*tropel*' (fight) pronounced by a Spanish native speaker

[5] There are other studies in OT which have described that vowel epenthesis are also a way to break up difficult sequences of multiple consonants (see Hall, 2011 for discussion) and that inserting an epenthetic vowel may be a response to solve difficult or unfamiliar consonant sequences so they can be produced faster and more accurate (i.e., Bradley & Schmeiser, 2003; Davidson, 2006; Hall, 2011; Kuipers et al., 1996).

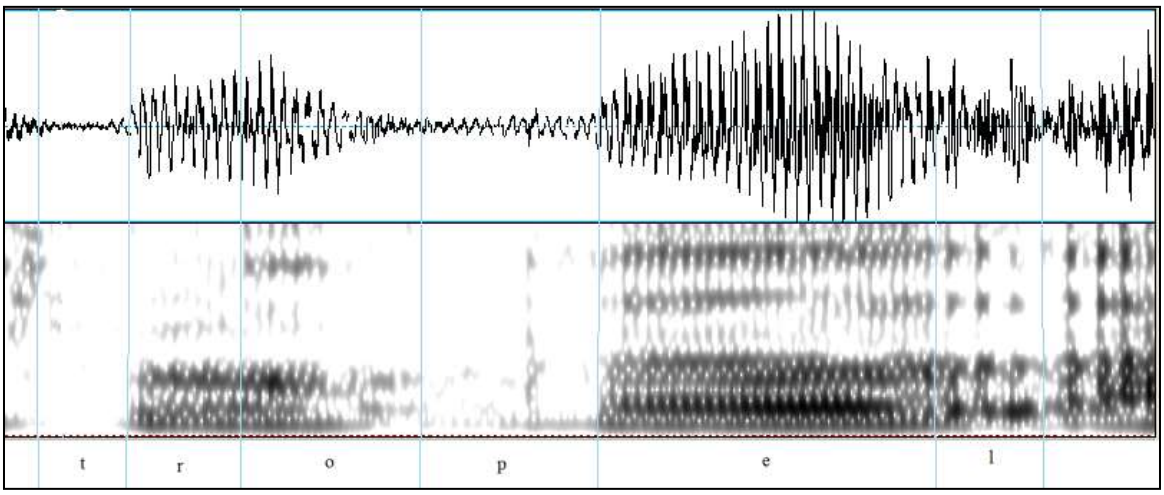


Figure 12. Spectrogram and waveforms of the word '*tropel*' (fight) pronounced by a Spanish learner

All the procedures followed from the beginning of the study aimed to provide as many data as possible in order to corroborate or discourage the proposed hypotheses. Based on other studies, the results expected after the trainings are received by the participants are: shorter VOT for all voiceless stops, reduction of the iR to almost zero and reduction of the rhotic length as well. Consequently, if the iR was close or equal to zero (meaning that the rhotic is a flap and not an approximant) we would see the epenthetic vowel.

CHAPTER 4. RESULTS

In the following chapter, the results obtained in each experimental group (Visual Feedback, Drills and Control) for the tokens containing SRC (/pr/, /tr/, /kr/) will be discussed based on the type of utterance: Carrier Sentences (Section 4.1) or Novel Phrases (Section 4.2). Each section will be subdivided to provide a wider analysis of each one of the three targeted phonetic features: VOT (Section 4.1.1 and Section 4.2.1), Rhoticity and rhotic duration (Section 4.1.2 and Section 4.2.2), and Epenthesis (Section 4.1.3 and Section 4.2.3).

A total of 21,060 tokens (39 Participants [13 Visual Feedback Group, 13 Drills s Group and 13 Control Group] x 36 tokens x 3 Drills per training x 5 trainings [Pre-Test, Training 1, Training 2, Training 3 and Post-Test] = 21,060 tokens [7,020 tokens per group]) were measured and analyzed by the investigator via Praat (Boersma & Weenink, 2001). The boundaries of the aspirations, intensity Ratio (iR), length of the rhotic segment and the epenthetic vowel (if present) were annotated by hand and each one of these measurements, including the iR were extracted by running a script in all the files recorded.

In order to determine if the gains obtained were significant, the data were submitted to a LMEM (Linear Mixed Effect Model). For each of the models conducted, fixed effects included both experimental groups (Visual Feedback Group and Drills Group) as well as the Control Group along with time of recording (Pre-Test, Training 1, Training 2, Training 3 and Post-Test). Random effects included both subject and initial phoneme (i.e., /p/, /t/, /k/) with both random slopes and intercepts[6]. Separate models were conducted for each of the features under study. Significance criterion was set at $|t| > 2.00$. Post hoc analysis, where relevant, was conducted using multiple comparisons with a Tukey adjustment.

Other phonemic errors like those in Munro & Derwing (1995a) such as deletion, insertion of an unexpected segments, or transposition of segments (i.e., ‘per’ instead of ‘pre’), were not included in the data analysis. From a total of 36,720 tokens collected, a

[6] As place of articulation was not of interest, it was included as a random effect. Results are presented for each phoneme for clarity and completeness.

total of 4,860 tokens were excluded from the study for these reasons. This resulted in a total of 21,060 tokens being included in the final analysis (7,020 per group).

4.1 Carrier Sentences

4.1.1 Voice Onset Time (VOT)

Results of the initial model conducted on VOT revealed a significant interaction between Group and Time, as illustrated by the difference between the intercept (Control Group- Pre-test) and both experimental groups at the Post-test (Drills Group: $\beta = -0.034$, $t = -13.563$; Visual Feedback Group: $\beta = -0.43$, $t = -17.047$). Important for the current analysis, post-hoc analysis revealed no significant differences between either of the experimental groups and the Control Group for Pre-test performance (Drills Group: $p = .7687$; Visual Feedback Group: $p = 1.000$). At the Pre-test, participants in the Visual Feedback Group produced average VOTs of 62ms for /p/, 92ms for /t/, and 84ms for /k/. Similarly, Drills Group produced average VOTs of 56ms for /p/, 83ms for /t/, and 75ms for /k/ while participants in the while the Control Group produced average VOTs of 59ms for /p/, 94ms for /t/, and 84ms for /k/. As such, the three groups of participants represent a homogenous group at the onset of the study.

Considering the impact of the interventions and the potential for change over time, both the Drills Group ($p < .0001$) and Visual Feedback Group ($p < .0001$) demonstrated significant improvement in VOT production between Pre-test and Training 1 and Pre-Test and Post-test. The patterns indicating the significant gains that both experimental groups obtained after the first and after the last intervention can be observed Figures 13, 14 and 15 as well as see a comparison on the significant gains versus the non-intervention group (Control Group). Surprisingly, the Control Group also evidenced some degree of positive change in their VOT production ($p = .009$) but it was not as significant as the results obtained with the experimental groups. Examining the average VOTs, the Visual Feedback Group produced in the Pre-test a VOT of 62ms for /p/, 92ms for /t/ and 75ms for /k/. After Training 1, these VOTs got reduced in at least a 33.9%, 48.9% and 26.7% respectively which means that these same segments scored an average of 41ms, 47ms and 55ms just after the participants received the first intervention. Similarly, results obtained from the

Drills group show that the VOT average of 56ms, 83ms and 75ms for /p/, /t/ and /k/ was reduced after Training 1 as well. The aspiration of /p/, /t/ and /k/ reduced in 39.3%, 53% and 29.3% represented in 34ms, 39ms and 53ms respectively. There was an interesting behavior in the Control Group's results as seen in Figures 13, 14 and 15. There was a significant increase in VOT between Training 1 and Training 2 ($p < .0001$) which was reduced again by Training 3. This increase in VOT was still significantly different than the VOT obtained in the Pre-Test ($p < .0003$) and significantly different from the Post-Test results ($p < .0001$). Control group, vis-à-vis, did not show a significant difference between the Pre-test and Training 1 as results show that /p/ had the same VOT of 59ms in both Pre-test and after Training 1. There was no variation at all. For /t/ and /k/, results show a VOT decrease of just 8.5% and 8.3% going from 94ms and 84ms to 86ms and 77ms respectively, corresponding to a reduction of only 8ms and 7ms.

When comparing the Pre-Test and Post-tests scores, the Visual Feedback Group produced a mean VOT of 62ms, 92ms and 75ms for /p/, /t/ and /k/ respectively in the Pre-test; however, the results obtained in the Post-test showed a reduction of 58.1%, 75% and 61.3% corresponding a final VOT of 26ms ($SD=0.008$, $SE=0.9$), 23ms ($SD=0.007$, $SE=0.8$) and 29ms ($SD=0.013$, $SE=1.5$) for these segments. Results for the Drills Group indicate that the trainings were effective for /p/, /t/ and /k/ as well as in the Visual Feedback Group. The VOTs had a significant reduction of 51.8%, 63.8% and 56% when comparing results from the Pre-test to the Post-Test ($p < 0.0001$). The initial VOT average was equal to 56ms ($SD=0.027$, $SE=2.5$), 83ms ($SD=0.023$, $SE=3.1$) and 75ms ($SD=0.028$, $SE=2.7$), for /p/, /t/ and /k/ and the final average VOT was of 27ms ($SD=0.008$, $SE=1.1$), 30ms ($SD=0.015$, $SE=1.3$) and 33ms ($SD=0.017$, $SE=1.6$) respectively. The Control Group, in contrast, did not exhibit a significant difference between performance between the Pre- and Post-test. The Control Group produced mean VOTs of 59ms, 94ms and 84ms in the Pre-test and mean VOTs of 54ms, 87ms and 75ms in the Post-test. These results do not show mush variance as the percentages of improvement related to each token (/p/, /t/, /k/) is less than 12%, more specifically, 8.5% for /p/, 7.4% for /t/ and 11% for /k/. All of the results described above show that the participants who were part of the pronunciation instruction groups did improve for this targeted phonetic feature (VOT).

Subsequent analysis, comparing group performance at the Post-test reveals that while the Control Group did evidence some improvement, both experimental groups produced significantly shorter VOTs than the Control Group at the Post-test ($p < .0001$).

Lastly, observing Figures 13, 14 and 15, it is relevant to examine the change in VOT that occurred after only one training for both the Drills and VF group. This improvement was statistically significant for both groups ($p < .0001$). Interestingly, the experimental groups evidenced similar VOTs at the Post-test ($p = .999$), there was no significant difference between experimental groups at the Post-Test. These results could be taken to suggest that while either type of phonetic instruction is successful in reducing the VOT duration relative to a control group that did not receive any specific training, both methods proved to be equally successful to improve this feature even after one intervention.

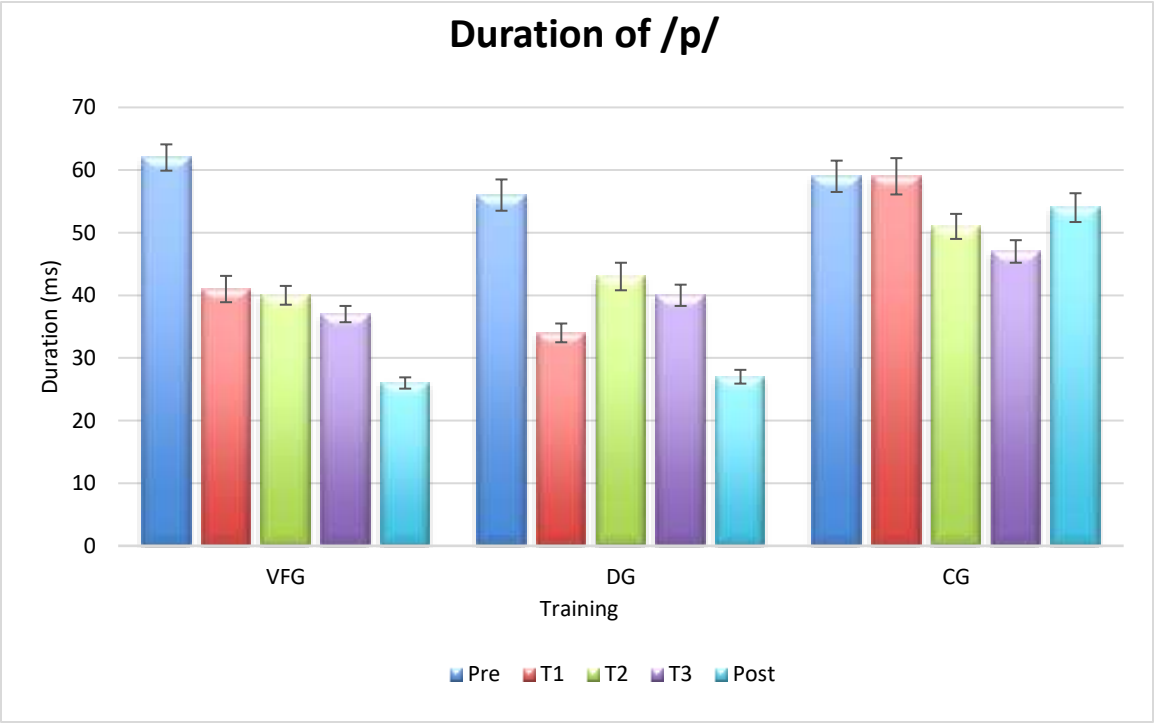


Figure 13. VOT values for /p/

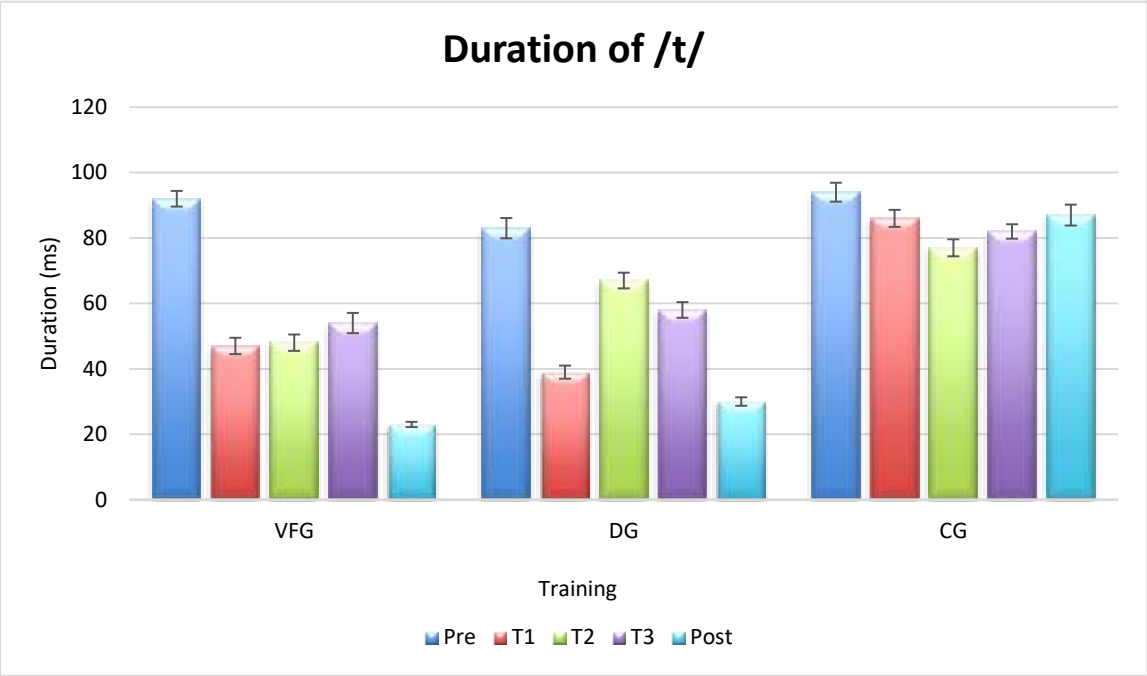


Figure 14. VOT values for /t/

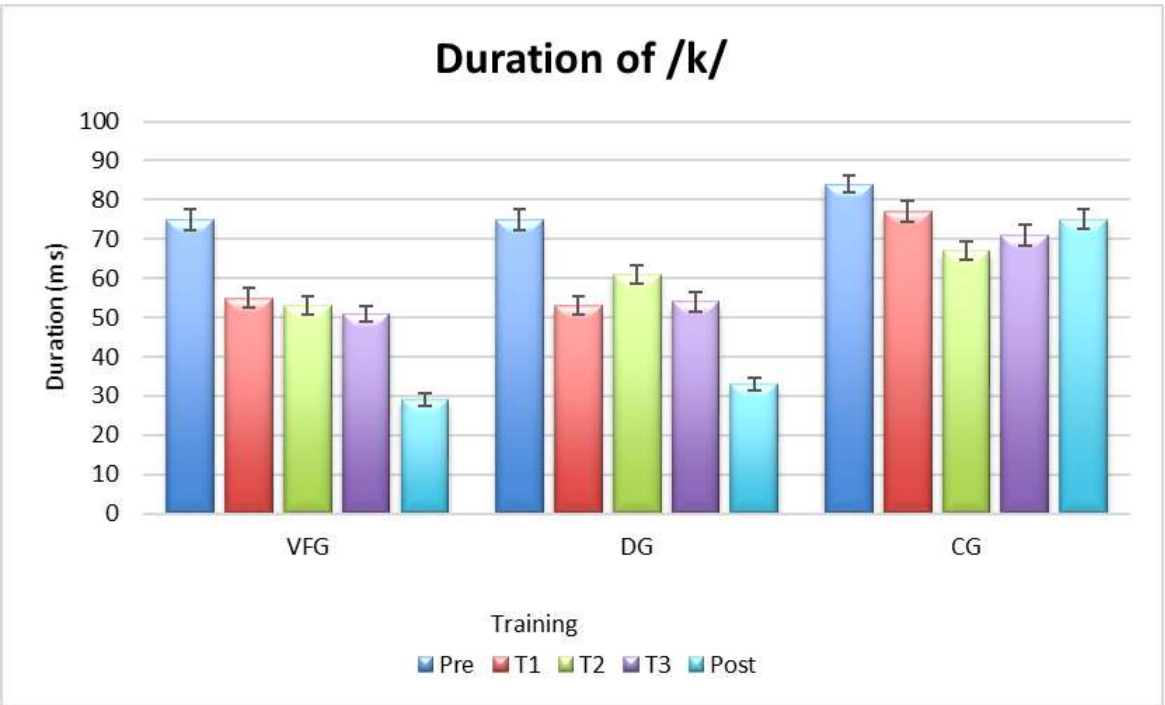


Figure 15. VOT values for /k/

4.1.2 Rhoticity: Intensity Ratio (iR) and Duration

Paralleling the analysis for VOT, a linear mixed effect model was conducted on iR and rhotic duration. An analysis of performance at the Pre-test showed no difference between the Control Group and the Drills Group (p = .998) nor the VF Group (p =.996). Again, the main model showed a significant difference between the intercept (Control Group- Pre-test) and the Drills Group, Post-test (β = -0.013, t = -2.53), although there was no difference between the intercept and the VF Group, Post-test (β = 0.004, t = 0.71). A comparison of each group's performance between the Pre- and Post-test revealed a significant change for all three groups: Drills Group (p = .067), VF Group (p < .0001) and Control Group (p < .0001). As mentioned above (section 3.4.2), an iR closer to 0 represents a more Spanish-like production of the rhotic segment due to the closure produced by the tip of the tongue touching the alveolar ridge in the articulation of the Spanish tap /ɾ/, a value closer to 1 represents a more English-like rhotic production. An analysis of Figure 16 reveals that the change was in the opposite direction than the one expected.

Even though the change of the iR was to the opposite direction (it increased instead of reducing it), results showed that for the intercept and Drills Group there was not significant difference (p= .676). For the Control Group, there was not a significant difference between Pre-Test and Training 1 and 3 (T1: p=1.000; T3: p=1.000) but Pre-Test and Training 2 did show a significant difference (p=*.094*) as well as Pre-Test versus Post-Test (p < .0001). There was not a significant difference between Pre-Test and after Training 1 for Control Group and Drills Group (Control Group: p= 1.000*;* Drills Group: p= .967) whereas the Visual Feedback Group presented a significant difference between Pre-Test and after Training 1 (p <.0001). The most significant finding was that there was not a significant difference between the Pre-Test and each of the trainings for the Drills Group (T1: p= .966; T2: p= *.872*; T3: p= .994) or between each training and the Post-Test (T1: p= 1.000; T2: p= 1.000; T3: p= 1.000). The Visual Feedback Group presented a significant difference between Pre-Test and each of the trainings (T1: p < .0001; T2: p < .0001; T3: p= .045). Moreover, post-hoc analysis reveals no significant difference between the Control Group and either of the two experimental groups at the Post-test (Drills Group: p = .968; Visual Feedback Group = 1.000). In short, none of the groups evidenced any improvement with respect to iR, and they all actually produced

significantly more English-like is at the Post-test. It is worth noting the relatively small degree of change in iR.

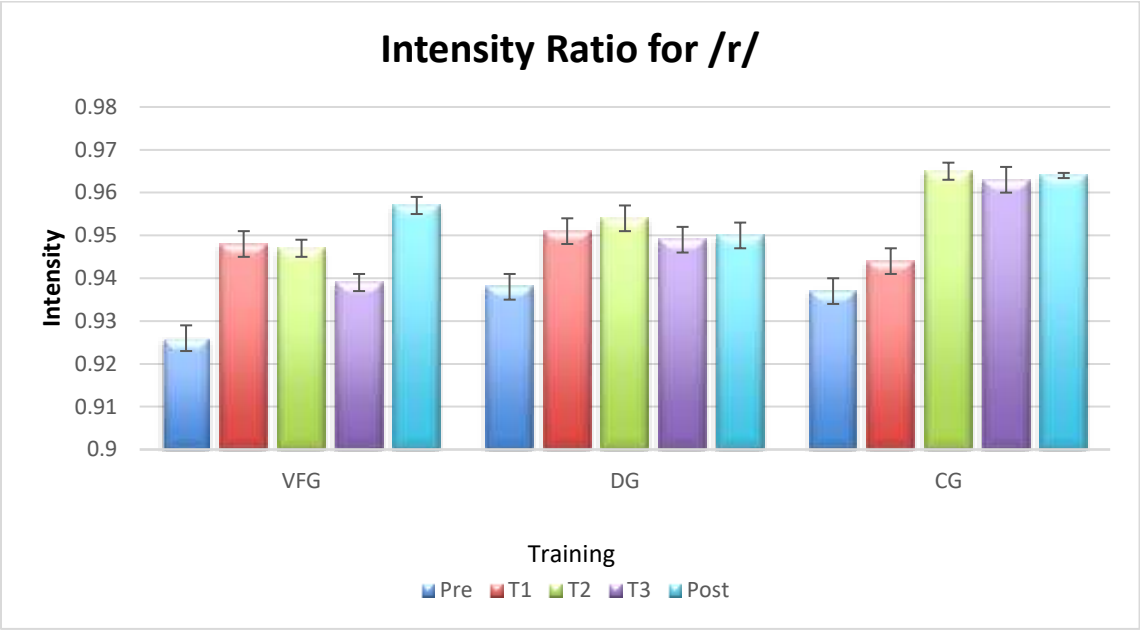


Figure 16. Rhotic Intensity Ratio

While the main model did not show an interaction between the main factors, parallel post-hoc tests were conducted on rhotic duration. Again, English approximant rhotics have a longer duration than the Spanish flapped rhotic. There was no difference between the Control Group and either of the experimental groups at the Pre-test (Drills Group: p = 1.000; VF Group: p = .894), with all groups producing rhotics of similar durations (Control Group: 37ms; Drills Group: 42ms; Visual Feedback Group: 37ms).

An examination of each group at the Pre-and Post-tests revealed that *all* groups produced a significantly shorter rhotic duration at the Post-test than the Pre-test (Control Group: p = .925; Drills Group: p < .0001; Visual Feedback Group: p < .0001). Moreover, there was no significant difference between any of the groups at the Post-test (p > .429). In Figure 17, the results for rhotic duration showed a trend towards the reduction of its lag, more Spanish-like. Again, neither group differed from the Control Group at the Pre-test (Drills Group: p = 1.000; Visual Feedback Group: p = 1.000). However, contrasting with results for iR, both experimental groups demonstrated a significant improvement between

the Pre- and Post-test performance for rhotic duration (Drills Group: $p < .0001$; Visual Feedback Group: $p < .0001$). The Control Group showed a significant improvement as well from its Pre-Test to the Post-Test (Control Group: $p < 0.925$). The duration of /r/ for the Visual Feedback Group decreased in a 29.7% going from 37ms to 26ms (*SD*= 0.010; *SE*= 0.0006). The second experimental group, Drills Group, reduced its duration in 25% from its Pre-test to the Post-test reducing the former average of 40ms to 30ms (*SD*=0.009; *SE*=0.0006). Interestingly, the Drills Group was the only one to show an effect of training and a significant difference between Pre-Test and Training 1 (p= .168). Similarly, the Control group showed a significant improvement since the decrease was of 21.4% passing from 42ms to a final average of 33ms between Pre-Test and Post-test. As with VOT, neither method proved to be more effective than the other but hypothetically, exposure to the target language and no training directed to improve this segment can effectively work to improve some of the SRC phonetic features, more specifically, rhotic duration.

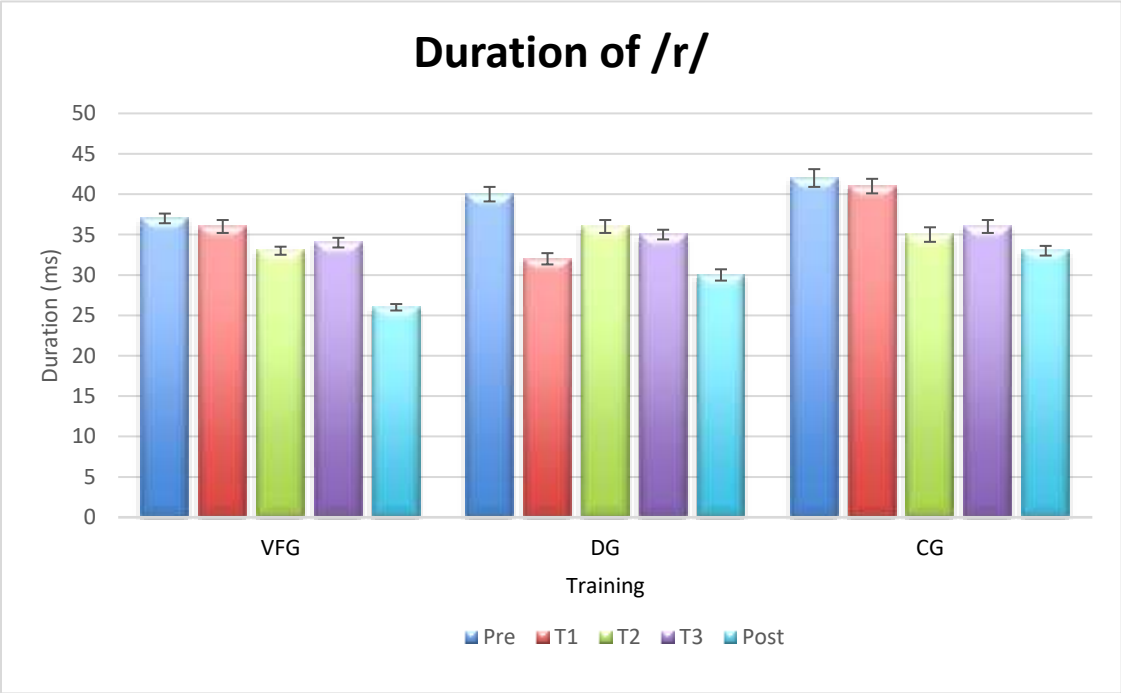


Figure 17. Rhotic duration in ms

4.1.3 Epenthesis

As epenthesis was coded as a categorical variable, either present or absent, data was analyzed with a linear regression. However, with less than 1% of tokens containing an epenthetic vowel, there was insufficient data to allow the model to converge. As such, discussion of epenthesis will rely on a brief description of the data.

The number of epenthetic vowels produced by each group is summarized in Table 9. The Visual Feedback Group started with 0 epenthetic vowels and ended up having 1 epenthetic vowel on the Post-Test. The Drills Group had 1 epenthetic vowel versus a total of 13 epenthetic vowels in the Post-Test produced by 5 different participants. The Control Group produced a total of three epenthetic vowels. Surprisingly, these segments appeared in the Pre-Test and following two trainings but there were no epenthetic vowels in Training 3 or in the Post-Test.

Table 9. Number of Epenthetic Vowels in ‘Carrier Sentences’

	Pre	T1	T2	T3	Post
VFG	0	1	2	1	1
DG	1	0	3	1	13
CG	1	1	1	0	0

Although the results are not promising or encouraging for the acquisition of the epenthetic vowel by L2 learners of Spanish, Figure 18 show a very incipient tendency towards improvement, moreover for the group which was exposed and performed Drills exercises and repetitions. Results show that at least 3 participants produced an epenthetic vowel in the Post-Test in the Drills Group. Participant D05 produced 5, participants D07 and D15 produced 3 each and two other participants produced one each too.

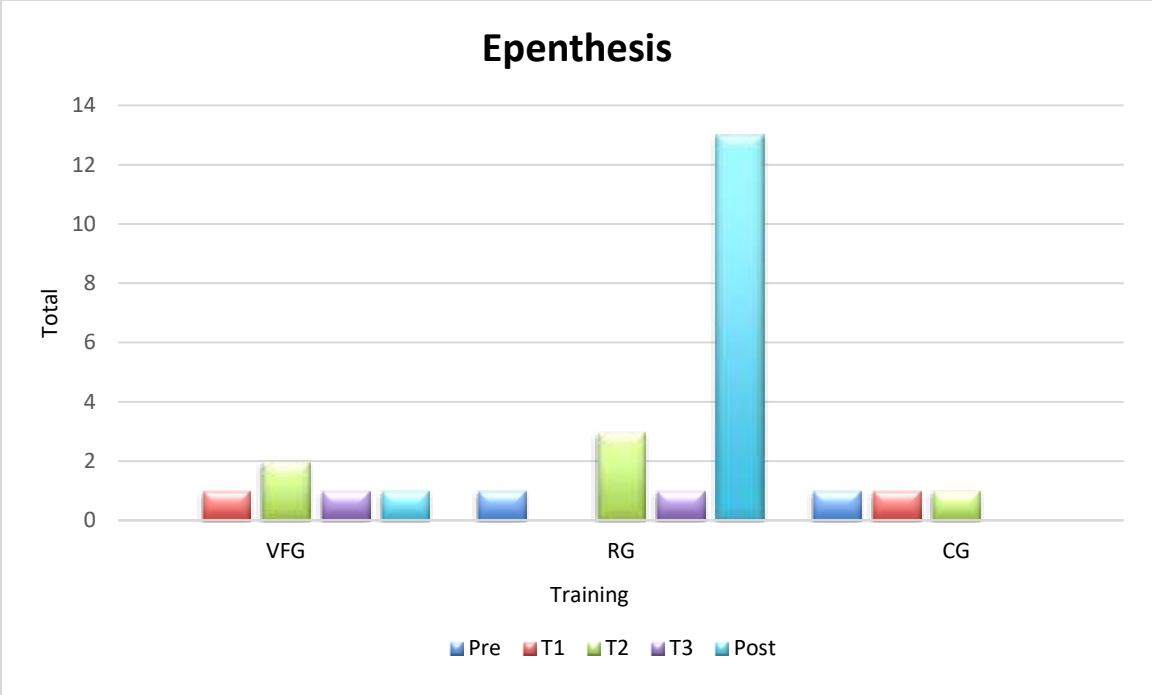


Figure 18. Number of Epenthetic Vowels for "Carrier Sentences"

As a whole, it can be concluded that none of the groups made any observable gains with respect to the production of the epenthetic vowel. While these results will be discussed more in Chapter 5 (Discussion), this is not an unexpected finding considering the general lack of improvement in the production of the Spanish-like tap rhotic segment. In brief, as claimed by researchers such as Colantoni and Steele (2006) and Blecua (2001), the epenthetic vowel in Spanish is used to allow the articulation of the flap but without a flap, there is virtually no production of the epenthetic vowel.

4.2 Novel Phrases

4.2.1 Voice Onset Time (VOT)

Results of the initial mixed model revealed a significant interaction between the two main factors, group and time, as illustrated by the significant difference between the intercept (Control Group, Pre-test) and Drills Group, Post-test (β = -0.043, t = -14.593), as well as between the intercept and Visual Feedback group, Post-test (β = -0.045, t = -15.057). Post-hoc analysis revealed that there was no significant difference between the Control Group and either of the experimental groups for the Pre-test (Control-Drills:

p =.873; Control-VF: *p* = .784). At the Pre-test, participants in the Visual Feedback group produced average VOTs of 64ms for /p/, 75ms for /t/, and 73ms for /k/. Similarly, participants in the Drills s group produced average VOTs of 65ms for /p/, 70ms for /t/, and 81ms for /k/ while the Control produced average VOTs of 73ms for /p/, 84ms for /t/, and 81ms for /k/. As such, these groups were comparable at the start of the treatment. This lack of difference during the Pre-Test can be seen in for each of the phonemes examined in Figures 19, 20 and 21 illustrating results for /p/, /t/, and /k/.

Considering the overall improvement of each of the experimental group (Visual Feedback Group and Drills Group), subsequent comparisons were conducted between the performance of each group at the Pre-Test and Post-test Phases. Results showed that there was a relatively significant difference in the Pre-tests between the two experimental groups and the Control Group (Visual Feedback Group: *p= 0.78*; Drills Group: *p=0.87*). Results also demonstrated a significant difference between the Drills group at the Pre- and Post-test (p < .0001) as well as between the Visual Feedback Group at the Pre-and Post-test (p <. 0001). Notably, there was no significant difference between the Control Group performance at the Pre- and Post-test (p = 1.000). These results illustrate that both experimental groups produced significantly shorter VOTs following the implementation of the intervention. Again examining the average VOTs, the Drills group produced mean VOTs of 25ms, 25ms, and 33ms for /p/, /t/, /k/, respectively. For the same segments, the VF group produced mean VOTs of 25ms, 32ms, and 34ms. Broadly, this represents an improvement of 61%, 67% and 55% for the Drills group and an improvement of 61%, 54% and 58% for the VF Group regarding the segments /p/, /t/ and /k/ respectively. The Control Group, in contrast, showed no significant difference between performance at the Pre- and Post-test. The Control Group produced mean VOTs of 64ms, 100ms and 78ms. The patterns showing the significant gains that both experimental groups obtained after the first and after the last intervention can be observed Figures 19, 20 and 21.

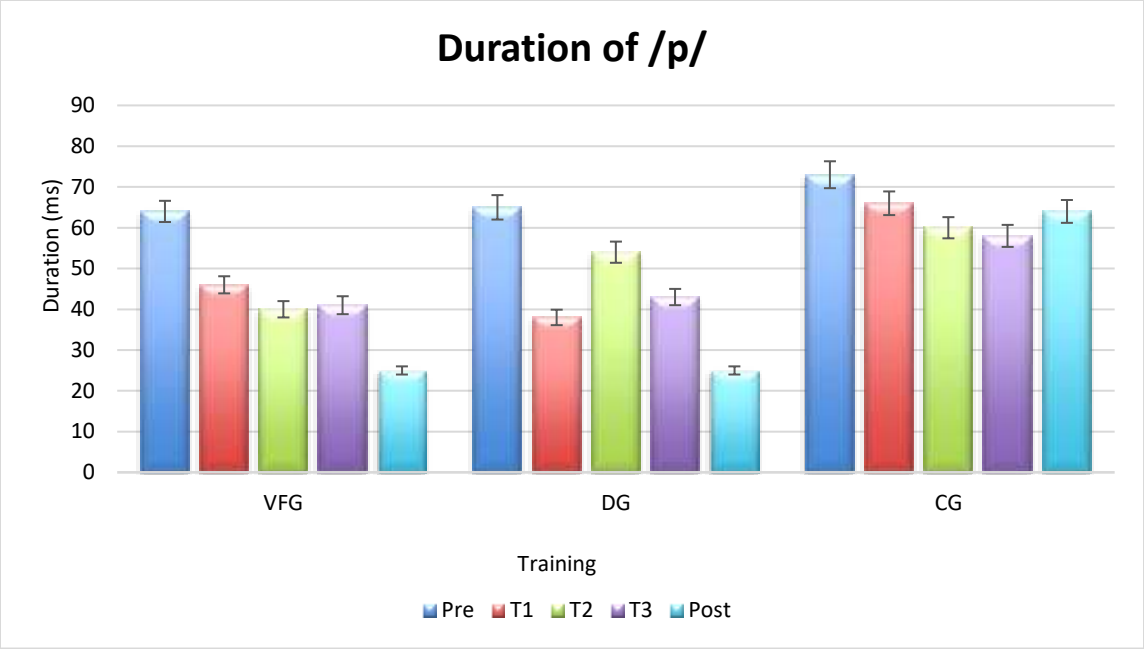


Figure 19. VOT values for /p/

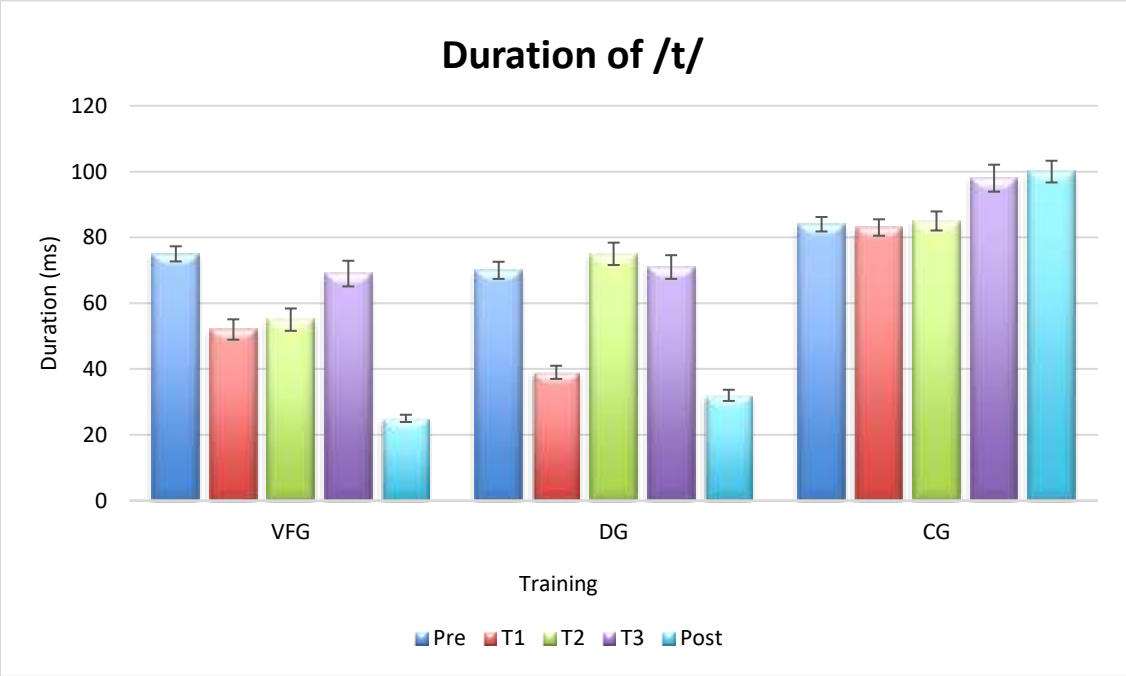


Figure 20. VOT values for /t/

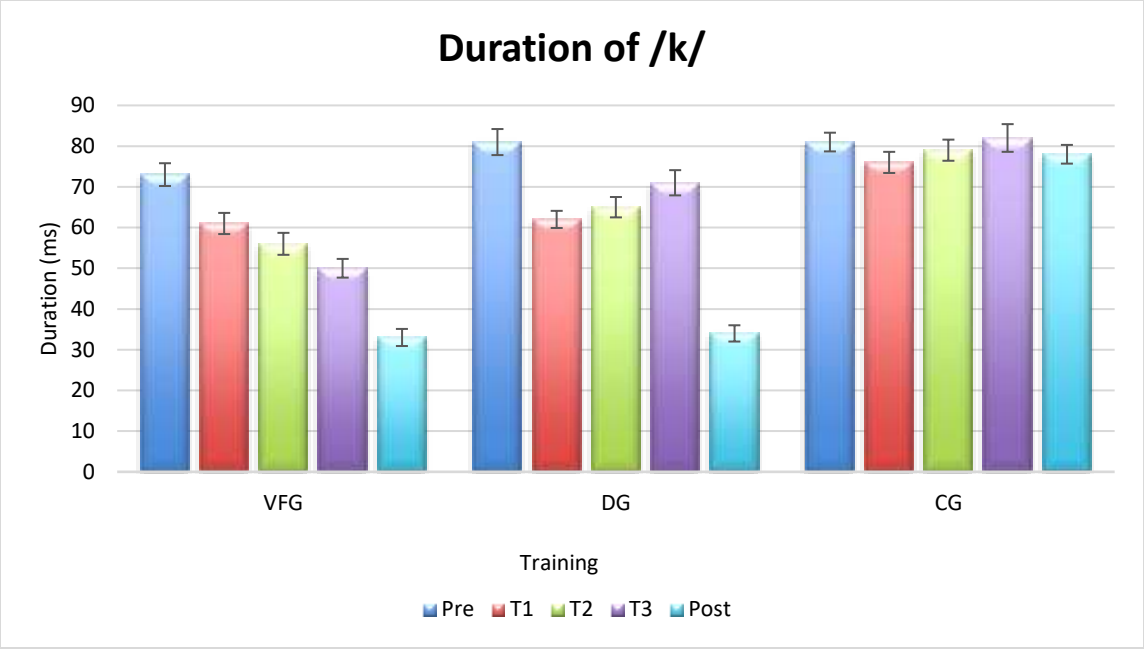


Figure 21. VOT values for /k/

Worth noting, is the nearly immediate effect of the training on VOT production. Both experimental groups showed significant improvement between the Pre-test and following the first intervention (T1) (Visual Feedback Group: $p < .0001$; Drills Group: $p < .0001$). It could be seen in Figures 19, 20 and 21 that both Visual Feedback Group and Drills Group showed a significant reduction over time in the length of the segments. For the VFG, from an overall VOT average of 64ms (*SD*= .022, *SE*=2.6), 75ms (*SD*= .20, *SE*= 2.3) and 73ms (*SD*= .45, *SE*=2.8) in the Pre-Test for p/, /t/ and /k/, there was a significant reduction of the aspiration to an average of 25ms (*SD*=0.008, *SE*=1), 25ms (*SD*=0.010, *SE*=2), and 33ms (*SD*= .010, *SE*= 2.1) for these same segments in the Post-Test. With respect to the Drills Group, the overall VOT average in the Pre- test for p/, /t/ and /k/ was of 65ms (*SD*=0.026, *SE*=3), 70ms (*SD*=0.022, *SE*= 2.6) and 81ms (*SD*=0.028, *SE*=3.2) and the Post-test results show that the final VOT scores were of 25ms (*SD*=0.008, *SE*=1), 32ms (*SD*=0.015, *SE*=1.7), and 34ms (*SD*=0.017, *SE*=2) respectively.

Results from the Control Group stand out in contrast as there was not significant improvement from the Pre-Test to the Post-Test and presented a significant difference even in the Post-test *Standard Deviation* (*SD=0.024* for /p/, *0.024* for /t/ *and 0.019* for /k/) and *Standard Error* (*SE=2.8* for /p/, *3.3* for /t/ *and 2.3* for /k/) results. Of note, results

demonstrate that there is a similar trend towards improvement in the reduction of the duration of the voiceless stop VOT when receiving pronunciation training despite the methodology used.

In short, three points should be noted: first, the experimental groups showed significant improvements in reducing the VOT for the three phonemes (/p/, /t/ and /k/) after Training 1 and when compared Pre-Test and Post-Test data. Second, none of the trainings (Visual Feedback Paradigm or Drills proved to be more effective than the other one, experimental groups did not show significant differences in the Post-Test results despite the treatment received. Third and last, CG indicated no significant gains in reducing the aspiration duration of the phonemes after being exposed to the Drills Group language but no training. Lastly, comparing the effectiveness of the two types of interventions, post-hoc analysis showed no difference in the performance of the two experimental groups at the Post-test (p = 1.000). As such, we can conclude that both treatments were equally effective in shortening voice onset time and that having pronunciation instruction significantly improves L2 phonetic features compared to traditional classroom instruction.

4.2.2 Rhoticity: Intensity Ratio (iR) and Duration

Paralleling the analysis for VOT, a linear mixed effect model was conducted on intensity Ratio (iR). As mentioned above (section 4.1.2), an iR closer to 0 represents a more Spanish-like production of the rhotic segment, while a value closer to 1 represents a more English-like rhotic production. Again, an analysis of performance at the Pre-test showed no difference between the Control Group and the Drills Group (p = .998) nor the Visual Feedback Group (p = 1.000). Interestingly, both Visual Feedback Group and Control Group showed a significant difference between the Pre-Test and each one of the trainings as well as with the Post-Test (Visual Feedback Group: T1: p= .0025; T2: p= .0001; T3: p= .0178, Post-Test: p=<.0001; Control Group: T1: p= .011; T2: p < .0001; T3: p< .0001, Post-Test: p=<.0001). More interesting, there was a significant difference between Pre-Test and Training 1 (p= .0357) but not between Pre-Test and trainings 2 and 3 (T2: p< .893; T3: p= 1.000) and there is an intermediate difference between Pre-Test and Post-Test leading towards a non-significant difference (p= 0.578). The main model showed a significant difference between the intercept (Control Group- Pre-test) and the Drills

Group, Post-test (β = -0.029, *t* = -5.236), and a near significant difference between the intercept and the Visual Feedback Group, Post-test (β = -0.011, *t* = -1.867). A comparison of each group's performance between the Pre- and Post-test revealed a significant change for both the Control Group ($p < .0001$) and Visual Feedback Group ($p < .0001$), and no significant change for the Drills Group ($p = .578$). However, an analysis of Figure 22 reveals that the change was actually in the opposite direction to the one expected. The iR was expected to be reduced through the interventions to zero or close to zero in order to provide evidence of the effectiveness of these methods to acquire this phonetic feature. Participants produced rhotic segments with an average of 1 (which is the highest value that could be possibly obtained) which means that their rhotic did not converge into a flap but it rather strengthened the features of the approximant rhotic characteristic of English. Post-hoc comparisons show that although there is not significant difference among the groups, the Drills Group maintained slightly 'positive' results versus the Visual Feedback group and the Control group which got significantly 'worse' results.

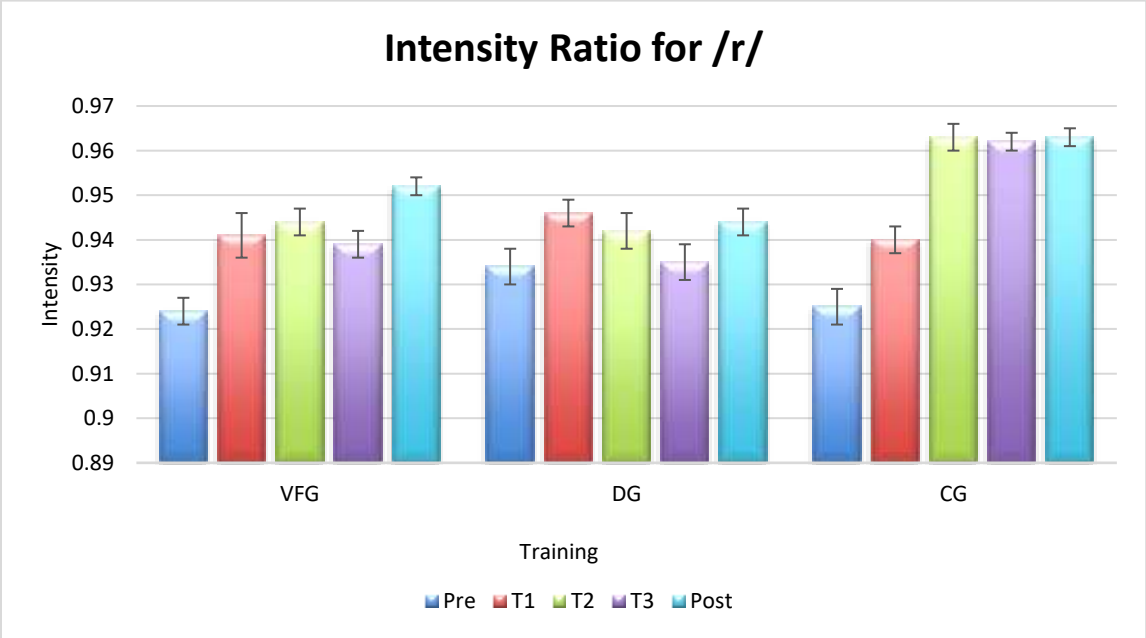


Figure 22. Rhotic Intensity Ratio

Similar to the results obtained in the Carrier Sentences, regarding the rhotic duration, results showed a trend towards improvement for the two experimental groups as seen in Figure 23. Again, neither group differed from the Control Group at the Pre-test

(Drills Group: p = 1.000; Visual Feedback Group: p = 1.000). However, contrasting with results for iR, both experimental groups demonstrated a significant improvement between the Pre- and Post-test performance for rhotic duration (Drills Group: $p < .0001$; Visual Feedback Group: $p < .0001$). The Visual Feedback Group decreased the duration of <r> by 27.5% while the Drills Group scored a 22.5%. Control group did not show a significant improvement since the Pre-Test versus Post-Test results indicated an overall decrease in the rhotic duration of just 4.9% ($p < .0001$). These results contrast with the results obtained in the Carrier Sentences for the same group and the same times of testing where the Control Group improved in rhotic duration by 21.4% ($p<.0001$). It should be noted that the overall change in duration for Phrases, still resulted in significant differences between the experimental groups and the Control Group at the Post-test (Drills Group: $p = .0007$; Visual Feedback Group: $p <.0001$). Again, as with VOT, neither method proved to be more effective than the other, experimental groups did not differ at the Post-test ($p = .9989$).

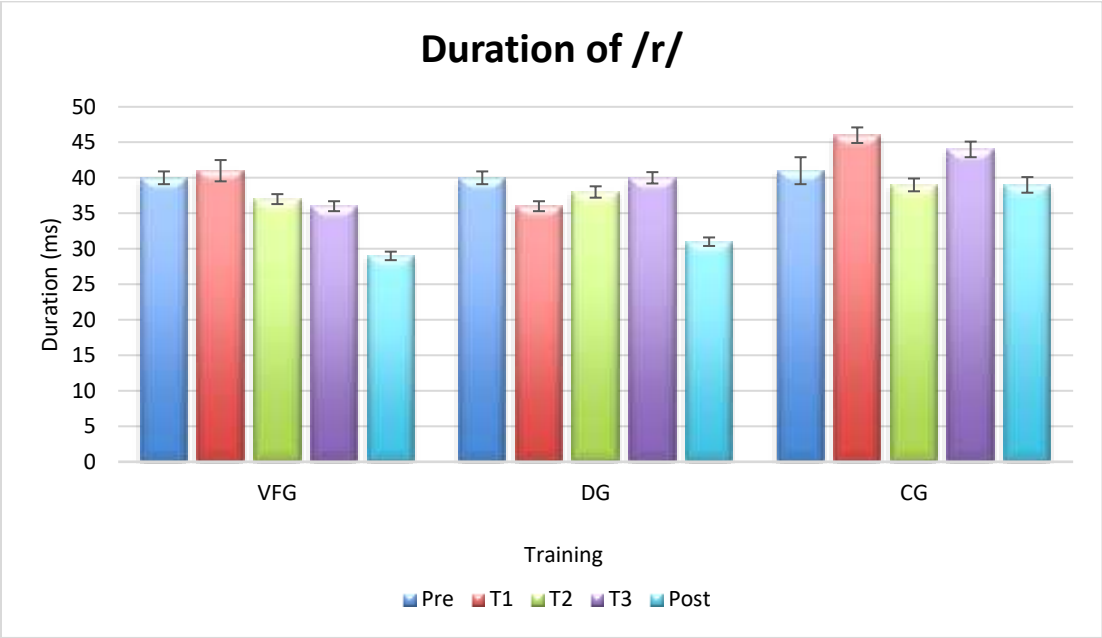


Figure 23. Rhotic duration in ms

4.2.3 Epenthesis

As epenthesis was coded as a categorical variable, either present or absent, data was analyzed with a linear regression. However, with less than 1% of tokens containing an epenthetic vowel, there was insufficient data to allow the model to converge. As such, discussion of epenthesis will rely on a brief description of the data. The number of epenthetic vowels produced by each group is summarized in Table 10. The Visual Feedback group started with 0 epenthetic vowels and ended up having 1 epenthetic vowel on the Post-Test. The Drills Group had one epenthetic vowel versus a total of 14 epenthetic vowels in the Post-Test. The Control group produced a total of two epenthetic vowels but after the first recording session there were no epenthetic vowels in any of the following training sessions.

Table 10. Number of Epenthetic Vowels in 'Novel Phrases'

	Pre	T1	T2	T3	Post
VFG	0	0	0	1	1
DG	0	3	13	8	14
CG	2	0	0	0	0

Interestingly, Figure 24 shows that even though the results are not conclusive and the data is insufficient to state that the Drills s group improved more than the VF group or the Control group, the figures show that this group produced more epenthetic vowels at the end of the trainings than the other two groups. However, it is worth noting that from the 14 epenthetic vowels obtained for the Drills Group in the Post-test, three participants (D01, D8 and D19) had one, two and three epenthetic vowels respectively. Most important, 8 of these epenthetic vowels were produced by the same participant in the Drills Group (D07) who had also produced 3 epenthetic vowels in the Post-test for the set corresponding to "Carrier Sentences". This could be an indicator of the partial effectiveness of this pronunciation method. These results resemble and support Colantoni & Steele's (2006) findings which proposed that only a very limited number of learners (usually highly skilled or more proficient) would be able to produce this features located at the end of the difficulty hierarchy.

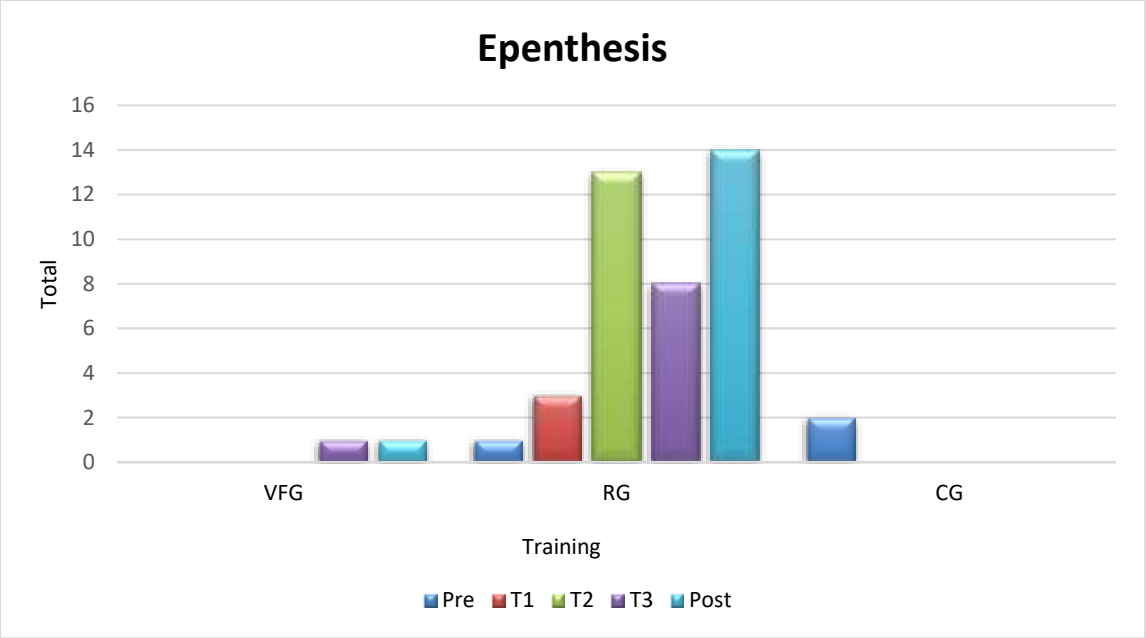


Figure 24. Number of Epenthetic Vowels in 'Novel Phrases'

4.3 Conclusions

Results obtained in the present study are consistent with those obtained by Colantoni and Steele (2006). In their study, participants' VOT improved before rhotic manner and before epenthesis although "for the most part, learners failed to master at least one and often all- of the phonetic properties necessary for target-like realization of Spanish SL (stop-liquid) clusters" (p. 70). However, in the present study, some interesting conclusions can be posed:

1) Receiving training in pronunciation lead to significant improvements in accuracy and speech production of stop-rhotic clusters (SRC) compared to traditional instruction. Pre-Test versus Post-Test results showed that the Visual Feedback Group and the Drills Group outperformed the Control Group in the production of SRC in low and high controlled environments (Novel Phrases and Carrier Sentences). Even though the Control Group improved significantly in reducing the rhotic duration in the most controlled environment (Carrier Sentences), the

remaining phonetic features did not improve significantly (less than 6%) compared to the experimental groups.

2) The use of a Visual Feedback Paradigm and Drills seem to be equally effective for improving the production of SRC's phonetic features. None of the experimental groups had significant gains respect to each other but their results highly contrasted with those obtained by the CG. Previous studies have shown that there is a drop in the VOT values after the first training but the results remained stable or had no significant differences in the upcoming treatments (i.e., Olson, 2014). In contrast, in the present study there was in fact a significant drop in the VOT values for the stops /p/, /t/ and /k/ as well as for the rhotic duration from the Pre-Test to the Post-Test. However, results did not remain stable; the variation continued throughout the trainings showing a trend towards improvement in each phonetic feature. Worth highlighting, after the last in-class training (Training 3), the VOTs and rhotic duration showed a second significant reduction as or equally significant as the length reduction seen after Training 1. One possible explanation for these results is that learners' acquisition of complex segments such as SRC depend on both perception (the way the input is provided) and production (articulatory constrains) (Colantoni & Steele, 2006) in spite of the type of intervention. The second explanation is that the less categorical the phonetic feature is, the more resistant it will be to the effects of training whether it is visual or auditory.

3) Colantoni and Steele (2006) claimed that there is a difficulty of acquisition hierarchy that is related to their degree of variability. They claimed that the learners' failure to acquire the targeted phonetic features might be because "learners' difficulty in parsing variability may inhibit accurate learning" (p. 71). The results in the present study seem to support such a claim as the rhotic manner and epenthesis (which according to the Colantoni and Steele's hierarchy have the greater degree of variability) were not subject to any improvement after the trainings and even the iR seemed to be in detriment (except for the DG). For the targeted phonetic features (VOT, rhotic duration, iR and epenthesis), results

showed that VOT and rhotic length are features less resistant to change when they are subject to pronunciation training. For both experimental groups, results showed a reduction in VOT in more than 60% and around 25% for the rhotic segments from Pre-Test to Post-Test.

4) As mentioned above, the iR seemed to be in detriment for the Visual Feedback Group as well as for the Control Group. Results for the Control Group are not as surprising as the fact that the Visual Feedback did not improve but it did show an increase (worsening) in the iR. However, this unexpected behavior could be explained by means that as "the difference between English [ɹ] and Spanish [ɾ] is perceptually very salient" (Colantoni & Steele, 2006, p.70), participants in the Drills Group could distinguish more clearly this difference due to being exposed to more auditory input.

5) As none of the groups actually improved in the acquisition or production of the Spanish flap, the possibilities of having epenthetic vowels in a systematic manner was practically ruled out and was confirmed by the results obtained.

Broadly, it can be concluded that the Visual Feedback and Drills groups approached to pronunciation instruction of SRCs was effective for VOT and rhotic duration. However, Drills Group performance was slightly better if considered it did not show a significant increase in the iR compared to both Visual Feedback and Control Group. However, the little to no improvement regarding the iR and epenthesis suggested that there is a difficulty acquisition hierarchy indeed for these type of complex segments. Reasons for this distinction are discussed below.

CHAPTER 5. DISCUSSION

The following chapter is structured as follows: an introduction of the research questions (RQ); introduction of the hypothesis; and answers to the RQs based on the results obtained in the experiment and previous research. The discussion of the results is introduced by experimental group (VFG, DG or CG) and also regarding the acquisition of each one of the three targeted phonetic features (VOT, rhoticity (rhotic duration and iR), and epenthesis) present in Spanish stop-rhotic clusters (SRC) /pr/ /tr/ and /kr/.

While pronunciation represents a key component of communicative competence (Morley, 1996), it has remained a largely under-studied aspect of language acquisition and teaching (Derwing & Munro, 2009). To date, several differing approaches to pronunciation have emerged, including both Visual Feedback (Ali, Lahrouchi & Ingleby, 2008; Chun, 2002; De Bot, 1980; De Bot & Mailfert, 1982; Levis & Pickering, 2004; Olson, 2014) and drills and repetitions (Acton, 1984; Paulston, 1970; 1971; 1976; Senel, 2006; Yoshida & Fukada, 2014). While preliminary research suggests that these approaches are beneficial for pronunciation instruction of segmental and suprasegmental features (Acton, 1984; Ali, Lahrouchi & Ingleby, 2008; Chun, 1998, 2002; de Bot, 1980; de Bot & Mailfert, 1982; Levis & Pickering, 2004; Olson, 2014; Paulston, 1971; Senel, 2006; Yoshida & Fukada, 2014), teachers still need systematic, empirically driven pedagogical approaches to implement within the L2 classroom (Arteaga, 2000; Derwing, Diepenbroek & Foote, 2012; Derwing & Munro, 2015; Ellis, 1997; Thomson & Derwing, 2004). Although the use of a VFP and Drills have provided evidence that they are useful to improve L2 learners' pronunciation (i.e., Olson, 2014 for VFP; Paulston, 1976 for Drills; Yoshida & Fukada, 2014 for Drills and Visual Feedback), to date no study has examined the potential of implementing a VFP and Drills into the L2 Spanish classroom to teach complex segments, such as SRC. As such, seeking to compare each of these treatments to "traditional" instruction, the first research question with its corresponding hypothesis is:

RQ1: Do Visual Feedback and Drills improve NNS accuracy on complex segments (e.g. SRC) more than “traditional” instruction? Specifically, does it improve for VOT, flapping and/or epenthesis?

Hypothesis 1: Participants will improve accuracy in the production of SRC with both types of interventions. Both interventions, Visual Feedback and Drills will improve learners’ production of the targeted segments and its phonetic features compared to the Control group which will be exposed to “traditional” in-class instruction.

The first hypothesis was supported by the current results. The results obtained in the present study provide evidence that teaching pronunciation using a VFP and Drills (and repetitions) has a positive effect on the accuracy of at least two of the targeted phonetic properties in SRC, more so than ‘traditional’ instruction. The effectiveness of the trainings and the evidence that pronunciation teaching is beneficial for L2 learners’ pronunciation could be observed after the first training. From the targeted phonetic properties (VOT, rhotic intensity and duration and epenthesis), the ones that received the highest influence of the trainings were VOT, followed by rhotic length as results showed that both experimental groups (VFG and DG) had a significant reduction in the aspiration of the voiceless stops (VOT) in both contexts (Carrier Sentence or Novel Phrases). Moreover, for each of these groups, significant improvements were seen immediately following the first (of three) trainings. Even though the CG demonstrated a slight improvement, Spanish L2 learners in the VFG and DG outperformed the CG in the production and acquisition of Spanish voiceless stops voicing.

Again, the Spanish flap evidences a lower iR with respect to its English counterpart given that the Spanish flap /ɾ/ is produced by an occlusion of a single contact of the tip of the tongue with the alveolar ridge. As a consequence, the airstream is blocked (Ladefoged & Maddieson, 1996, p. 232) and the intensity with respect to the following vowel would be reduced significantly. However, with respect to the rhotic segment, there are two aspects worth noting. First, contrary to what was expected, the intensity Ratio (iR) increased significantly from the Pre-Test to the Post-Test for the VFG and the CG, becoming less native-like. The DG did not improve in the production of the rhotic segment but slightly worsen. Even though, there was not a significant difference among groups at the Post-Test.

All of the groups seemed to rely on the orthography and L1 phonology for these complex segments rather than in making them more contrastive or less English-like. That is, participants in the VFG, DG and CG produced the approximant /ɹ/ that is characteristic of English productions of SRCs. Second, the rhotic duration was reduced as well, similar to the behavior of the voiceless stops. Reduction occurred following the first training and once more from the third training to the Post-Test.

Spanish learners seemed to have a domino effect in the SRC, the reduction in length of one of the segments spreads to other segments within the cluster, in this specific case, the rhotic. There does not seem to be a direct correlation between the iR and the rhotic length; the increase in the iR does not presuppose the increase of the rhotic duration but rather the decrease of it, according to the results obtained. The reduction of the iR would have provided evidence of the effectiveness of the trainings in the production of the flap /ɾ/ and it would have been a good predictor for the presence or absence or an epenthetic vowel. Results showed that neither group improved in the production of the epenthetic vowel. The absence of epenthetic vowels displayed in the data may be due to a direct effect from the lack of improvement of the iR. As such, a low iR (close or equal to zero) produced by the occlusion of the flap should lead the Spanish learners to epenthesize because the correct articulation of the Spanish flap /ɾ/ would promote the aerodynamic conditions to produce the epenthetic vowel. Therefore, the likelihood of producing an epenthetic vowel is highly reduced if L2 learners are not able to produce the flap. Also, the transference of English approximant /ɹ/ to Spanish SRC makes epenthesis unnecessary as a repair strategy to break these complex segments.

The fact that the VOT and rhotic duration were the only phonetic feature that had significant gains, more than those obtained for the other two phonetic features, suggest that there is in fact an acquisition difficulty hierarchy as Colantoni and Steele (2006) proposed. They proposed that the first phonetic features to be acquired should be voicing (VOT), followed by rhotic manner (flap /ɾ/) and finally, the most difficult feature would be the epenthetic vowel. In light of this, the results obtained in the present study confirm this acquisition hierarchy as both experimental groups (VFG and DG) improved significantly for the voicing feature, with aspiration reduced by more than 60% by the Post-Test. Rhotic

manner and presence of epenthesis were unaffected by either training; moreover, the measurement used for the rhotic segment (iR) got significantly worse for the VFG and slightly worse for the DG in spite of the voicing and rhotic duration shortening. However, the reduction in the length of the rhotic segment was still too long to allow a full closure and have any effect in the production of the epenthetic vowel as it was still voiced due to the transfer of the English approximant [ɹ] (also see Colantoni & Steele, 2006, p. 69). The improvement in voicing could be supported by Dalton and Seidlhofer (1994), who proposed that individual sound segments are relatively easy to teach because they may not affect comprehensibility and intelligibility even though the speaker presents a heavy accented production in the L2 (p. 73). This, to say that learners acquire the first segment that is easiest for them to produce (/p/, /t/, /k/) while complex segments such as SRC with more (at least three) and more marked features (for instance rhoticity and inter-consonantal epenthesis) "are particularly difficult to teach" and for L2 learners to acquire (Dalton & Seidlhofer, 1994, p.73). Roach (1991) also developed a similar idea regarding the ease or difficulty of acquisition of L2 features. His claim was that "the complexity of the total set of sequential and prosodic components of intonation...makes it a very difficult thing to teach" (p. 11). Therefore, and taking into consideration the results obtained in the present study, it could be said that complex segments such as SRCs should be expected to have a similar behavior. These "implicit" methods could improve individual sound segments with less marked (contrasting) features such as length. As the alveolar tap [ɾ] is already existent in the English phonology as an allophone of [t] and [d] in non-stressed syllables such in 'butter' /ˈbʌɾə/ or 'ladder' /ˈlæɾə/ (Blecua, 2001; Face, 2006; Hammond, 1999), (orthographically or not) it is probable that NSE will not associate the tap with a rhotic which could lead first to have "some difficulty of re-categorization of this sound in acquiring Spanish" (Face, 2006, p. 48). In addition, the use of explicit instruction could provide the researcher greater improvements in the acquisition (or re-association[7]) of not only the rhotic manner within the SRCs but potentially of the epenthetic vowel. For instance, the use of more explicit methods, such as mentioning items in pictures or narration in response to a written cue or reading passages (Major, 2001; Reeder, 1998;

[7] Note we do not use the term 're-categorization' proposed by Face (2006, p. 54) as the flap/tap /ɾ/ does not acquire a different category but the 'new' sound in Spanish is associated to the existing sound in English.

Tarone, 1979, 1983), have provided an improvement in accuracy as low as 3% after the first training session to 79% in the final session (Major, 2012) for the intervocalic tap /ɾ/.

While the previous research question and hypothesis detailed the potential for improvement for each of the two trainings (VFP and Drills) relative to a control group taking into account the targeted complex segments and its phonetic features, the second research question examines which of the two training groups evidenced the greatest improvement in the acquisition of the targeted segments:

RQ2: Does visual feedback or drills lead to greater improvements for NNS's accuracy on complex segments?

Hypothesis 2: Learners will produce more accurate target segments (compared with the repetitions group) when a VFP is implemented, based on the multimodal cues and increased noticing.

This hypothesis was not supported. Both trainings were equally effective in improving the accuracy of overall production of SRC phonetic features, especially the reduction of VOT and rhotic length. Positive points taking into account that learners "must learn to control length" in order to have more Spanish-like productions (Colantoni & Steele, 2006, p. 71). Many researchers have claimed that in order to acquire a L2 it is necessary for the learner to be aware of what s/he is learning. For instance, Schmidt (1990, 1993a, 1993b, 1994a), Schmidt and Frota (1986), Long and Crookes (1992) and Ellis (1993) have stressed the importance of noticing the L2 patterns in order to convert them into intake (Robinson, 1995, p. 285). Other authors have proposed that the awareness a L2 learner has, is based in pattern recognition and signal detection (Baddeley, 1986, p. 225; Robinson, 1995), and even that it presents different modalities in which auditory versus visual perception are included (Wickens, 1980, 1984, 1989). Therefore, noticing differences between L1 and L2 patterns could make a difference in acquiring effectively L2 speech patterns. None of the interventions had greater results than the other regarding VOT, rhotic length or epenthesis from Pre-Test to Post-Test. However, even though the iR did not show improvement and it actually produced results that were contrary to the ones expected, the DG could maintain similar values from the Pre-Test to Post-Test, not having a significant difference as the VFG and CG did regarding Pre-Test to Post-Test values. As

mention before, the rhotic length decreased significantly but it did not show any correlation to the improvement of the acquisition of the rhotic manner nor epenthesis. The last feature, epenthesis, was the phonetic feature with the least improvement.

Considering the above results, both trainings were equally effective (and/or ineffective) for various reasons: First, the SRCs contain phonetic features that present a greater difficulty for L2 learners of Spanish: VOT and rhotic length are more categorical while rhotic manner and epenthesis are less categorical. This distinction supports Colantoni and Steele's hierarchy which says that the voicing should be acquired before the rhotic manner before the epenthesis is produced. Second, the types of instruction focused on training and making the Spanish learners to be acquainted of different characteristics of the targeted segments: visual versus acoustic. For instance, a more categorical feature such as VOT was more noticeable in the spectrograms used by the VFG, while rhotic manner could have been more perceptible for the DG which was exposed to more auditory input. This could be a possible explanation on why the DG presented an intermediate (negative) difference for Carrier Sentences and Novel Phrases but not significantly detrimental as in the case of the Visual Feedback and Control groups. Colantoni and Steele (2006) concluded that for mastering the phonetic features of SRCs, the researcher must look at perception (i.e., input) and production (i.e., articulatory constraints). By means of the present study, one of the trainings focused more on visual perception (VFG) and the other one in production (DG) which could explain why both experimental groups obtained similar results for each phonetic feature and after each training (except for the iR).

Whereas the two previous research questions and hypotheses showed the benefits of teaching pronunciation within the L2 classroom using two different types of training (VFP and Drills), the following research question and hypothesis will examine accuracy change in each one of the phonetic features of the targeted segments.

RQ3: Does accuracy change apply to each of the segments of the complex cluster similarly (e.g. VOT, Rhoticity, Epenthesis)? Do these changes follow the difficulty acquisition hierarchy?

Hypothesis 3: Most learners in the experimental groups will produce closer VOT values compared to the Spanish stops in both groups. A small percentage of participants

will produce the flap ‘r’ and an even smaller percentage of participants will epenthesize; suggesting the existence of a difficulty acquisition hierarchy in Spanish SL clusters: VOT> Rhoticity > Epenthesis as proposed by Colantoni & Steele (2006).

This hypothesis was partially substantiated. Both experimental groups moved their VOT values towards more Spanish-like productions, reducing aspiration by more than 60% from Pre-Test to Post-Test for both Carrier Sentences and Novel Phrases. For rhotic length, the trainings also had a positive effect for both experimental groups and surprisingly, there was a significant difference for the Control Group in the Carrier Sentences. For rhotic iR, both Visual Feedback Group and Control Group presented a negative difference that was significant from Pre-Test to Post-Test. Although Drills Group did not have significantly negative difference from Pre-Test to Post-Test for the iR, the iR was not significantly different with regards to the Visual Feedback and the Drills Group. Drills Group showed a significant difference from Pre-Test to Training 1 for Novel Phrases but an intermediate difference from Pre-Test compared to the other trainings or Post-Test. Regarding the presence or absence of the epenthetic element, there was not a significant difference for either group although it could be seen a very small tendency for improvement in the Drills Group in both Carrier Sentences and Novel Phrases which can be related to the fact that the iR was slightly lower among the three groups. Results obtained in the study suggest that there is not a difficulty acquisition hierarchy for Spanish SL clusters as Colantoni and Steele claimed (2006). Instead, it seems to be a continuum in the acquisition of the SRC phonetic properties that parallel the order of acquisition proposed by Colantoni and Steele (2006). Results showed that Spanish learners acquire VOT along with rhotic duration, then rhotic manner and finally, the epenthetic vowel. However, if the hierarchy would follow a strict order of acquisition of the phonetic features, it would be basically impossible to see that the Control Group had significant gains for rhotic length in the Carrier Sentences while it did not improve in any other feature, or in Carrier Sentences or Novel Phrases.

To sum up, of the three research questions (RQs) made at the outset, the first two favored pronunciation training over “traditional” instruction for each one of the phonetic features proposed (VOT, rhoticity and epenthesis). The second RQ showed that neither VFP or Drills is more effective than the other, but both trainings seem to be equally

effective for acquiring the stop-rhotic clusters phonetic features. The final RQ provides evidence that Spanish L2 learners may follow a difficulty hierarchy when acquiring the aimed phonetic features for stop-rhotic clusters (SRC) due to language-specific constraints.

CHAPTER 6. CONCLUSION

The following chapter introduces the pedagogical implication (Section 6.1), limitations (Section 6.2), future directions (Section 6.3) and a general conclusion (Section 6.4) of the study.

6.1 Pedagogical Implications of the study

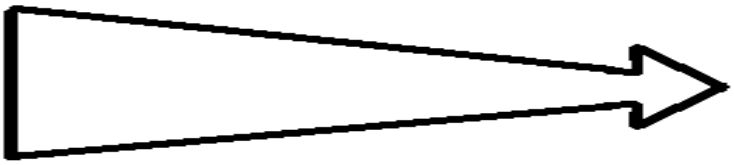
Some of the findings from this research project lead to conclusions regarding the application of different pronunciation teaching methods within the L2 classroom. pronunciation instruction. Three major considerations are detailed here.

Pronunciation training is effective as well as necessary in order to improve L2 learners acquisition and mastery of phonetic features in SRC. Some studies have previously indicated that the use of drills and repetitions are effective to improve the accuracy of L2 speech (Paulston, 1970, 1971; Yoshida & Fukada, 2014). Similarly, the use of VFP have provided positive results for acquiring suprasegmentals (Chun, 1989, 1998; De Bot & Mailfert, 1982; De Bot, 1983; Hardison, 2004; Weltens & De Bot, 1984) and segmentals (Motohashi-Saigo & Hardison, 2009; Carey, 2004; Olson, 2014). For this specific study, both Drills and VFP had positive effects in the acquisition and improvement of SRC phonetic features (VOT, rhotic manner and duration and epenthesis). Having established that pronunciation instruction is effective for acquiring SRC phonetic features, the efficiency in the acquisition of more native-like speech regarding SRC in Spanish did not seem to be conditioned by the type of training, but rather by the difficulty of acquisition of each phonetic feature; less categorical, more resistant, more categorical, less resistant to be affected by pronunciation training. Interestingly, the two phonetic properties that were more susceptible to the trainings are linked to durational parameters, this is VOT and rhotic duration, and it may be an effect of perceived auditory and visually differences. However, rhotic manner and epenthesis, tended to be less marked properties and were more resistant to change as consequence of the trainings.

Second, taking into account the difficulty acquisition hierarchy proposed by Colantoni and Steele (2006), the phonetic feature that is least resistance to change by means of the

trainings is the one that should be mastered at first (VOT), while the more resistant features are at the middle (rhotic manner) and bottom (epenthesis) of the hierarchy. Although Colantoni and Steele (2006) do not include rhotic length as part of the hierarchy, it could be included as part of the 'rhotic category' having two subcategories that are 'rhotic duration' and 'rhotic manner'. This may indicate that durational values (voicing or rhotic duration) should be acquired first before the rhotic manner and epenthesis can be mastered. More important, as a 'hierarchy' depends on ranked values and a 'continuum' is a progression of values, one could hypothesize that for SRC, there seems to be an 'acquisition difficulty continuum' rather than a 'acquisition difficulty hierarchy' (proposed by Colantoni & Steele, 2006) as VOT and rhotic duration improved significantly while iR and epenthesis did not show significant improvements, though DG showed a very small trend to improvement regarding these two features. Then, a hierarchy like Colantoni and Steele's (2006) (voicing> rhotic manner>epenthesis) could be presented in a SRC acquisition difficulty continuum such as:

Voicing -Rhotic duration - Rhotic manner - Epenthesis



Finally, it is worth noting that two distinct phases of improvement can be identified: first, after receiving the first training and second, before the last training session. Learners displayed significant improvements in the reduction of VOT and rhotic duration after the first training session but following that, the length of these features increased (slightly and considerably). However, before the last training session, the duration of the VOT and the rhotic segment shorten significantly once again and even more than after the first training session. It is possible that before achieving further greater gains in the acquisition of phonetic features, there could be a significant 'worsening' in the production of the targeted features. Therefore, for the results obtained in the present study regarding iR and epenthesis, it could mean that getting 'worse' may be a stepping stone or an articulatory

strategy to re-accommodation the L1 parameters to the new input (Ramírez, 2006; Norris & Ortega, 2000; White, Spada, Lightbown & Ranta, 1991) before producing the Spanish flap /ɾ/ and epenthesis.

6.2 Limitations

Even though the results presented above provide evidence that explicit pronunciation instruction is effective in acquiring some of the phonetic features in complex segments (SRC), that visual and auditory methods seem to be equally reliable for assessing the pronunciation of Spanish SRC and that there seems to be an acquisition difficulty continuum for this type of complex segments, some limitations have surfaced:

First, when preparing the tokens that were meant to be used in each one of the training sessions, there were a large number amount of words that are cognates in Spanish and English, especially with the vowel /a/. All the tokens containing a combination of a SRC plus 'a', were deleted in order to avoid the transfer of properties from the L1 to the L2. While it is unlikely that this would have impacted the outcome, a balanced design would contain a uniform number of tokens from each of the relevant conditions.

Second, while the current study employed a control group of native English speakers, who did receive any explicit instruction, it would have been useful to use a control group in the target language (Spanish) to be able to compare what is the degree of nearness, percentage of variation, and epenthesis between the experimental groups and the control groups.

Finally, duration was not normalized in the current study. However, in order to account for the differences in speech rate for the different speakers and have more precise results regarding the durational parameters for each segment with respect to the total length of the token, normalizing the duration would be appropriate. Normalization could shed light on the durational properties of the rhotic segment, in particular.

6.3 Future Directions

First, noting that there seems to be an acquisition difficulty continuum of the phonetic properties of SRC, the researcher would include some time in the first training session to provide explicit information about what are the features that differ in English and Spanish. Although there were positive effects for implementing pronunciation

instruction using Visual Feedback and Drills, it would be beneficial for students to not only implement a visual paradigm and drills within the classroom but also corrective feedback that would ensure that learners are coping with the aimed parameters. Although the goal of not telling students the targets of this study was to avoid overt pronunciation or any other negative effects related to awareness of the L2 speech, it seems necessary to define some basic parameters that allow the learners to focus on receiving smaller pieces of information. As such, including explicit instruction may ultimately better serve learners.

For future studies, the researcher would instruct the learners in the word-level rather than in the phrase level, thus learners have a less new information on which to focus. Regarding the amount of time spent in pronunciation instruction within the classroom, the researcher could propose two courses of action based on the positive results obtained in this studies: one local, one global. In the local level, the researcher could recommend that students have a specific time assigned to just receive pronunciation instruction. At the global level, it would be more than useful to include pronunciation in the syllabus design along with the activities that are being currently implemented.

6.4 General Conclusion

The present study has provided evidence that Spanish L2 learners benefit from receiving pronunciation instruction relative to a non-instruction group. For the production of SRC, the VFG and DG showed significant improvements in the reduction of aspiration of the voiceless plosives /p/ /t/ and /k/ represented in shorter VOTs even after the first training. Even though the iR and epenthesis showed more resistance to training (VFP or drills), the results still show that there is some influence of the trainings on the production of the targeted tokens when compared to the control group. Interestingly, there was no significant difference in the contexts where the tokens where placed (Carrier Sentences or Novel Phrases). Although the extent of the improvements may be limited to the complexity of the segments being taught and their phonetic variability, results indicated that both Visual Feedback and Drills are effective methods in teaching pronunciation for L2 learners and that explicit instruction is helpful for attaining more native-like speech.

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APPENDIX A: CROSS-LINGUISTIC EPENTHESIS

List of languages with presence of an epenthetic segment in Hall (2006)

Armenian	/tʰusdɹ/	tʰusdəɹ	'daughter'
Breton	/arxant/	ara̯xãnt	'silver'
Bulgarian	/gɐ̃ɹbav/	gəɹə̯baf	'hunchbacked'
Chamicuro	/tuʔlu/	tuʔu̯lu	'chest'
Dutch	/kalm/	kaləm	'quiet'
English (various dialects)	/arm/	arəm	'arm'
Finnish	/kalvo/	kala̯vo	'transparency'
German	/bʁatən/	bə̯ʁatən	'to fry'
Hausa	/kurkutu/	kʷuru̯kʷu:tu	'small drum'
Hocank	/ho:ʧɐ̃k-ra/	ho:ʧɐ̃gə̯rə	'the Hocank'
Hua	/okrumaʔ/	oku̯rumaʔ	'sky'
Imdlawn Tashlhiyt Berber	/smd-x/	smdə̯x	'I added'
Irish	/gʲlʲaun/	gʲi̯lʲaun	'valley'
Kekchi	/paʔt/	paʔa̯t	'twins'
Kera	/kɛtpɛŋ/	kɛtəpɛŋ	'chatted'
Lakhota	/wablake/	wabəlake	'I saw'
Mamainde	/mih+takxu/	mihi̯takxu	'it is cloudy'
Mokilese	/pwedla/	pwɛti̯la	'lucky'
Mono	/gɐ̃frũ/	gɐ̃fũ̯rũ	'mortar'
Moroccan Colloquial Arabic	/katb/	kata̯b	'write (ACT PART)'
Piro	/ʃjo/	ʃijo	'bat'
Popoluca	/itʔa/	ita̯ʔa	'your father'
Saami	/skuol:fi:/	skũol:ə̯fi:	'owl'
Sanskrit	/darʃata/	darə̯ʃata	(no gloss)
Scots Gaelic	/ʃaLk/	ʃaLa̯k	'hunting'
Sierra Popoluca	/miɲpa/	miɲə̯paʔ	'he comes'
Spanish (Chilean)	/kronika/	ko̯ronika	'chronicle'
Tiberian Hebrew	/ʃalaħ-t/	ʃalaħa̯t	'you (FEM SG) sent'
Upper Chehalis	/q'ʷoɬweʔ/	q'ʷoɬə̯weʔ	'maple'

APPENDIX B. TIME CONTROL FORM

Level SPAN 201		Section		
Name				
Class time				
Activity #	Group (VFG, DG, CG)			
Time of Beginning	Time of End	Total time	What did you do? (Recorded, screenshot, find the word)	How much time did it take you to find the six word to print?

APPENDIX C. PRE-TEST ACTIVITY FOR VFG

Actividad de Pronunciación #1

Pre-Actividad

Vas a grabar la lista completa de oraciones y frases y mandarlas al correo electrónico de tu profesor. Trae a la clase una imagen visual (screenshot) de tus palabras de la lista de frases (Phrases) que contienen ‘pr’, ‘tr’ o ‘cr’

1) Antes de Comenzar:

Descarga el programa PRAAT de internet para Macintosh o Windows (praat.org)
No tardará mucho ni ocupará mucha memoria de tu computador.

2) Durante la Grabación:

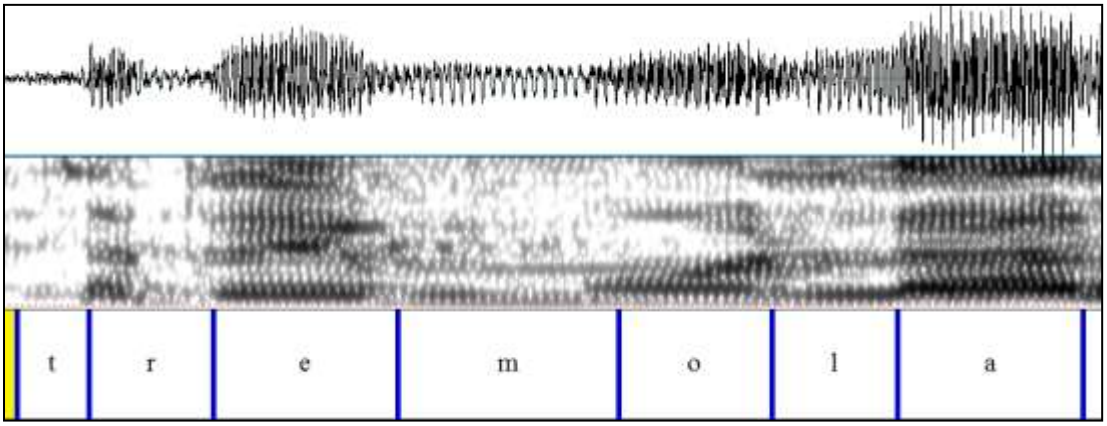
- Entra en una habitación silenciosa, de esto dependerá la calidad de tu grabación.
- Conecta tus audífonos o micrófono.
- Abre PRAAT y cierra la ventana de color rosa/rojo (Praat Picture) y deja abierta la venta ‘Praat Objects’.
- Haz click en el botón “New” y selecciona “Record Mono Sound”
- En la ventana “Sound Recorder” no hay que cambiar nada, sólo presiona “Record”
- Pronuncia de la forma más natural toda la lista de frases (Phrases).
- Repite la lista de frases tres veces en la misma grabación.
- Una vez termines con las frases (Phrases), presiona “Stop”.
- Presiona “Save to list”.
- Cierra la ventana “Sound Recorder” y vuelve a la ventana “Praat Objects”.

3) Después de la Grabación PASO I :

- Selecciona tu nueva grabación (Untitled Sound) y presiona “Save” y luego “Save as .WAV file”
  - Guarda tu grabación con tu nombre, seguido de “Phrases”
    Ejemplo: Hernandez**Phrases1**.wav
  - Asegúrate de adjuntar (make sure you attached) tu grabación y manda la grabación a tu profesor al correo: anajhernandezm@gmail.com

- Selecciona tu grabación (Untitled Sound) en la ventana de ‘Praat Objects’ y da click en “View and Edit”
  - Ahora tienes una imagen de las palabras que grabaste!
- Usando el botón ‘in’ y el ‘scroll bar’, mueve la imagen hasta que veas la imagen de tu primera palabra.
- Selecciona tu palabra y dale click a “Sel”. Las ondas (*waves*) de tu palabra deben llenar toda la ventana.
- Haz un ‘screenshot’ de la primera palabra por cada combinación de ‘pr’, ‘tr’ o ‘cr’. Ejemplo: ‘prestó’, ‘tremola’, ‘cronista’
- Pásalas a un documento Word e imprímelas.
- A MANO, marca cada una de las letras y vocales de cada palabra y establece los límites (*boundaries*) de cada una.
  - Debes escuchar MULTIPLES veces tu grabación, ¡eso te ayudará!
- Al final debes obtener una imagen como la siguiente:

Palabra ‘tremola’ en la oración ‘Ella *tremola* la bandera’



- **Trae las 6 imágenes a la clase el día 21 de Septiembre de 2015**

4) Después de la Grabación PASO II :

- Graba las oraciones (sentences) usando el mismo proceso de arriba.
- Selecciona tus grabaciones y presiona “Save as WAV file”.
- Guarda tu grabación con tu apellido seguido de “Sentences”
  Ejemplo: Hernandez**Sentences1**.wav
- Manda tu grabación a tu profesor al correo electrónico: anajhernandezm@gmail.com
- NO tienes que imprimir estas oraciones.

Lista # 1

Phrases	Sentences
La primavera es mi estación favorita	Digo presupuesto de nuevo
El cuervo crocita en el atardecer	Digo crisol de nuevo
Hubo un grande tropel allá	Digo tremola de nuevo
Ese tipo de crucigramas me gustan	Digo primera de nuevo
Un virus bastante pruriginoso causa una afección cutánea	Digo creciente de nuevo
No trepemos esa puerta	Digo triturar de nuevo

Actividad de Pronunciación #1

Actividad en Clase

1) En grupos de 3 o 4, mira la imagen de la palabra ‘primavera’ que tu grabaste.

- ¿Cómo decidiste marcar los límites de cada sonido?

  _____

  _____

- ¿Qué características visuales tienen tu ‘p’ y tu ‘r’?

  _____

  _____

- ¿Es tu ‘p’ más larga o más corta que la de tus compañeros?

  _____

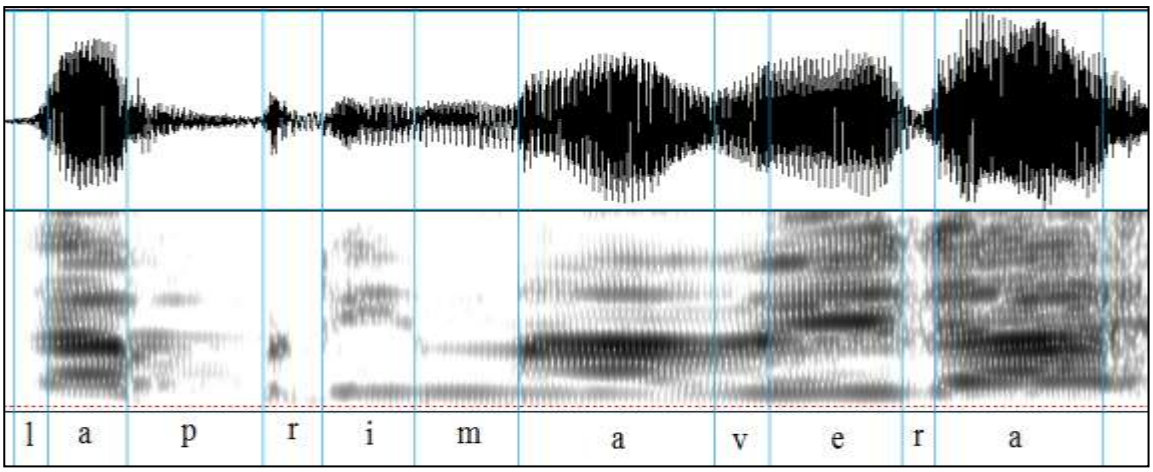
  _____

- ¿Es tu ‘r’ más larga o corta que la de ‘r’ de tus compañeros? ¿Es tu ‘r’ más clara (lighter) o más oscura (darker) que la de ellos?

  _____

  _____

1.1) Ahora vas a ver la palabra pronunciada por un nativo. Luego, compara tu palabra con la palabra de un nativo de español.



- ¿Qué características visuales tienen la ‘p’ y la ‘r’ del hablante nativo?

  _____

  _____

- ¿Es tu ‘p’ más larga o más corta que la del hablante nativo? ¿Es más clara o más oscura que la del hablante nativo?

  _____

  _____

- ¿Es tu ‘r’ más larga o más corta que la del hablante nativo? ¿Es más clara o más oscura que la del hablante nativo?

  _____

  _____

- ¿Crees poder identificar la diferencia auditoria entre tu ‘p’ y tu ‘r’ y la del hablante nativo? ¿Cómo?

  _____

  _____

1.2) Ahora escucha una grabación de la palabra “pruriginoso” producido por un hablante nativo de español y por un hablante nativo de inglés. ¿Cuál es la diferencia **auditoria**? ¿Cómo puedes describirla?

_____

_____

2) En grupos de 3 o 4, mira la imagen de la palabra ‘crocita’ que tú grabaste.

- ¿Cómo decidiste marcar los límites de cada sonido?

  _____

  _____

- ¿Qué características visuales tienen tu ‘c’ y tu ‘r’?

  _____

  _____

- ¿Es tu ‘c’ más larga o más corta que la de tus compañeros?

  _____

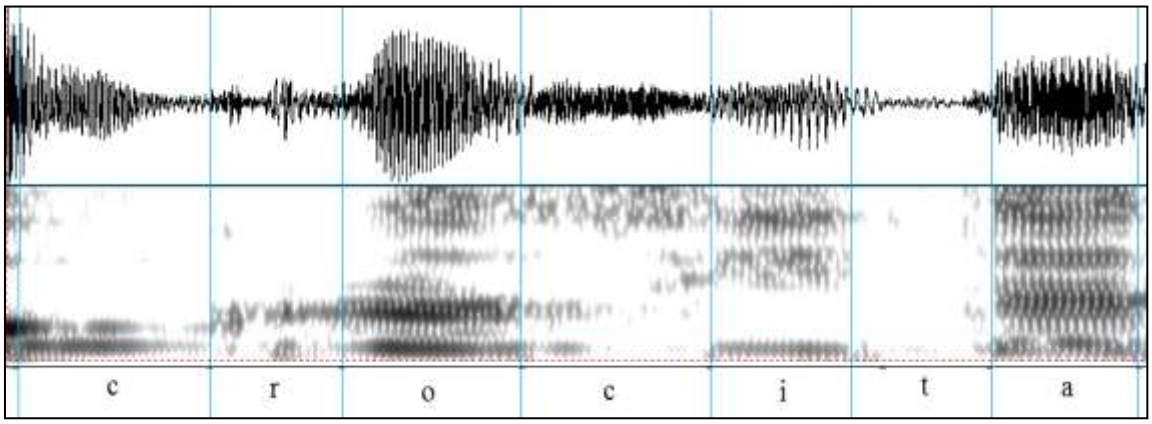
  _____

- ¿Es tu ‘r’ más larga o corta que la de ‘r’ de tus compañeros? ¿Es tu ‘r’ más clara (lighter) o más oscura (darker) que la de ellos?

_____

_____

2.1) Ahora, vas a ver la palabra ‘crocita’ pronunciada por un nativo. Luego, compara tu palabra con la palabra de un nativo de español.



- ¿Qué características visuales tienen la ‘c’ y la ‘r’ del hablante nativo?

_____

_____

- ¿Es tu ‘c’ más larga o más corta que la del hablante nativo?

_____

_____

- ¿Es tu ‘r’ más larga o más corta que la del hablante nativo? ¿Es más clara o más oscura que la del hablante nativo?

_____

_____

- ¿Crees poder identificar la diferencia **auditoria** entre tu ‘c’ y tu ‘r’ y la del hablante nativo? ¿Cómo?

_____

_____

2.2) Ahora escucha una grabación de la palabra “crucigrama” producido por un hablante nativo de español y por un hablante nativo de inglés. ¿Cuál es la diferencia **auditoria**? ¿Cómo puedes describirla?

_____

_____

3) En los mismos grupos de 3 o 4, mira la imagen de la palabra ‘tropel’ que grabaron.

- ¿Cómo decidiste marcar los límites de cada sonido?

_____

_____

- ¿Qué características visuales tienen tu ‘t’ y tu ‘r’?

_____

_____

- ¿Es tu ‘t’ más larga o más corta que la de tus compañeros?

_____

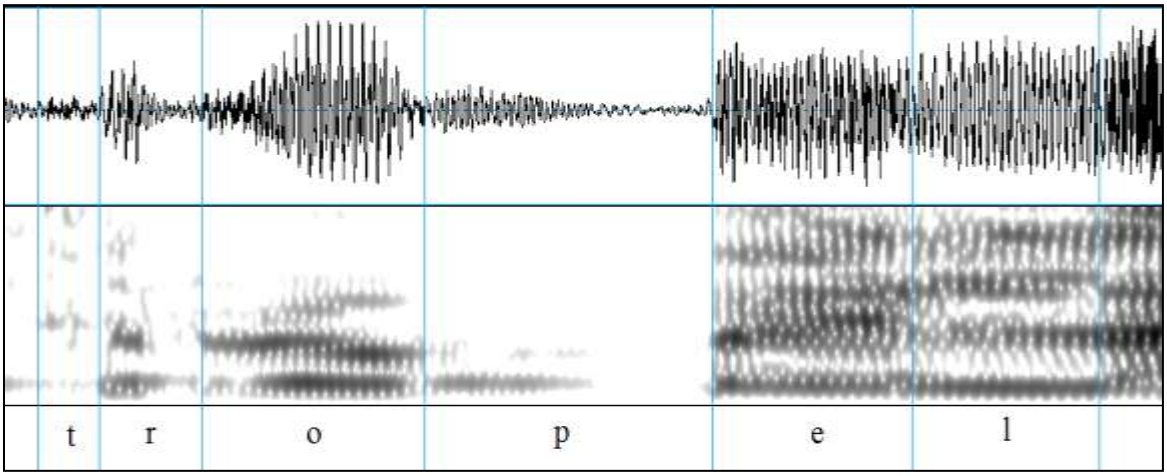
_____

- ¿Es tu ‘r’ más larga o corta que la de ‘r’ de tus compañeros? ¿Es tu ‘r’ más clara (lighter) o más oscura (darker) que la de ellos?

_____

_____

2.3) Ahora, vas a ver la palabra ‘tropel’ pronunciada por un nativo. Luego, compara tu palabra con la palabra de un nativo de español.



- ¿Qué características visuales tienen la ‘t’ y la ‘r’ del hablante nativo?

  _____

  _____

- ¿Es tu ‘t’ más larga o más corta que la del hablante nativo?

  _____

  _____

- ¿Es tu ‘r’ más larga o más corta que la del hablante nativo? ¿Es más clara o más oscura que la del hablante nativo?

  _____

  _____

- ¿Crees poder identificar la diferencia **auditoria** entre tu ‘t’ y tu ‘r’ y la del hablante nativo? ¿Cómo?

  _____

  _____

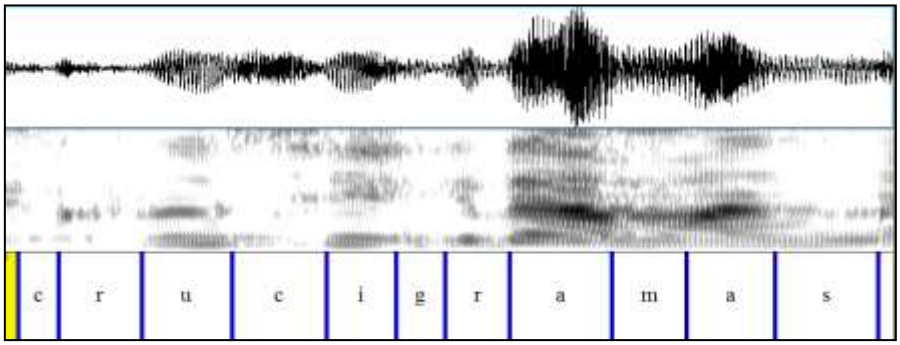
3.1) Ahora escucha una grabación de la palabra “trepemos” producido por un hablante nativo de español y por un hablante nativo de inglés. ¿Cuál es la diferencia **auditoria**? ¿Cómo puedes describirla?

_____

_____

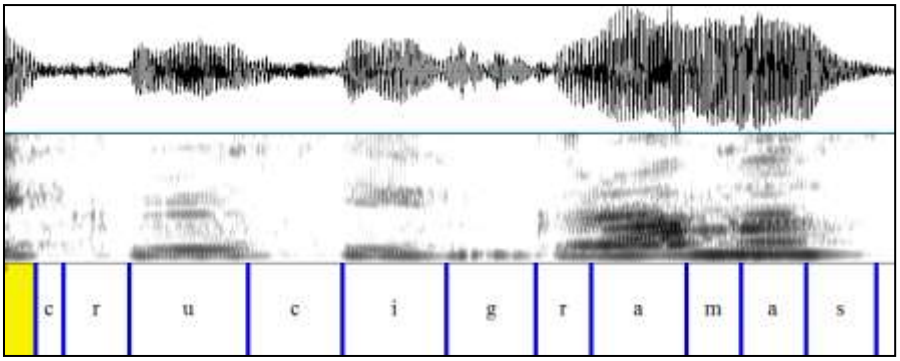
4) Pruebita.

Identifica las palabras que han sido pronunciadas por un hablante nativo de español y cuales han sido pronunciadas por un hablante nativo de inglés**.**



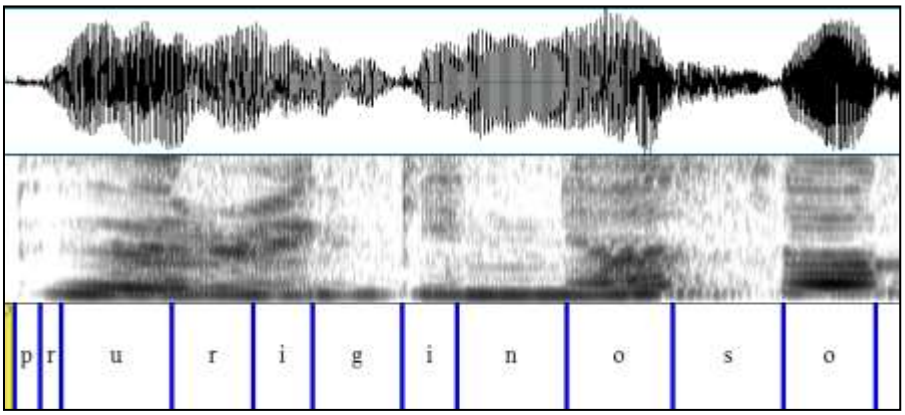
Nativo ☐

No nativo ☐



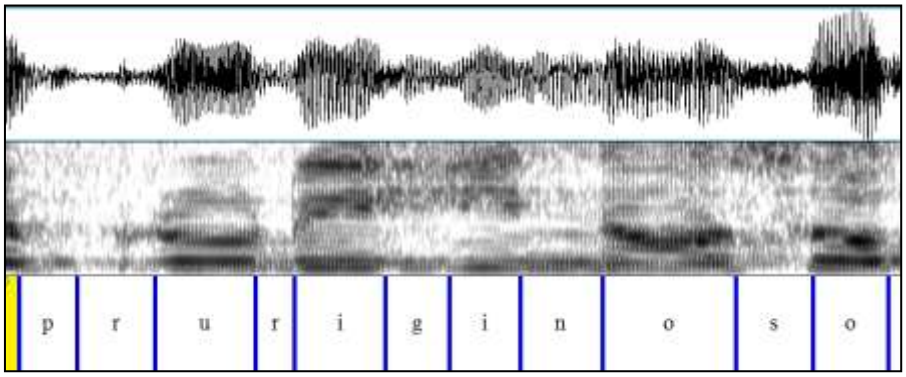
Nativo ☐

No nativo ☐



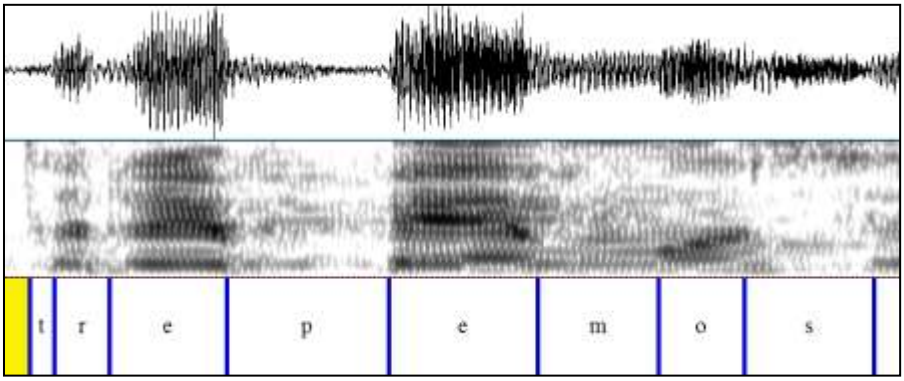
Nativo ☐

No nativo ☐



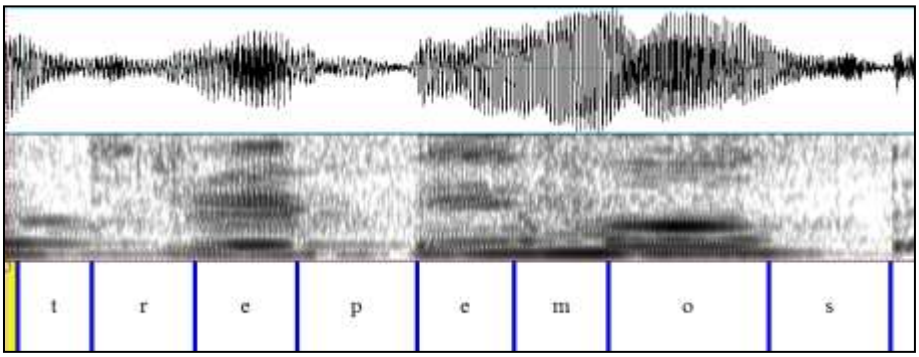
Nativo ☐

No nativo ☐



Nativo ☐

No nativo ☐



Nativo ☐

No nativo ☐

Actividad de Pronunciación #1

Tarea en casa

1) Graba la lista #2 de frases y oraciones (Phrases and Sentences), intentando imitar a los hablantes nativos de español. Puedes intentarlo múltiples veces y mirar las imágenes. Graba cada lista tres veces en el mismo archivo de sonido.

2) Sigue las instrucciones de la 'Primera Sesión'.

- Guarda tu grabación con tu nombre, seguido de "Phrases2"
  Ejemplo: Hernandez**Phrases2**.wav
- Manda la grabación a tu profesor al correo electrónico anajhernandezm@gmail.com y asegúrate de adjuntar tu grabación (make sure you attach the file!).

3) Trae las siguientes 6 imágenes a la clase el día 14 de Octubre de 2015

4) Graba las Oraciones (Sentences) usando el mismo proceso de arriba.

- Selecciona tus grabaciones y presiona "Save as WAV file".
- Guarda tu grabación con tu nombre seguido de "Sentences2"

Ejemplo: Hernandez**Sentences2**.wav

5) Manda tu grabación a tu profesor por correo electrónico a anajhernandezm@gmail.com.

Lista #2

Phrases	**Sentences**
La sal de prusiato está compuesta de elementos químicos	Digo crisol de nuevo
Si no trepáis el árbol, no alcanzaréis al gato	Digo presupuesto de nuevo
La bebé probeta nació hoy	Digo creciente de nuevo
La casa crujió en la tormenta	Digo triturar de nuevo
Me tronché el dedo	Digo tremola de nuevo
Uso la red para cribar el trigo	Digo primera de nuevo

APPENDIX D. PRE-TEST ACTIVITY FOR DG

Actividad de Pronunciación #1

Actividad en Clase

PASO I:

1) Vas a escuchar la grabación que tu profesor va a reproducir. Es necesario que escuches con atención y que pronuncies en voz alta (out loud) las oraciones.

2) Ahora en grupos de 3 o 4 estudiantes respondan las siguientes preguntas:

- ¿Cuál es el sonido/ segmento que más se dificulta pronunciar? ¿Por qué?

  _____

  _____

- ¿Qué características auditorias tiene tu ‘p’? ¿Es larga o corta?

  _____

  _____

- ¿Qué características auditorias tiene tu ‘r’? ¿Cómo la describes?

  _____

  _____

- ¿Qué características auditorias tiene la ‘p’ del hablante nativo?

  _____

  _____

- ¿Qué características auditorias tiene la ‘r’ del hablante nativo?

  _____

  _____

- ¿Cuál es la diferencia entre tu ‘pr’ respecto a la del hablante nativo? ¿Cómo reconoces esta diferencia?

  _____

  _____

- ¿Crees que tu ‘p’ es igual o diferente que la del hablante nativo? ¿Cómo?

  _____

  _____

- ¿Crees que tu ‘r’ es igual o diferente que la del hablante nativo? ¿Cómo?

  _____

  _____

- ¿Qué crees que podrías para mejorar la pronunciación de tu ‘pr’¿Cómo?

  _____

  _____

3) Ahora repite de nuevo las frases y oraciones teniendo en cuenta lo que escribiste anteriormente.

- ¿Crees que pronunciaste la ‘p’ y la ‘r’ más similar a las del nativo esta vez?¿Por qué?

  _____

  _____

PASO II:

1) Vas a escuchar la grabación que tu profesor va a reproducir. Es necesario que escuches con atención y que pronuncies en voz alta (out loud) las oraciones.

2) Ahora en grupos de 3 o 4 estudiantes respondan las siguientes preguntas:

- ¿Cuál es el sonido/ segmento que más se dificulta pronunciar? ¿Por qué?

  _____

  _____

- ¿Qué características auditorias tiene tu ‘t’? ¿Es larga o corta?

  _____

  _____

- ¿Qué características auditorias tiene tu ‘r’? ¿Cómo la describes?

  _____

  _____

- ¿Qué características auditorias tiene la ‘t’ del hablante nativo?

  _____

  _____

- ¿Qué características auditorias tiene la ‘r’ del hablante nativo?

  _____

  _____

- ¿Cuál es la diferencia entre tu ‘tr’ respecto a la del hablante nativo? ¿Cómo reconoces esta diferencia?

  _____

  _____

- ¿Crees que tu ‘t’ es igual o diferente que la del hablante nativo? ¿Cómo?

  _____

  _____

- ¿Crees que tu ‘r’ es igual o diferente que la del hablante nativo? ¿Cómo?

  _____

  _____

- ¿Qué crees que podrías para mejorar la pronunciación de tu ‘tr’¿Cómo?

  _____

  _____

3) Ahora repite de nuevo las frases y oraciones teniendo en cuenta lo que escribiste anteriormente.

- ¿Crees que pronunciaste la ‘t’ y la ‘r’ más similar a las del nativo esta vez?¿Por qué?

  _____

  _____

PASO III:

1) Vas a escuchar la grabación que tu profesor va a reproducir. Es necesario que escuches con atención y que pronuncies en voz alta (out loud) las oraciones.

2) Ahora en grupos de 3 o 4 estudiantes respondan las siguientes preguntas:

- ¿Cuál es el sonido/ segmento que más se dificulta pronunciar? ¿Por qué?

  _____

  _____

- ¿Qué características auditorias tiene tu ‘c’? ¿Es larga o corta?

  _____

  _____

- ¿Qué características auditorias tiene tu ‘r’? ¿Cómo la describes?

  _____

  _____

- ¿Qué características auditorias tiene la ‘c’ del hablante nativo?

  _____

  _____

- ¿Qué características auditorias tiene la ‘r’ del hablante nativo?

  _____

  _____

- ¿Cuál es la diferencia entre tu ‘cr’ respecto a la del hablante nativo? ¿Cómo reconoces esta diferencia?

  _____

  _____

- ¿Crees que tu ‘c’ es igual o diferente que la del hablante nativo? ¿Cómo?

  _____

  _____

- ¿Crees que tu ‘r’ es igual o diferente que la del hablante nativo? ¿Cómo?

_____

_____

- ¿Qué crees que podrías para mejorar la pronunciación de tu ‘cr’¿Cómo?

_____

_____

3) Ahora repite de nuevo las frases y oraciones teniendo en cuenta lo que escribiste anteriormente.

- ¿Crees que pronunciaste la ‘t’ y la ‘r’ más similar a las del nativo esta vez?¿Por qué?

_____

_____

APPENDIX E. ‘PRUEBITA’ FOR THE DG

Frases	Nativo de español	Nativo de inglés
La primavera es mi estación favorita		
El cuervo crocita en el atardecer		
Hubo un grande tropel allá		
Ese tipo de crucigramas me gustan		
Un virus bastante pruriginoso causa una afección cutánea		
No trepemos esa puerta		

APPENDIX F. EXTRA-ACTIVITY 1 FOR CG

Homework to send via email. Don Quijote de la Mancha: Capítulo 1

Nombre: _____ **Fecha:** _____

1) Mira este video con mucha atención: https://www.youtube.com/watch?v=ZAOIITBbHzY
2) Luego utilizando el vocabulario y el presente simple responde a las siguientes preguntas:
   - Describe físicamente al personaje principal
     _____
     _____
   - ¿Cómo es el escritorio del personaje principal?
     _____
     _____
   - ¿Qué acostrumbra hacer Don Quijote en sus momentos de ocio y en su rutina diaria?
     _____
     _____
   - Escribe la rutina diaria de Don Quijote.
     _____
     _____
   - ¿Qué le sucedió al Quijote por dormir poco y leer mucho?
     _____
     _____
   - ¿Por qué se quiere convertir en un caballero?
     _____
     _____
   - ¿A dónde quiere ir Don Quijote y cuál es el objetivo de su viaje?
     _____
     _____
   - ¿Cómo se llama el caballo de Don Quijote? Escucha con atención cuando lo dice.
     _____
     _____
   - ¿Por qué crees que escogió el nombre para su caballo?
     _____
     _____